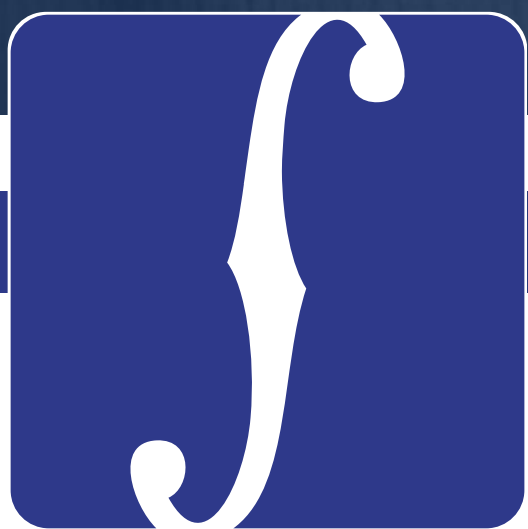




CELLO

THE U.S. HOUSING AND
MORTGAGE MARKET COMPANION

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Preface

The Housing Crisis of 2007-2009 refocused attention on the central role that housing and mortgage markets play in the economy. The *Companion* was born out of a need for a digestible account of core aspects of the housing and mortgage markets without being overwhelmingly encyclopedic in scope. Needless to say, there are many ways to present the material and the slant chosen here is inevitably colored by a markets-oriented focus, since this is Cello Capital's area of expertise. Another distinct bias in the coverage of the *Companion* is its focus on the single-family residential mortgage market. This is by far the largest sector of the overall mortgage market but the multifamily residential and commercial property markets also play an important role in the economy. It is hoped that subsequent versions of the *Companion* will devote more space to these and other areas.

Although the *Companion* is focused on the U.S., some attempt is made at placing the country's housing and mortgage markets within a broader global context so that some of the choices and trade-offs that have been made in terms of the design of the U.S. housing finance system can be better understood. Having said that, at the time of writing, it appears quite likely that the architecture of housing market is likely to undergo a fundamental reshaping in future years. The two most significant U.S. mortgage market entities, Fannie Mae and Freddie Mae, are currently under conservatorship, and the prevailing consensus is that a return to the quasi-private model that these two institutions followed prior to the Housing Crisis is not a viable option.

The organization of the *Companion* follows a top-down approach which begins by emphasizing the importance of housing and mortgage markets from a macroeconomic standpoint (Chapter 1). The goal here is to delineate how the health (or lack thereof) of these markets flows through to the rest of economy (and vice-versa). The next objective is to survey the institutional infrastructure that supports residential real estate mortgage markets (Chapter 2). It is somewhat challenging to come to grips with this collection of institutions without placing them in some sort of historical context so the chapter also adds a brief narrative account of how the architecture of the U.S. system of housing finance has evolved over time. As part of the journey towards understanding the mortgage markets in greater detail, Chapter 3 characterizes some of the most commonly utilized mortgages. Chapter 4 provides a detailed overview of the U.S. mortgage markets. From the vantage point of an investor in mortgages, the most important feature of these loans is the homeowner's prepayment option. Chapter 5 has a detailed review of homeowner prepayment behavior. Given this general backdrop, Chapter 6 takes an initial step into exploring the intricacies of the market for securities

backed by collections of mortgages by reviewing how securitizations work.

Three additional chapters in varying stages of development that were not completed in time for this draft are:

- Chapter 7: U.S. MBS Markets
- Chapter 8: Linking Primary and Secondary Markets
- Chapter 9: Pricing and Hedging MBS

Chapter 1

The Importance of Housing

Housing has played a critical role in human history with archaeological evidence for the earliest homes dating back at least 700,000 years.¹ These earliest structures appeared to have mostly been used for shelter and as a convenient location to store tools and to bring food to. In modern life, these core functions of housing (offering shelter, a center for domestic life) remain while being supplemented by other equally important roles: offering proximity to critical services (schools, supermarkets, transportation to jobs), signaling individual tastes and preferences (through customization), and serving as vehicles for investment.

Given that virtually all humans use some form of housing, it is perhaps not surprising that another role that housing plays is its interaction with the environment. In fact, residential heating, cooling and electrical consumption accounted for 18% of all greenhouse gas emissions in 2007, and household-related travel also generates a substantial amount of greenhouse gas emissions.²

While the aspects of housing we have discussed above are inescapably important, our main focus here is on delineating housing's contribution to the economy. Economic history shows that this contribution has been very significant; here are some statistics that drive home this point:

- According to the U.S. Census Bureau, the homeownership rate averaged about 66% over 2010-2014.
- Housing is the largest asset for most households with the primary residence accounting for 29% of total family assets in 2010.³
- Most houses have mortgages and a mortgage is the largest liability of most households. As of 2010, debt secured by the primary residence comprised 75% of all household debt.⁴

¹[Moore12]

²[Schwartz10]

³[Bricker12]

⁴Ibid.

- Housing-related economic activity (construction, renovation, maintenance, services related to real estate transactions and financing - mortgage banking, real estate agents, appraisers, movers, notaries, etc.) represents a significant share of GDP and employment. Historically, residential investment has averaged roughly 5% of GDP while housing services have averaged between 12% to 13%, for a combined 17% to 18% of GDP.

Taking all of these factors into account, it therefore comes as little surprise that the institutional structure for providing affordable and adequate housing to the community (the **housing finance system**) is a topic of considerable government policy focus. At a social level, it is thought that homeowners are better stakeholders in their community. Additionally, as economic history has repeatedly demonstrated, changes in the housing market and/or in the structure of the housing finance system can have profound implications for individuals and institutions. Concretely:

- The securitization⁵ of mortgages has introduced strong linkages between the housing market and the broader financial market. In the U.S., approximately 85-90% of all newly originated mortgages are securitized.⁶
- Since several key financial intermediaries (particularly commercial banks) have large and levered exposures to mortgages (and real estate in general), the housing and mortgage markets play an important role in macroeconomic stability.
- The valuation of mortgages and housing (which in turn crucially depend on the trajectory of interest rates and house prices) can have profound consequences for institutional and individual behavior. This in turn means that Central Banks will often carefully monitor trends in home prices and mortgage rates and may from time to time attempt to influence either of these two.
- Rents are an important of the consumer price index (CPI) and need to be closely monitored in order to track inflationary pressures.

Given this backdrop, it is clearly important to more formally articulate how housing impacts the economy. The next section attempts to put together a coherent framework for the various routes through which this takes place.

1.1 Housing and the Macroeconomy

Housing affects the economy mostly through the channel of home prices. Both increases and decreases in the value of housing have a spillover impact not only on the service sector

⁵**Securitization** can be described as a financial technology that transforms a collection of illiquid but homogeneous cash flow generating assets into a liquid financial instrument that can be traded in the capital markets. In our context, the assets consist of residential mortgage loans. See Chapter 6 for further details.

⁶The statistic is for the 2010-2014 time period. Source: *Inside Mortgage Finance*.

surrounding housing, but also on consumer behavior and the stability of financial institutions. A high-level categorization of the different ways in which house prices impact the economy is itemized below.⁷ Each of these mechanisms is explored in further detail in what follows.

1. A direct impact through home building and remodeling (in economic jargon, **residential fixed investment**);⁸
2. An indirect impact through personal consumption;
3. A monetary policy impact in that changes in house prices are sensitive to interest rates and therefore react to changes in monetary policy.
4. A **credit channel** effect in which changes in property prices impact the ability of households and firms to borrow more and that consequently turns out to have broader implications for financial stability;
5. A price stability effect as housing markets can impact inflation through the rental market; and,
6. An impact on labor mobility.

It is also important to determine the underlying factors behind the change in house prices since different causal factors for the change might impact the economy in different ways. For example, if home prices increase because rental rates have risen (thinking of owner-occupied housing prices as the present value of a stream of future rental payments), the positive effect on an existing homeowner may be balanced out by the negative impact on a prospective first-time homeowner who finds the house more expensive. On the other hand, if home prices have risen because mortgage rates are lower then this should help both the homeowner and the first-time homebuyer who might still be able to afford the more expensive house.⁹

1.1.1 The Direct Impact: Residential Investment

An increase in home prices means that existing homes become more expensive relative to their replacement cost (all else being equal) which in turn stimulates new construction. This effect may take time to work itself through the economy because construction needs long cycles to take place and faces other constraints like the availability of land, competitive

⁷See [ECB2003] for further details.

⁸According to the U.S. Bureau of Economic Affairs (BEA): “Investment in residential structures consists of new construction of permanent-site single-family and multi-family units, improvements (additions, alterations, and major structural replacements) to housing units, expenditures on manufactured homes, brokers’ commissions on the sale of residential property, and net purchases of used structures from government agencies. Residential structures also include some types of equipment that are built into residential structures, such as heating and air-conditioning equipment.”

⁹In general, we can decompose changes in house prices as occurring for one of the following reasons: (1) Higher value of service flows (and rents); (2) Increased income expectations; (3) Changes in financial asset prices; and (4) Expectations of future home price increases and speculative activity.

conditions, availability of mortgage financing etc. The contribution of construction to the economy whether through new structures or improving existing ones is measured through **residential investment**. Residential investment has averaged between 3-6% of nominal GDP and consists of 3 components:

1. Construction of new single-family homes ($\approx 57\%$);
2. Construction of new multifamily housing ($\approx 6\%$); and,
3. “Other structures”: Improvements and brokers’ commissions ($\approx 37\%$)

The quickest way to summarize the impact of residential investment on the GDP is by noting that it often serves as a leading indicator of business cycle developments. In other words, while such expenditures do not comprise a large nominal fraction of GDP, their contribution can be quite large in downturns and upturns of the economy.¹⁰

1.1.2 The Indirect Impact: Housing Wealth and Consumption

Some of the varied ways in which housing affects consumption include the impact of mortgage interest expenses on consumer balance sheets, the ability of homeowners to extract home equity and use these funds for various purposes, and the “wealth effect” (the permanent increase in consumer spending associated with an increase in home values). Consumption plays a very significant role in the U.S. GDP. Recent estimates of the effect of housing wealth on consumer consumption suggests that each dollar of increase in housing wealth leads to between 4.5 to 16 cents of spending. The reduction in housing wealth coming from a 10% decrease in home prices could result in a negative impact on consumption as a share of GDP of about 1%.¹¹

Whether **home equity withdrawal (HEW)** itself affects consumption is a topic of debate.¹² Is it HEW that is driving consumption or is HEW a proxy for the wealth effect created by rising property prices? Current research suggests that HEW has a temporary effect on consumption while the housing wealth effect is longer lasting. Current estimates suggest that each dollar of HEW withdrawal is expected to lead to between 10 to 50 cents of consumption expenditure.¹³

¹⁰See [Leamer07].

¹¹See [Menegatti07] for a useful summary of this research.

¹²There are various different ways in which a homeowner can withdraw equity from their homes [Klyuev06]: (1) Take out a mortgage with a value in excess of that of the house; (2) Exercise negative amortization options, thereby increasing debt; (3) Refinancing existing mortgage into one with a higher loan balance (“cash-out refinance”); (4) Take out a second mortgage or home equity loan; (5) Increase the size of the mortgage when moving into a new home of similar value; (6) Trade down to a lower-value house when the homeowner has no mortgage or while maintaining the current level of secured debt; and (6) Sell a house and repay the mortgage to move into a rental.

¹³[Menegatti07].

1.1.3 Housing and Monetary Policy

The significant impact of residential investment and consumer spending on GDP is hard to discount. Both investment and spending are affected by house prices so a natural area of focus for the Federal Reserve is determining to what extent house prices are impacted by the prevailing monetary policy stance. In principle, lower policy rates lower mortgage rates, which affects demand for houses. Lowering policy rates may also change expectations for future house price increases and can affect construction and the supply of housing by influencing short-term rates.¹⁴ Besides affecting housing supply/demand conditions, lower mortgage rates also allow existing mortgage holders to lower their monthly payments through refinancing. This in turn leads to an increase in their disposable income.

Monetary policy is just one item in the tool kit of regulators and the Federal Reserve in terms of being able to influence the housing sector. **Macroprudential policy** aims to increase the soundness of the financial system by responding proactively to housing booms that are believed to be “bubble”-like.¹⁵ Key macroprudential tools in this context include the imposition of limits on loan-to-value (LTV) and debt-to-income (DTI) ratios.¹⁶ Another approach is to impose stricter capital requirements on loans to the real estate sector, thus discouraging banks from taking on heavy exposure to the sector.

1.1.4 Housing and the Credit Channel

An increase in property prices (commercial and residential) permits firms and households to borrow more to finance consumption and investment, which in turn further increases property prices. This effect also works in reverse when property prices are decreasing. This feedback loop constitutes the so-called **financial accelerator** effect. This mechanism enforces a strong link between the mortgage market (which is used to finance most properties) and house prices. The price and availability of mortgage credit influences the demand side picture and therefore affects home prices. Also, the performance of mortgage credit (at the macro level) influences provision of credit. This is another financial accelerator effect.

Figure 1.1 provides a visual illustration of the financial accelerator when applied to the mortgage and housing markets.

1.1.5 Rents and Inflation

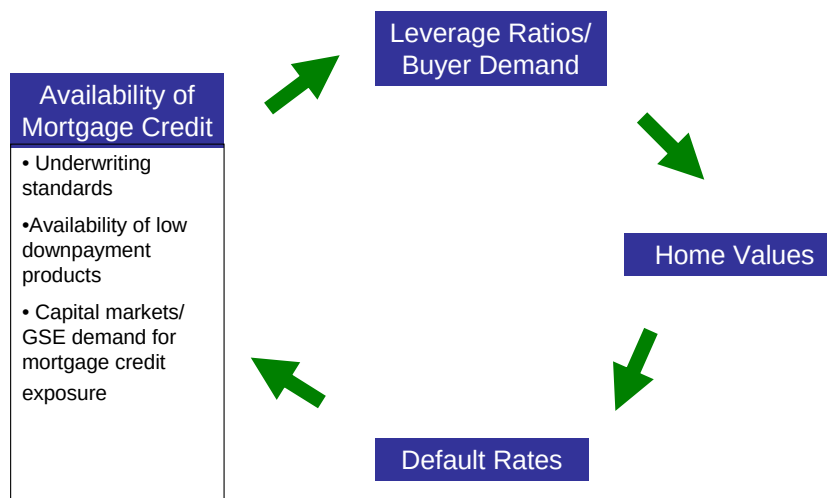
Housing markets can impact inflation through the rental market. Rents are an important component of the consumer price index, and changes in rents can have a potentially sizable

¹⁴Rates on construction loans are often tied to short-term rates.

¹⁵**Microprudential policy** aims to ensure the resilience of individual financial institutions. Such policy is necessary for a sound financial system but may not be sufficient; sometimes, actions suitable at the level of individual institutions can destabilize the system as a whole.

¹⁶For example, Hong Kong SAR has imposed caps on loan-to-value and debt-to-income ratios since the 1990s and Korea since the 2000s.

Figure 1.1: The Link Between Mortgage and Housing Markets



Source: Cello

impact on consumer price inflation. In more detail, the Consumer Price Index (CPI) measures the average change in price of a fixed basket of goods and services purchased by urban consumers. Housing is included in this basket but is tricky to price because it is a durable good whose service contributions need to be measured over time. Also, housing is an investment good and since CPI is a measure of consuming a fixed market basket, investment purchases should not appear in the index. The Bureau of Labor Statistics (BLS) tackles these issues by employing a special approach just for owner-occupied housing. Essentially, the BLS attempts to measure the cost of consuming owner-occupied housing by comparing it to what it would cost to rent the same housing. This approach is called **owners' equivalent rent of primary residence** (OER). OER can be defined as the change in what a homeowner would have to pay in order to rent their current home in a competitive market, and is calculated by surveying nearby rental units.

The importance of OER in terms of impacting inflation statistics can be gauged from the fact that it accounts for roughly 25% of the overall CPI and is approximately four times as large as the next most important component. The weight of this component swells to nearly one-third in the mostly commonly cited inflation statistic – the **core CPI** which excludes food and energy. Given its enormous weight, OER has a substantial effect on readings of the core CPI.

Still, some shortcomings associated with using this measure as a metric to estimate the inflationary pressures faced by consumers surfaced in the housing crisis of 2007-2009. Specifically, the substantial decline in house prices over this period led to a glut of foreclosed homes, and forced the homeowners living in these properties into rentals. On the other hand, the foreclosed properties vacated by homeowners have only very gradually made their way back into the housing market and this supply/demand mismatch led to a decrease in rental vacancy and consequently a pickup in rental prices. All of this has led to an increase in CPI over the same period but arguably this does not necessarily reflect a substantial or permanent price increase for many households. As the foreclosed properties make their way back into the housing market this trend would be expected to reverse.

1.1.6 Housing and Labor Mobility

When viewed from a geographical perspective, labor mobility speaks to the ability of workers to permanently relocate to areas that offer better employment prospects.¹⁷ This mobility depends crucially upon the housing market in many ways. For example, being in a mortgaged house with negative equity¹⁸ essentially means that the house is unsellable without the homeowner potentially absorbing a large out-of-pocket cost. Similarly, if the current residence were to be mortgaged at a lower rate than what is currently available, the homeowner has to be able to afford the payment shock associated with financing a new house. It follows

¹⁷We could also have a situation in which the household temporarily moves out of the house and rents it for a period of time before returning again.

¹⁸Defined as a condition in which the homeowner's mortgage-related indebtedness is greater than the value of the house.

that having large transaction costs associated with moving and/or the presence of negative equity can hinder labor mobility and the adjustment of the labor markets following economic shocks which cause broad-based declines in house prices. For example, a recent academic study estimated that (a) owners with negative equity were one-third less likely to move than otherwise similar owners without negative equity; and (b) For every \$1,000 in higher in mortgage debt service costs, mobility was about 12% lower.¹⁹ However, attempts to connect lower mobility to higher unemployment rates have yielded much more mixed results.²⁰ Finally, it should also be said that efficient rental markets also play a key role in ensuring labor mobility.

¹⁹[Ferreira11]

²⁰[Valletta13]

Chapter 2

Housing Finance Systems

Given that housing is a critical part of the economy, the natural question arises: How do we pay for it? Houses are expensive – in the U.S., the average home costs approximately 2-3x median annual income. So, most people will need substantial loans or a very significant amount of savings in order to purchase a home. As a matter of fact, loans for financing housing are not only substantial, they also have to be for the long-term in order to ensure homeowners are able to stay in their homes for an extended period of time without having to “renew” the loan. The traditional loan used to finance housing is a **mortgage** – a long-term loan secured (or **collateralized**) by the property itself. In other words, the mortgage lender has the right to seize the borrower’s house if they do not repay their mortgage in a timely fashion.

A **housing finance system** (HFS) can be viewed as a collection of institutions that provide funding for both owner-occupied and rental housing. An HFS provides funds for both the producers (for example, construction companies) and consumers (prospective homeowners) of housing. The core functions provided by these institutions revolve around the **origination**, **funding**, **servicing**, and **investing** of mortgages. A brief summary of each activity is given below; they will be explored in much greater detail in further chapters.

- **Origination.** Mortgage origination is a three-step process consisting of **application**, **approval**, and **settlement**. The process commences when a borrower files a loan application with a lender either to refinance a previous loan or to purchase a new home. This step often involves assembling a comprehensive package of documents, which the lender uses to determine whether the borrower qualifies for mortgage credit. The loan approval process is carried out or coordinated by the lender based on documents submitted by the borrower with the mortgage application. The settlement process allows the lender to ensure that the borrower has legal title to the property and to complete other formalities related to funding the loan. Lenders also require the establishment of an **escrow** account. This is used to ensure that borrowers can pay off their real estate tax bills, hazard, and flood insurance premiums, and mortgage insurance premiums (if applicable) on a timely basis.

- **Funding.** In order to provide borrowers with mortgages, one needs to find a large and relatively inexpensive pool of capital. This capital is said to fund these mortgages. Funds can be raised directly from within an institution (through its deposits or its other sources of borrowed money) or through the capital markets. In the latter case, funds are generated by selling the mortgages directly to investors in the so-called **secondary mortgage market**.
- **Servicing.** Servicing a mortgage loan consists of collecting monthly mortgage payments from borrowers, remitting these payments to the investors in these loans, paying property tax and hazard insurance bills from the escrow funds collected from borrowers in their monthly mortgage payment, and performing collection, loss mitigation and foreclosure activity with respect to delinquent borrowers.
- **Investing and Risk Management.** The ultimate holder of the mortgage, whether a depository institution or a mutual fund, takes on the risks associated with the mortgage. These risks can be broadly categorized as: **interest-rate risk** – the risk that the price of a mortgage will change as market interest rates move around, and **credit risk** – the risk that the borrower may default on their mortgage payments.

The goal of this chapter is to broadly understand the nature and function of HFS institutions. By broadly, we mean by isolating some of the common features found across HFS institutions from all over the world. Since we will largely focus on the U.S. housing and mortgage markets in this book, we also provide a brief account of the historical evolution of the U.S. housing finance system. The last section locates the U.S. HFS within a global context by comparing it to HFSs around the world with respect to some key characteristics.

2.1 The General Architecture of an HFS

There are several ways to describe a housing finance system. The approach here is to focus on three crucial “design decisions” that underpin the design of a HFS: the nature of the institutions that provide mortgage funding, the prevalence of securitization in funding mortgages, and the degree to which the mortgage finance industry is vertically integrated.¹

2.1.1 Mortgage Lenders

In most countries, the predominant mechanism for financing housing is a depository institution that gathers savings (in the form of deposits) from households and firms and makes loans to homebuyers. Several types of deposit-taking institutions exist: **commercial banks** offer a complete range of banking services; **savings banks** deal mostly with the household sector;

¹See Chapter 2 in [Chiquier2009] for a deeper exploration of the structure of HFSs.

and specialist housing finance institutions such as **building societies**² or **savings and loan associations** focus their lending primarily on housing.

Figure 2.1 shows the flow of funds in a very simple mortgage lending setup where savers deposit their money in a bank or a savings institution, which then in turn lends that money to homebuyers. The figure illustrates a classic instance of **financial intermediation** in which funds are channeled from people who have a surplus of money (**Savers**) to those who have a deficit (**Homebuyers**) through a financial institution.

A somewhat different type of mortgage lending arrangement takes place through **mortgage banks**. These specialized institutions originate and service portfolios of mortgage loans that are funded by securities they issue. These securities (also referred to as **covered bonds**) are general obligations of the mortgage bank and are typically purchased by institutions with long-term sources of funds (for example, pension funds and insurance companies). The flow of funds found in a mortgage banking lending system is depicted in Figure 2.2. The mortgage bank system dates back to the late 1700s and has been extensively used in Continental Europe (particularly in Germany and Scandinavia). The bonds issued by mortgage banks are considered very high quality as a result of conservative underwriting, strong regulation, priority rights of investors in the event of bankruptcy, and transparent operations. While mortgage banks are transparent and efficient producers of mortgage assets, they are in decline because they are limited by specialization – their funding is constrained by product selection and their inability to provide other types of financial services. One other issue with a mortgage bank is that the covered mortgage loans remain on the balance sheet and, as such, require capital against the credit risk that they pose.

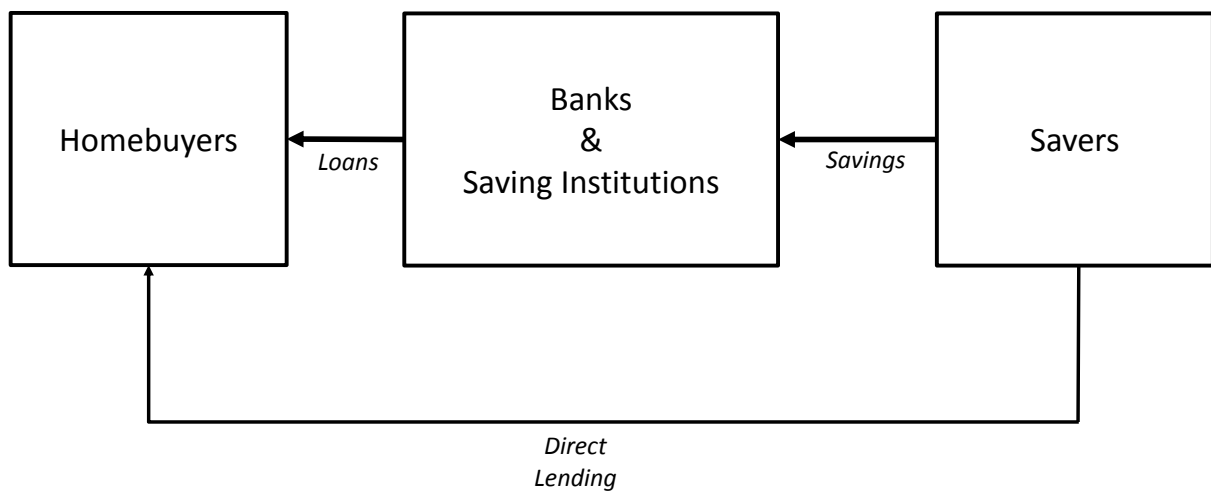
2.1.2 Primary and Secondary Mortgage Markets

The **primary mortgage market** defines the context for the transaction between a mortgage borrower and a mortgage lending institution. In the **secondary mortgage market**, loans from a number of borrowers are “pooled” into mortgage securities and then sold to a third party.³ A defining characteristic of secondary market-based systems is the widespread securitization of newly originated loans. These may be sold to specialized institutions called **conduits** or through **special purpose vehicles (SPVs)**. These entities raise funds through issuance of securities backed (or collateralized) by the loans. The vast majority of residential loans in the U.S. are currently funded through the secondary market.

²Building societies were the initial models for a HFS institution and were founded in England in 1775. The early versions of a building society cobbled together the savings of households for the purposes of home construction. Members would agree to contribute regularly to the society, build houses together, and allocate houses by lottery until each member was housed. Once members were provided housing and had repaid their loans, surplus assets, if any, would be distributed among members and the society would be terminated. During the mid-19th century, societies developed into permanent institutions, attracting funds not just from borrowers but also from other savers, and also started lending for the purchase of existing houses as well as building new ones.

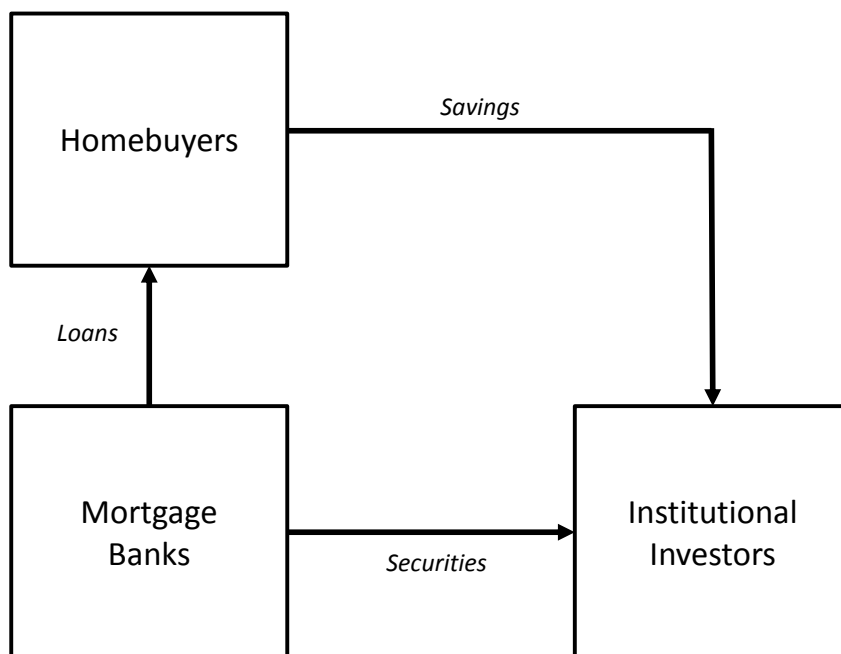
³Secondary markets for mortgage loans do exist – it is not necessarily true that all loans are sold into securities.

Figure 2.1: Depository and Direct Lending



Source: [Chiquier2009]

Figure 2.2: The Mortgage Bank System



Source: [Chiquier2009]

2.1.3 Bundling and Unbundling Mortgage Finance

In a bundled model, a single financial institution performs the major functions of origination, servicing, funding, and investing although other third-parties such as mortgage insurers, appraisers and credit agencies are involved in some of these activities. On the other hand, in an unbundled system, origination, servicing, portfolio management and funding are distributed over various specialized entities. There has been a significant movement towards unbundling in the U.S. HFS starting from the 1980s driven by increasing technical sophistication, regulatory change, information technology and standardization.⁴ Unbundling is particularly prevalent in activities that lend themselves to economies of scale such as servicing and securitization. Figure 2.3 illustrates how the unbundling takes place in practice. Specifically:

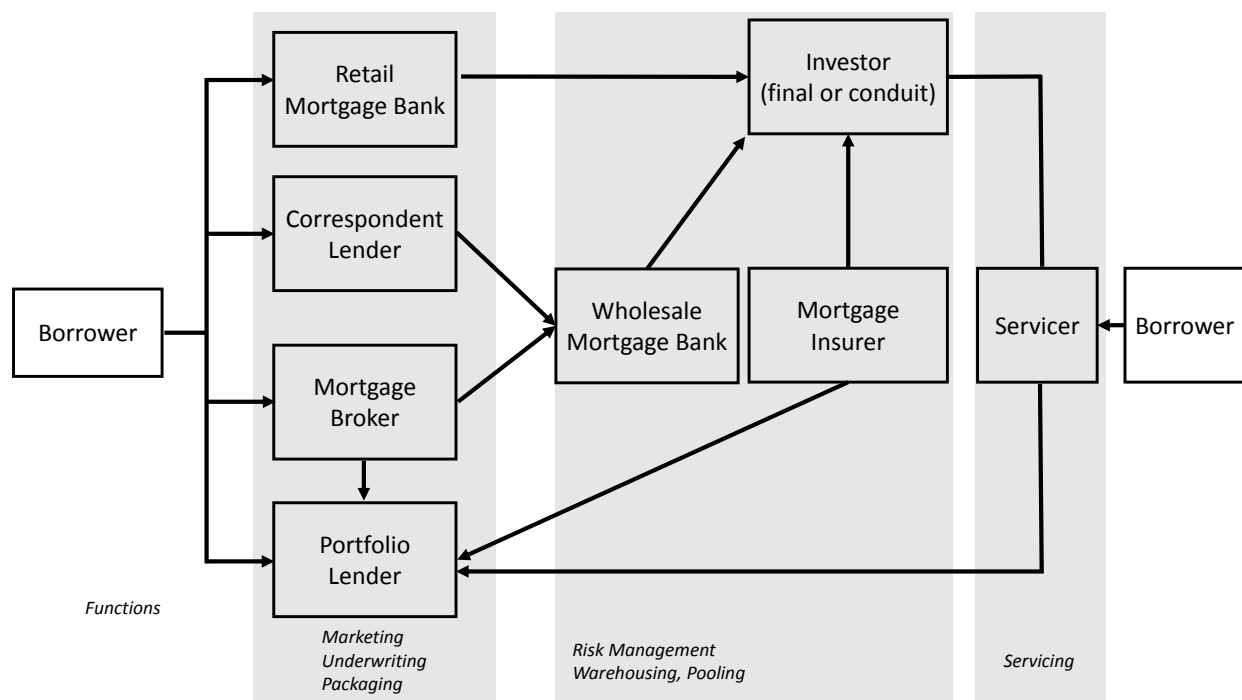
1. **Mortgage origination.** Retail branches of financial institutions or correspondent lenders close loans in their own name (but underwritten to the specifications of the ultimate investor) and then immediately sell them to larger, wholesale lenders that can get better execution in the secondary market.
2. **Mortgage servicing.** The mortgage servicer may not be the same entity as the originator. In the U.S., there has been significant consolidation in servicers and the top 5 mortgage servicers administer approximately 45% of all outstanding mortgage loans.⁵
3. **Investors.** Investors in mortgages can be depositories, mutual funds, hedge funds etc. Investors provide funds to the housing market by funding whole loans or investing in mortgage-backed bonds or securities. These investors may be of domestic or foreign origin. Credit risk management is often specialized as well, provided by third parties such as mortgage or bond insurance companies for the benefit of investors.

It is worthwhile mentioning here that there have been some significant concerns raised about the trend towards increased unbundling in recent years. In particular, the 2007-2009 Housing Crisis exposed some fundamental flaws in this model. With the rise of the secondary market as the primary source of funding for most residential mortgages as part of the unbundling trend, many primary market participants migrated towards a business model in which they retained little if any of the credit risk associated with these loans and were essentially fee-driven. For example, loan brokers received fees to originate loans, lenders to sell and service loans, rating agencies to rate securities, and investment banks to structure and sell securities. The only entities bearing the risk in terms of the credit performance of these loans were the investors in securities backed by these mortgages. Consequently, the fee-driven entities were somewhat incentivized to focus on quantity at the expense of quality in terms of the loans they were originating since they had “no skin in the game.” This development will be explored in a little more depth in the next section.

⁴[Jacobides03]

⁵As of Q1 2014, according to *Inside Mortgage Finance*.

Figure 2.3: The Unbundling of Mortgage Finance



Source: [Chiquier2009]

2.2 The Evolution of the U.S. Housing Finance System

Of course, housing has a long history in the U.S. but our account is focused on housing finance systems and thus more or less begins when mortgage loans become broadly available.⁶

2.2.1 Pre-history (pre-1930s)

The first institutional arrangement to finance housing in the United States was a **Terminating Building Society (TBS)**, which originated in England in 1775 and played a dominant role in the U.S. in the early to mid-19th century. TBSs evolved into more formal lending institutions such as **Permanent Building Societies, Buildings and Loans**, and, eventually, **Savings and Loans (S&Ls)** or **Thriffs**. All of these institutions worked in essentially the same way – they pooled together deposits to fund loans primarily for member-depositors.

Another early institution dedicated to mortgage lending was a **mortgage bank**. These banks were established in the 1870s to lend in the expanding Midwestern and Western states. Those institutions originated and serviced loans with the funds raised by selling mortgage-backed bonds (MBBs), modeled after the practice in France and Germany at that time. Under this intermediation process, investors took on the credit risk of bond issuers and were compensated through a premium in the interest rate. This setup can be described as an early form of a secondary mortgage market. Initially, the MBB market grew significantly as both issuers and buyers of the security benefited from scale economies in loan origination, servicing, and funding. During the recession in the 1890s, MBBs defaulted in large numbers because of their lax underwriting standards.

In this “pre-historical” era, **balloon mortgages**⁷ were the dominant source of financing for single-family homes. These mortgages had maturity terms between 3 to 10 years, semiannual payments with no or partial amortization of principal, variable interest rates, and a maximum loan-to-value ratio of about 50%. Furthermore, since a significant amount of the loan was due at maturity, the homeowner had to rely on being able to repay or refinance their debt in order to maintain possession of their homes. It was fairly common for borrowers at this juncture to take on second, third and even fourth mortgages at effective interest rates ranging from 18-20% since they could not afford the downpayment associated with the first mortgage. Low homeownership rates prevailed ($\approx 37\%$ in 1900) and borrowers employed a multiplicity of different sources such as land subdividers, builders, brokers, local investors, or friends and relatives to obtain mortgage financing. Given these idiosyncratic sources of funding, the absence of secondary markets in any kind of scale, interstate banking restrictions, and a lack of uniform standards for lending, it should come as no surprise that the market was intensely

⁶There is, as yet, no completely satisfactory single volume history of the U.S. Housing and Mortgage Markets. The following account relies heavily on [HUD06, Herzog09, Levitin12, Levitin13, Weiss89, Wright05].

⁷A balloon mortgage does not fully amortize over the term of the mortgage note and leaves a balance due at maturity. This final payment is called a **balloon** payment because of its size relative to the regular monthly payments. It is now (as of 2016) rare to find residential real estate financed by balloon mortgages although they remain very common in commercial real estate lending.

local in that interest rates and the availability of financing varied significantly by region. For example, the average mortgage rate on first mortgages ranged between 6% to 6.50% over the 1920-1993 period but state-level variation in mortgage rates was as high as 5.90% to 10%.

The booming economy in the 1920s bolstered both the real estate and consumer credit markets and new players such as **mortgage insurance** companies entered the HFS.⁸ By 1930, the homeownership rate was hovering around 45%.

2.2.2 Institutionalization (1930s to 1960s)

The Great Depression (1929-1940) led to a fundamental reshaping of the institutional landscape for mortgage lending. A combination of bank failures, rising unemployment, deposit withdrawals, and tightening credit led to widespread defaults on mortgages. Borrowers were not able to afford their current monthly payments as a result of unemployment and decreasing wages, and also because of an inability to pay the mortgage off when it came to term. Lenders, who were already running substantial foreclosure inventories, exacerbated the cycle by either dumping properties on the market or refusing to lend at all. The statistical trends associated with this period attest to the severity of the downturn:

- The amount of annual mortgage lending dropped by 80% over the 1930-1933 period from \$3.4 billion to \$663 million.
- Private residential construction dropped by more than 80% from 1930 to 1933.
- By spring 1933, 50% of all home mortgages were in default and more than 1,000 mortgages were being foreclosed each day.
- The unemployment rate was 25% in 1933.
- The number of banks fell by 35% and the number of S&Ls by about 13% between 1930 and 1935.

The Hoover and Roosevelt governments made strenuous attempts to revive the housing markets. Initially, the Hoover government tried to use the States to pass mortgage moratorium legislation that granted borrowers relief through delaying foreclosure, extending redemption periods, and preventing inflated deficiency judgments.⁹ When this failed to have much of an impact, the next step was the creation of several federal housing-related institutions. The **Reconstruction Finance Corporation (RFC)** and the **Federal Home Loan Banks (FHLBanks)** were Hoover-era creations that were only effective to a limited extent although

⁸Mortgage insurance companies guarantee the lender against the risk of foreclosure and the first such company was established in 1887.

⁹The **right to redemption** provides a homeowner with the ability to buy back their house after the foreclosure sale within a fixed period of time (the **redemption period**). A **deficiency judgment** corresponds to the lender extracting a payment from a defaulted borrower that is equal to the difference between the outstanding mortgage amount and the foreclosure sale price plus default-related costs.

they did create a framework for increasing government involvement in the housing sector. With the New Deal, Roosevelt kicked off a significantly higher degree of Federal assistance to the housing market through **HOLC** which bought delinquent mortgages; the **Federal Housing Administration (FHA)**, which established a government-backed mortgage; and **Fannie Mae (FNMA)**, which provided a source of demand for government mortgages. A capsule summary of the chronology and functions associated with each of these institutions is as follows:¹⁰

- **Reconstruction Finance Corporation (1932).** This government-backed corporation granted loans to financial institutions (and, in particular, banks) who in turn extended loans to industrial and agricultural enterprises. The hope was that this would promote economic recovery and ease unemployment. The institution was dissolved in 1953.
- **Federal Home Loan Banks (1932).** By 1932, a wave of banking failures had led to deposit runs and banks were increasingly reluctant to allow depositors to withdraw cash or to extend mortgage credit. The 12 FHLBanks established during this period tried to add liquidity to the system by extending loans to S&Ls that were backed by the mortgages these institutions had in portfolio. These loans were in turn backed by FHLB-issued bonds that were purchased by a diverse base of investors thus removing some of the geographic variation prevalent in the mortgage market. Unfortunately, these institutions were unable to make a dent in the most significant issue facing the mortgage market – a torrential wave of foreclosures.
- **Home Owner's Loan Corporation (1933).** HOLC issued debt to purchase delinquent mortgages and refinanced existing borrowers into affordable rates. HOLC ended up refinancing approximately 20% of all qualifying home mortgages in the country and was wound down by 1951. The average borrower that HOLC refinanced was two years delinquent on the original mortgage and three years behind on property taxes. In the course of helping banks clear their balance sheets, HOLC played an important role in introducing standards for many aspects of the mortgage market such as providing borrowers with long-term, fixed-rate, fully amortized mortgages and uniformizing processes involving origination and foreclosure.
- **Federal Housing Administration (1934).** The FHA established a government-backed mortgage with the goal of stimulating residential construction activity. With FHA insurance, mortgage lenders were protected from default risk. The FHA was probably the most important government institution to come out of the housing crisis for the following reasons:

¹⁰Also worth mentioning in this context are the two deposit insurance companies that were established – the **Federal Deposit Insurance Corporation (FDIC)** (1933) for commercial banks and the **Federal Savings and Loan Insurance Corporation (FSLIC)** (1934) for S&Ls. Deposit insurance helped make deposits less liable to “runs” from bank customers and thereby enabled depository institutions to better manage their interest rate risk (of which mortgages were a significant part) without keeping significant liquid assets on hand.

1. It further standardized the types of mortgage that HOLC introduced by introducing the 30-year level-payment self-amortizing mortgage.
 2. It increased the maximum LTV to as much as 93%.
 3. It established minimum standards for housing construction that were widely adopted by the construction industry.
 4. By freeing lenders from most of the costs of mortgage default and insurance, FHA helped ease the credit crunch and lowered mortgage rates. Mortgage rates fell by as much as 200-300bps because of the introduction of FHA.
- **Federal National Mortgage Association (1938).** FNMA created liquidity for FHA loans by providing a steady source of demand for government-backed mortgage loans. FNMA was chartered as a private institution and issued debt to buy mortgages. Fannie Mae played a relatively minor role in the mortgage market during the Depression and early post-War years. Over a longer time period, it laid the ground for three structural features that continue to characterize the modern day U.S. HFS:
 1. A secondary mortgage market that links capital markets to mortgage borrowers;
 2. A nationally-integrated mortgage market with little regional variation in mortgage rates; and,
 3. A 30-year amortizing fixed-rate mortgage as a national standard.

An important U.S. mortgage institution does not appear on the list above since it was established after the Great Depression: the **Veteran's Affairs Home Loan Guaranty Program (VA)** (1944). VA was authorized by the GI Bill of Rights to help veterans returning to the United States from WWII purchase single-family loans and thus help stimulate the post-war economy.¹¹ In 1948, Fannie Mae was effectively reborn as Congress rechartered it under the authority of the Federal Housing Administration. The recharged institution was authorized to purchase FHA-insured as well as VA-guaranteed mortgages.

The housing markets ran pretty smoothly for a while until the U.S. encountered its first post-war credit crisis in 1966. The interest rates that depository institutions could pay on deposit accounts were capped by federal regulation, whereas Treasury bond rates were not. At the same time, since the economy was expanding more rapidly than the Federal Reserve Board believed to be prudent, the Fed refused to raise the Regulation Q ceiling on interest rates payable by depositories in order to keep pace with unregulated interest rates on commercial paper and Treasury securities. The net result was a massive outflow of capital from depository institutions to Treasury bonds and commercial paper resulting in a severe shortage of funds for mortgage lending. Residential construction declined by 23% between the first quarter of 1966 and the first quarter of 1967 but Fannie helped stabilize the market by continuing to buy FHA/VA mortgages.

In 1968, the Johnson administration, eager to clear room in the federal budget for Great Society spending and the Vietnam War, split up Fannie Mae into two entities. One entity,

¹¹The program was later expanded to include veterans of all succeeding wars, peacetime veterans, men and women on active military duty, surviving spouses and reservists.

continuing to bear the name Fannie Mae, was privatized as a **government-sponsored entity (GSE)**.¹² The other entity remained government-owned and was christened the **Government National Mortgage Association (GNMA)** or **Ginnie Mae**. Ginnie Mae's mission was restricted to the securitization of FHA-insured and VA-guaranteed mortgages. Fannie Mae, under a revised federal charter, became privately capitalized, but under government regulation, and with a third of Fannie's board of directors appointed by the President of the U.S.

The newly privatized Fannie Mae continued to conduct secondary market activities. It was originally restricted, however, to purchasing only FHA/VA mortgages, even as Ginnie Mae, the new governmental entity, took over Fannie Mae's affordable housing functions. Ginnie Mae also began the first mortgage securitization in the U.S. by securitizing FHA-insured mortgages. FHA insured the mortgages but Ginnie Mae guaranteed the timely payment of principal and interest on bonds backed by FHA-insured or VA-guaranteed mortgages thus virtually eliminating any exposure to credit risk for the end investor. These bonds are known as **mortgage-backed securities (MBS)** and played a central role in the history of the housing starting with the 1980s.

Table 2.1 recaps some of the key government interventions into the mortgage market in this era.

Table 2.1: Key Housing-related Federal Legislation:1933-1969

1933	FDIC and HOLC established.
1934	FHA created.
1938	FNMA created to provide secondary market for FHA loans.
1944	VA loan program created as part of the Veterans Bill of Rights.
1948	FNMA begins to purchase VA loans.
1968	GNMA created. FNMA becomes a shareholder-owned GSE.

Source: Cello

2.2.3 Securitization (1970s to 1980s)

By the mid-1960s, the U.S. HFS faced the challenge of rising inflation and interest rates. The inflationary environment resulted from a confluence of factors including the federal government's large budget deficits in the 1960s during its involvement in the War on Poverty and the Vietnam War; the first and second oil shocks during the 1970s, which resulted in the

¹²A GSE is an amalgam of a private sector corporation and a government entity and is created by Congress to improve the functioning of credit markets. This is primarily achieved through the GSEs providing some form of credit guarantee that limits the risk of capital losses to those supplying the funds. While GSE-backed securities do not carry an explicit government guarantee of creditworthiness, investors generally assume that an "implicit guarantee" is present since the government would not allow such an important institution to fail. The first GSE was the **Farm Credit System** created in 1916. The first housing finance-related GSE was the FHLBank.

ramping up of energy prices; and the change in monetary policy in the mid- to late-1970s that adopted money aggregates instead of interest rates as the policy target. Large movements in interest rates highlighted another risk that investors in mortgages face distinct from credit risk, namely, interest rate risk. In the U.S., homeowners are typically allowed to repay their mortgages at will, this is the so-called **prepayment option** that we will discuss in much more detail in Chapter 5. In a nutshell, the prepayment option causes the effective maturity (**duration**) of mortgage loans to lengthen when interest rates rise because prepayments slow down, and to decrease when interest rates fall. Mortgage investors will typically attempt to hedge this risk but sharp movements in interest rates can lead to substantial duration mismatches between mortgage loans and the instruments that they are hedged with (deposit funds, for example).

Returning to the S&Ls, the inflation shocks caused them several problems. First, they had relied on short-term deposits to fund long-term, fixed-rate mortgages. In the face of rising interest rates, this “borrow-short-lend-long” operational approach squeezed profit margins. Interest rates on deposits increased but the only place to earn higher returns was on new mortgages. Second, high interest rates dampened housing demand, which had an adverse impact on new mortgage origination volume for S&Ls. Third, as investors, S&Ls also experienced a mismatch in the maturity of their assets (mortgages) versus their liabilities (deposits) because of the upward shift in interest rates. Fourth, money market mutual funds (MMMFs) came into existence in the early 1970s, opening up a new investment channel for small savers. Rates paid by the MMMFs were not regulated while Regulation Q limited the rates that banks and thrifts could pay on time deposits (their primary source of funds). These new vehicles resulted in a significant drop in deposit inflows to thrifts while MMMF assets increased from \$3.5bb in 1977 to \$180bb in 1981, more than a 50-fold increase in 4 years.

In 1969-1970, another interest rate spike caused a further round of disintermediation, resulting in a decline in funding for depositories and a substantial decline in housing starts. As interest rates (and inflation) rose and construction declined, home prices increased and mortgages became less and less affordable. Congress responded in 1970 with the Emergency Home Finance Act. The Act authorized Fannie Mae to purchase **conventional** (non-FHA/VA) mortgages and also created the **Federal Home Loan Mortgage Corporation (FHLMC)** (also known as **Freddie Mac**), which was also similarly authorized to purchase conventional mortgages. Freddie Mac began to purchase conventional mortgages in 1971, and Fannie began to do so in 1972.¹³

Although both Fannie and Freddie were authorized to issue MBS, as well as to hold loans in portfolio, Freddie Mac initially operated quite differently from Fannie Mae. Freddie Mac operated primarily as a securitization operation, like Ginnie Mae, offering guaranteed pass-throughs, but for conventional rather than FHA/VA mortgages. Freddie Mac did not originally hold loans in portfolio in order to avoid competing with the thrifts from which it bought mortgages (and which owned it indirectly). Through the 1970s, Fannie, in contrast,

¹³Fannie Mae and Freddie Mac are often loosely collectively referred to as “the GSEs,” even though there are several other such entities currently in existence including the Federal Home Loan Bank, the Farm Credit System, the Financing Corporation, and the Resolution Funding Corporation.

held loans in portfolio and issued long-term bonds and short-term notes. This difference meant that Fannie was exposed to both interest rate and credit risk, while Freddie held only credit risk.

As rates rose dramatically in 1974-1975 and 1979-1981, Fannie's long position in mortgage debt placed it under severe financial pressure. It had to finance itself at higher rates than the yield on the mortgages it held. Freddie Mac did not face this interest rate risk because it was only issuing mortgage-backed securities and had no portfolio beyond what was in its securitization pipeline. As a result of interest rate pressures, Fannie Mae began to engage in securitization in 1981. With the rechartering and privatization of Freddie Mac in 1989 and as part of the reform of the thrift industry and the FHLBanks, the Fannie and Freddie models converged.

It was within this institutional and economic backdrop that the modern form of securitization was first introduced in the 1970s. The initial motivating force behind the introduction of this technology was to increase the supply of capital available for homeownership given that the S&Ls were not believed to be able to fully meet housing demand. A second phase of securitization commenced when S&Ls began to struggle because of the various interest rate-related pressures they were facing (as described above). The GSEs increased their role in their marketplace by securitizing seasoned loans on S&L balance sheets as MBS, thus providing these institutions with a valuable source of liquidity. Critical to the development of the securitization market in these years was the market's perception that GSE guaranteed MBS held little to no credit risk because these institutions were implicitly backed by the federal government.¹⁴ An investor in GSE MBS was consequently focused not on how much of their principal would be returned but when it would be returned.

Table 2.2 highlights some of the key housing related legislation from this era of securitization.

Table 2.2: Key Housing-related Federal Legislation: 1970-1990

1970	FHLMC created.
1981	S&Ls allowed to invest in ARMs, deposit ceilings removed.
1982	S&Ls securitize and sell-off below-market-rate mortgages.
1986	Tax Reform Act of 1986 eliminates all interest-related personal deductions except for mortgages and home equity loans.
1989	FHLMC restructured as public corporation.

Source: Cello

¹⁴The events of the Housing Crisis demonstrated that this assumption was indeed correct.

2.2.4 Rise of the GSEs and the Private Market (1990s to mid-2000s)

Into the late 1980s, the GSEs still had fairly small market share and most mortgages were still held in portfolio, particularly by S&Ls. The main effect of the GSEs prior to the 1980s was to provide reassuring liquidity if the market needed it. It was only with the collapse of the S&Ls in the 1980s that the GSEs – now both privatized – truly emerged as market giants. The creation of a robust secondary market for non-FHA/VA mortgages under the then-privatized Fannie Mae and the eventually privatized Freddie Mac, meaningfully loosened regulatory control over housing finance. The significance of this deregulation through privatization was not immediately apparent, but it set the stage for later developments in the 2000s.

The privatization of Fannie Mae and Freddie Mac meant that their management would be subject to pressure from shareholders, who were not particularly concerned about the policy goals embodied in the GSEs. The GSE's loan purchases were also subject to **conforming loan limits**¹⁵ and LTV limitations¹⁶ absent mortgage insurance. Otherwise, however, underwriting was left up to the GSEs.

Within this backdrop, the GSEs grew in strength throughout the 90s and dominated the secondary market until the 2003-2006 time period. In the meanwhile, a completely private, unregulated secondary mortgage market also began to emerge starting in 1977. This was the **private-label** securitization market (also known as the **Non-Agency** market). The Non-Agency market began with the securitization of ultra-high quality mortgages that were too large to meet the GSEs' conforming loan limits but gradually expanded to include borrowers with poor credit or more generally borrowers who fell outside the underwriting guidelines of the GSEs. While the Non-Agency market remained quite small for many years, it began to take off in the mid-1990s as a result of the S&L crisis and because of the extension of credit to ever riskier borrowers. By 2006, almost 50% of all mortgage originations were non-traditional products and private label securitization had grown to 56% of the securitization market.

Another notable development in the 1990s was the introduction of **automated underwriting (AU)** systems in the U.S. HFS. **Underwriting** refers to the process of estimating a borrower's ability and willingness to repay a loan. An AU system is piece of computer software that evaluates the likelihood that a borrower will repay their loan based on data summarizing how borrowers with similar loan, property and credit characteristics have repaid their loans in the past. Freddie Mac's AU system is called **Loan Prospector (LP)** and Fannie Mae's AU system is known as **Desktop Underwriter (DU)**.¹⁷ Following the introduction of AUs by Fannie Mae and Freddie Mac, large lenders followed suit by developing their own proprietary system and the general use of AUs in mortgage origination decisions exploded. For example,

¹⁵Fannie Mae and Freddie Mac are prohibited from purchasing mortgages with balances greater than a specified amount called the conforming loan limit. A complex set of calculations regulates these limits on a year-to-year and region-by-region basis; as of 2016, the loan limit for most regions was \$417,000.

¹⁶By charter, Fannie Mae and Freddie Mac are not allowed to purchase residential mortgages with LTVs over 80% without credit enhancement. This credit enhancement usually assumes the form of mortgage insurance.

¹⁷LP was formally launched in February 1995 and DU in April 1995.

the share of Fannie Mae acquisitions processed through its AUs increased from less than 10% in 1997 to about 60% in 2002. AU systems save time and money in the refinancing process for some mortgage borrowers. These efficiencies are obtained by automating parts of the loan approval process that formerly involved a human underwriter.

2.2.5 Housing Crisis (2007-2009)

The housing crisis of 2007-2009 and the time period directly before it, is perhaps best described in terms of the evolution of house prices. Home prices were relatively flat through most of the 1990s. According to the S&P/Case-Shiller Index, home prices increased by about 8.3% from Q1 1990 to Q1 1997. At this point, home prices began a rapid increase – peaking in the 2nd quarter of 2006 over 132% higher than they had been in Q1 1997. Then, by Q1 2009, home prices had decreased by over 32% from their 2006 peak. This suggests that the home price trends before the crash of 2007-2009 were part of a housing “bubble.” We briefly discuss four of the most commonly cited causes of this bubble:

1. **Low mortgage interest rates.** Mortgage interest rates in the U.S. peaked at 18% in 1982 as the Fed hiked rates in an attempt to squeeze inflation out of the economy. Mortgage rates generally fell over the next 20 years and were in the 6-7% range from 2001 to 2007. The primary reason for these low mortgage rates is generally held to be an influx of savings entering the U.S. from countries with high savings rates such as Japan and rapidly growing economies such as China and Brazil. Investors in these countries sought investments providing relatively low risk and good returns and channeled their savings towards MBS, thus driving MBS prices higher and mortgage rates lower.
2. **Low short-term interest rates.** From 2002 to 2004, the Fed pushed the federal funds rate down to what were then historically low rate levels in an attempt to strengthen the recovery from the 2001 recession. Low short-term rates encouraged the use of adjustable-rate mortgages (ARMs). As home prices rose faster than household incomes, many prospective home buyers were unable to afford house payments under fixed-rate mortgages. But, ARMs whose rates are often indexed to short-term interest rates, could provide the buyer with lower monthly payments initially since short-term interest rates were lower than long-term interest rates. As the housing market heated up, mortgage lenders became more creative with ARMs, developing ARMs in which one could choose to make only payments of interest on the loan, or could choose to make payments of only a portion of the interest due (thus increasing the balance outstanding on the loan each month).

Low short-term rates also contributed to the housing bubble by encouraging the use of **leverage** – investing with borrowed money. The practice of leveraging increased the financing available for mortgage lending and thus contributed to rising home prices.

3. **Relaxed standards for mortgage lending.** Standards for mortgage loans were relaxed as a result of the following factors: new government policies aimed at fostering

an increase in homeownership rates among lower-income households, greater competition in the mortgage loan market, the increasing securitization of home mortgage debt, and speculative activity by all parties in the mortgage lending process.

Standards for mortgage loans were relatively consistent in the decades prior to the development of the housing bubble. Most mortgages were 30-year fixed-rate loans requiring a down payment of at least 20% or mortgage insurance if the 20% downpayment requirement was not met. Borrowers also had to demonstrate that their income was sufficient to ensure that the monthly mortgage payments would be manageable. In the mid-90s, the Community Reinvestment Act was modified to compel banks to increase their mortgage lending to lower-income households. To meet the new requirements of the Act, many banks relaxed their mortgage lending standards. This pressure also filtered through to Fannie Mae and Freddie Mac. Beginning in 1996, the Department of Housing and Urban Development began to increase the percentage of mortgage loans to lower-income households that Fannie and Freddie were required to hold in their portfolios. This caused Fannie and Freddie to relax the standards that mortgages had to meet to be classified as “conforming.” Finally, with the advent of the Internet came greater competition in the mortgage market which led to lowering of fees and relaxing of standards by mortgage lenders. More conservative lenders had to choose between either lowering their standards or losing market share.

The increased securitization of home mortgage debt also contributed to relaxed mortgage standards. As discussed previously, securitization can undermine the responsibility for maintaining lending standards since the originator is following an “originate-to-distribute” model, as opposed to the traditional practice of “originate-to-hold.”

The relaxation of mortgage standards was exemplified by the increase in subprime originations. Subprime mortgages are home loans given to persons who are considered a poor credit risk. Historically, subprime mortgage rates have a foreclosure rate that is much as 10 times higher than prime mortgages. Subprime mortgages increased from 5% of new home loans in 1994 to 20% in 2006.

4. **Irrational Exuberance.** All participants (government regulators, mortgage lenders, investment bankers, credit rating agencies, foreign investors, insurance companies, and home buyers) acted on the assumption that home prices would continue to rise. In particular, it was widely held as a fundamental “law” that home prices could not fall on a national basis (as opposed to a local one). This assumption led to unchecked activities by a number of parties. Government regulators felt no need to try to control rising home prices, which they did not recognize as a bubble. Mortgage lenders continued to make increasing number of subprime mortgages and adjustable rate mortgages. Investment bankers continued to issue highly leveraged mortgage-backed securities. Credit rating agencies continued to give AAA ratings to securities. Foreign investors continued to purchase billions of dollars into highly rated mortgage-backed securities. Insurance companies continued to sell insurance contracts to investors in mortgage-backed securities and home buyers continue to purchase homes (often for speculative purposes).

Home prices reached their peak in the second quarter of 2006 and did not fall drastically at first – home prices fell by less than 2% from the second quarter of 2006 to the fourth quarter of 2006. However, foreclosure rates increased by 43% over these two quarters and increased by 75% in 2007 compared to 2006. This dynamic was driven by speculators walking away from properties as prices fell and homeowners with subprime adjustable-rate mortgages not being able to refinance because the decrease in home prices meant that they had negative equity in their homes. As the monthly payments on these mortgages adjusted upwards, they were no longer affordable.

Just as rising home prices reinforced the continuing rise in home prices, falling home prices reinforced the continuing decrease in home prices. The increase in foreclosures added to the inventory of homes available for sale. This further decreased home prices putting more homeowners into a negative equity position and leading to more foreclosures. The increase in foreclosures also decreased the value of mortgage-backed securities. This made it difficult for investment banks to issue new mortgage-backed securities, eliminating a major source of financing for new mortgage loans and contributing to the continuing decline in home prices.

The bursting of the housing bubble caused significant losses for the following institutional participants in the housing market:

1. **Mortgage lenders:** A third of the top 30 mortgage lenders were either acquired (e.g., Countrywide Financial by Bank of America), filed for bankruptcy (e.g., New Century Financial), or were liquidated.
2. **GSEs:** Both of the GSEs entered conservatorship in September 2008 because of the losses they experienced on their investment portfolios and guarantee business.
3. **Investment banks:** The 5 largest U.S. investment banks either filed for bankruptcy (Lehman Brothers), were acquired by other firms (Bear Stearns and Merrill Lynch), or became commercial banks subject to greater regulation (Goldman Sachs and Morgan Stanley).
4. **Foreign investors:** Banks and governments who had invested in mortgage-backed securities.
5. **Insurance companies:** Sellers of credit default swaps such as AIG. Credit default swaps are a type of contract that insures against the default of debt instruments, such as mortgage-backed securities.

2.2.6 The Post-Crisis Era (2010 - Present)

The Great Recession of 2007-2009 locked many borrowers out of the refinancing market. Two main factors were responsible for this lack of access:

- Weakened household balance sheets, reduce income and widespread unemployment left many borrowers unable to qualify for a refinancing because of restrictions based

on affordability tests (for example, debt-to-income ratios). For some borrowers, the constraint was a too high loan-to-value (LTV) ratio.

- Lenders and mortgage guarantors, including Fannie Mae, Freddie Mac, and the FHA instituted more stringent underwriting requirements, leaving many borrowers unable to qualify for a new loan even when they were current on their existing mortgage.

The government responded to these constraints by introducing new refinance programs, such as the GSEs' **Home Affordable Refinance Program (HARP)** in early 2009. HARP provided borrowers who would not otherwise qualify for refinancing because of declining home values or reduced access to mortgage insurance, the ability to refinance their mortgage into a lower interest rate. We can think of HARP as being a streamlined refinancing program for low- or no-equity mortgages where the borrower meets certain qualifications and has their existing mortgages guaranteed by Fannie Mae or Freddie Mac. An essential feature of HARP was that it allowed borrowers to carry forward into the new loan any existing private mortgage insurance (PMI) from the previous loan; also, if no mortgage insurance existed on the original loan, then none was required for the refinanced loan.

As one would expect, the introduction of a large-scale refinancing program such as HARP stimulated refinancings but the overall level of activity was significantly smaller than the 4 to 5 million homeowners that were estimated to meet the basic eligibility criteria for HARP. Potential sources of friction that prevented borrowers from refinancing include:

- **Loan-level pricing adjustments (LLPAs).** LLPAs are upfront fees that are added to the refinancing costs of loans that are judged to have higher default risk characteristics, such as high loan-to-value ratios. The fees increased the cost of refinancing and thus discouraged borrowers from participating in the HARP program.¹⁸
- **Putback risk.** Putback risk is the possibility that the loan originator will have to repurchase the loan from the GSEs because underwriting violated GSE guidelines.¹⁹ Although the streamlined HARP guidelines do not require lenders to verify all aspects of a borrower's application, lenders who process the HARP refinancings have putback risk both from the refinancing and from the original underwriting, even if the refinancing lender did not underwrite the original loan. The risk is real as has been manifested by the GSEs aggressively pursuing legal action against lenders in many cases. The aggressive posture by the GSEs may have made lenders reluctant to refinance loans originated by other lenders and thus limited participation in the program.
- **Junior lienholders.** In some cases, holders of junior liens refused to allow their loans to remain subordinate to a proposed new refinance loan, thus holding up the HARP process.

¹⁸Information on the magnitude of LLPAs and their on various loan- and borrower-level attributes can be found at <http://www.freddiemac.com/singlefamily/pdf/ex19.pdf>.

¹⁹The liability associated with closing a loan for the GSEs or FHA is generally referred to as **representation and warranties**.

- **Mortgage insurers.** Some mortgage insurers did not agree to reunderwrite their policies.

Given the original HARP's shortcomings, a further set of program enhancements (colloquially referred to as **HARP 2.0**) that addressed these shortcomings was announced on October 2011. This iteration of HARP was much more successful than the previous iteration and is scheduled to be sunset in December 2016 after helping millions of underwater borrowers refinance their mortgages.

Despite its maturity and sophistication, the U.S. HFS is currently in a deeply unsettled state. Of particular note is the fact that the conservatorship of the GSEs is still in place with a deeply polarized Congress making it unlikely that the either of the two existing proposals on the table (see Figure 2.4) will gain traction any time soon. On the positive side, the retained portfolios that the GSEs hold have been steadily shrinking and a program to transfer credit risk from the GSEs to private hands has been somewhat successful.

The Non-Agency MBS remains moribund with only \$13.7 billion of issuance in 2015 compared to more than a trillion in 2006. Banks are placing high quality non-agency loans in portfolio while, on the secondary market side, investors are still not willing to accept new MBS. Essentially, the Housing Crisis revealed some structural issues in terms of how private label MBS are put together and investors are not yet convinced that all these issues have been satisfactorily addressed.²⁰

²⁰[Goodman16] has a useful summary of what remains to be accomplished in terms of increasing the current level of investor comfort.

Figure 2.4: Legislative GSE Reform Proposals

H.R. 2767 Protecting American Taxpayers and Homeowners Act of 2013 (PATH Act)		S. 1217 Housing Finance Reform and Taxpayer Protection Act of 2014 (Corker-Warner Bill, amended as Johnson-Crapo Bill)	
Overall objective	End federal control of and privatize the housing finance system	A reduced, explicit role for federal government paid for by mortgage borrowers	
Ultimate GSE footprint	Full phase-out of the GSEs within five years	Complete wind-down over five years	
Path for GSE portfolio	GSEs put into receivership and their assets sold off	Cease issuance of new guarantees and other new business within five years after enactment, reduce portfolio at a rate of 15 percent per year in the interim	
Risk sharing	Private sector bears full risk	Government backstop for catastrophic losses for MBS that meet statutory qualifications to be provided by a new entity—Federal Mortgage Insurance Corporation (FMIC), first-loss position no less than 10 percent	
Infrastructure and role for a public utility	FHFA to establish and supervise the National Mortgage Market Utility – a nongovernment, nonprofit entity that develops standards for servicing, pooling, and securitizing and operates a securitization platform	FMIC to develop risk sharing mechanisms, oversee and supervise the Common Securitization Platform, and set standards for approved market Participants	
Rules on private sector activity	Establish rules and regulations for a covered bond market place, risk retention rules repealed, increased disclosure to investors, improved processes for investors to establish the right to act on a security, gradually reduced conforming loan limits, requirements for approved issuers, aggregators, and servicers	Specified criteria for "covered securities" and "eligible mortgages", conforming loan limits remain the same (FMIC has no power on these limits), requirements for approved guarantors, issuers, aggregators, servicers, and insurers, FMIC to establish and capitalize a mutually owned company to facilitate access to secondary market by smaller Lenders	
Social goals	Limited to FHA and VA/RHS loans, FHA guarantee subject to tighter rules	Housing Trust Fund, Capital Magnet Fund, and Market Access Fund supported by fees on outstanding securities (average 10 basis points), FMIC to support the primary markets to ensure access to mortgage Credit	
Other government role (e.g., liquidity safety net)	FHFA's mandate for community investment eliminated	FHLB supervision transferred from FHFA to FMIC	

Source: [IMF15]

2.3 A Global Tour of Housing Finance Systems

Thus far, we have reviewed the overall architecture of housing finance systems and also studied how the U.S. HFS has evolved over the last 100+ years. To wrap up the discussion, we take a brief tour of housing finance systems over the world, to get a sense for their similarities and differences. Of course, a crucial point of reference for us is the U.S. housing finance system, which we will explore in depth in succeeding chapters. Here, we just highlight some of the top-level features that distinguish the U.S. from the rest of the world.

The U.S. is among the largest housing market in the worlds with 130 million units, only China and India have more. In most countries, the majority of units are owner-occupied. The national government plays a role in housing supply in some countries but not the U.S., where most housing construction is driven by the private sector. Figure 2.5 provides a snapshot of the institutional features of housing finance systems over the world. Some of the features that make the U.S. system unique are:²¹

- The prevalence of the 30-year fixed-rate mortgage: Most countries use variable-rate mortgages. In general, even when fixed-rate mortgages are available, they are generally offered for a much shorter time period (one to five years).
- Mortgage interest is not broadly deductible from income taxes: In addition to the U.S., the only other countries that allow the deductibility of mortgage interest are India, Denmark, the Netherlands, and, for five years, Ireland.
- Relatively consumer-friendly mortgage contracts: Loan terms are often stricter in other countries, and the U.S. is one of the very few countries where the majority of loans are **non-recourse** (i.e., the borrower's other assets are protected in the event of a default). For example, loans are full recourse in European countries. In China, loans are linked to compulsory savings and retirement fund schemes, providing an added incentive to stay current. Many other countries allow mortgage lenders to charge prepayment penalties when loans are prepaid.
- Presence of the GSEs in the Secondary Market: Most governments support mortgage lending through various types of mortgage insurance programs or by offering covered bonds. Governments participate much less in secondary markets, and in some countries the secondary capital market for mortgages are much smaller relative to the primary market than in the U.S. Thus, lending institutions are more likely to hold the loans that they make in such countries.

²¹[Bardhan 11]

Figure 2.5: Comparative Data on Global Housing Finance Systems

	Loan-to-Value Ratio	Variable or Fixed Rate (Majority)	Mortgage Deductible	Fee-Free Prepayment	Government Guarantee or Mortgage Insurance	Capital Market Role	Recourse or Non-Recourse
Australia	70%	Variable	No	Partial; full from mid-2011	For lenders	Moderate	Recourse
Canada	90%	Short-Term fixed	No		GMHC	Moderate, government presence	Recourse
China	60%	Variable	No	Yes	Insurance required for HPF loans	None	Loans linked to compulsory savings
Germany	74%	Variable with fixed limits	No	No	Covered bonds	Bonds	Recourse
Spain	70-80%	Variable	Only for Low income	No	No	Covered bonds 64%; ARM 36%	Recourse
U.K.	74%	Short-term fixed, then turn variable	No	Yes, if rate not disconnected	Insurance available for mortgagor	Grew 2000-2007; now negligible	Recourse
U.S.	80-90%	Both, currently mainly fixed	Yes	Primarily; details vary by state	Multiple agencies involved	Highly developed	Non-recourse

Source: [Bardhan11]

Chapter 3

Introduction to Mortgages

Thus far we have discussed mortgages at a fairly high level. This chapter goes into some of the nitty-gritty details of the different types of mortgages while also providing some background information on how borrowers choose mortgages and how one goes about computing monthly payments on a mortgage.

A mortgage is a loan that is collateralized by real estate. Like any loan, a mortgage can be described by a collection of basic characteristics:

- **Loan Amount.** The amount of money borrowed.
- **Loan Term.** The length of time over which the loan is scheduled to be repaid.
- **Repayment Schedule.** Typically, mortgage loans get paid off in monthly installments. Biweekly terms are also possible but are rarely available.
- **Contract Interest Rate.** The rate charged to the borrower for borrowing the money.

Several different property types can serve as collateral for mortgages:

- **Residential real estate.** **Single-family** (one- to four-family homes) or **multifamily** (more than four families) homes. Examples of multifamily properties include condos and co-operatives.
- **Non-residential real estate.** **Commercial** (office buildings, shopping centers, hospitals, apartment buildings) and farm properties.

In the U.S., as Table 3.1 demonstrates, single-family housing and the mortgages on these houses (“home mortgages”) form the most important part of the housing market.

Apart from the type of property that can serve as collateral for mortgage loans, there are several other ways to categorize them:

Table 3.1: Share of Property Types in the U.S. Mortgage Market: Q2 2014

Property Type	Amount (\$000s)
Single-family	9,855,419
Multifamily	957,581
Commercial	2,330,846
Farm	178,775

Source: Federal Reserve Flow of Funds

1. **Loan Purpose.** Mortgages are often categorized by whether they are used to purchase property (sometimes known as **purchase money mortgages**) or to refinance an existing loan. Refinance loans can themselves be of two types: **rate-term** or **cash-out**. Other loan purposes include short-term loans taken out – **construction loans**, **bridge loans** etc. A final loan purpose is a **home equity** loan, in which the borrower can elect to receive funds to use for any purpose. Reverse mortgages are available to older homeowners to borrow against the equity they have in their homes.
2. **Lien position.** Lien position determines which mortgage loan receives priority over other loans in the event of a foreclosure or bankruptcy. A mortgage that is in **first lien** position, sometimes called a **senior** loan, has priority for payment over a mortgage in a **junior lien** position if there is a foreclosure or bankruptcy proceeding. A **simultaneous second lien** is a second lien originated at the same time as a first lien mortgage, which may allow a consumer to borrow an amount up to 100% of the value of the home. Sometimes lenders have allowed consumers to borrow an amount greater than the value of the property, although this practice is not common in today's mortgage marketplace.
3. **Closed-end or Open-end.** Most mortgages are considered **closed-end** in that they consist of installment financing where the amount borrowed and the repayment schedule are set at the outset of the transaction. Closed-end mortgages can take first or junior lien positions. In contrast, **home equity lines of credit (HELOCs)** are **open-end credit**, extended to a consumer under a plan in which the borrower plans to draw against a credit line multiple times.

Over time, a wide variety of mortgage instruments have been introduced in various housing markets all over the world in order to meet the varying needs of borrowers and lenders. There is a fundamental tension in designing the “perfect” mortgage in that the needs of borrowers and lenders are often in opposition. Borrowers are mostly interested in how affordable the mortgage is, both at inception and over time. On the other hand, the lender will want to ensure that they obtain an appropriate return over the life of the loan.¹ Given that the effective life of a mortgage can often cover several years and consequently span a number

¹Here we are assuming that the lender is the ultimate holder of the loan, although the same argument applies whoever the end investor is. As discussed in Chapter 2, the return on the mortgage needs to compensate the mortgage investor for both the interest rate and credit risk inherent in the instrument.

of economic environments, it is sometimes the case that either the borrower or the lender faces an unexpected “shock” in terms of their respective exposures. Two such examples have already been discussed in Chapter 2 with regards to the experience of lenders of balloon mortgages in the Great Depression and the S&L’s with long-term fixed-rate mortgages in the 1980s, and some other historical examples will be briefly discussed below. Another point worth noting here is that, as we shall shortly see, different mortgage designs can lead to very different risk exposures for the borrower and lender.

3.1 Types of Mortgages

There are two core design principles at work for most mortgages – how the interest rate on the mortgage evolves over the time and how mortgage principal is paid back (**amortized**) over time. Let’s consider some of the risks associated with respect to both design principles. First, note that a mortgage rate that is allowed to evolve depending on prevailing interest rate levels is correlated with the level of inflation in the economy, with the opposite conclusion holding if the rate is fixed. In other words, a borrower with a fixed-rate mortgage is insulated from any inflationary pressures while the investor in an adjustable-rate mortgage has significant exposure to this risk factor.

Another key aspect of mortgage design is the amortization formula. The standard mortgage instrument is a level-payment, fully amortizing loan which calls for equal monthly payments over the life of the loan. Variations include the **constant amortization mortgage**, which has a level principal payment and declining total payment over time, and the **graduated payment mortgage**, which has a rising payment over the first few years before leveling out at a constant payment higher than on a standard mortgage loan for the remaining life. The constant amortization mortgage has a higher initial monthly payment which reduces affordability for the borrower. The graduated payment mortgage enhances affordability at the beginning of the loan but holds the possibility of a payment shock for the borrower. In general, mortgages which allow the monthly payment to change can pose significant affordability issues for borrowers in stressed economic environments as we will discuss below.

3.1.1 Fixed-rate Mortgages

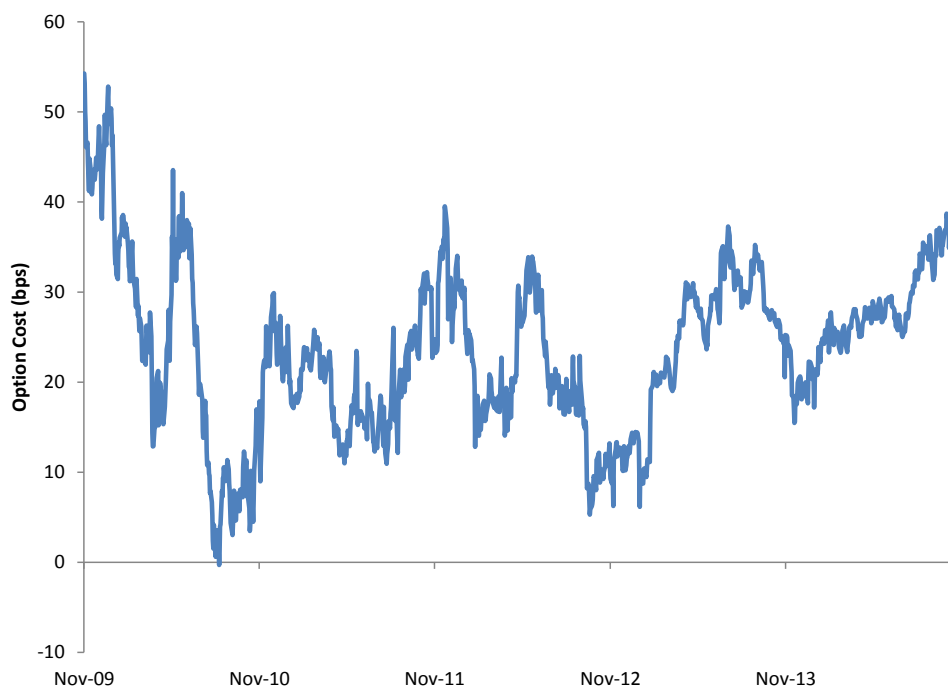
In a fixed-rate mortgage (FRM), the contract rate is held constant for the term of the mortgage. FRMs are most suitable for stable inflation and interest-rate environments and can become either prohibitively expensive for borrowers or too risky for lenders to offer in more volatile environments. This is because longer-term FRMs present a significant risk-management challenge for lenders – apart from credit risk, there is substantial interest rate risk arising from the long term of the mortgage and prepayment risk arising from the interest-rate sensitivity of borrower repayment.² Interest-rate risk led to the bankruptcy of a

²For example, if interest rates fall, borrowers will refinance their loans, shortening the maturity and exposing the lender to **reinvestment risk**. If rates rise, borrowers may keep their loans longer, creating

majority of the savings and loan industry in the United States in the 1980s.

In order to mitigate some of the interest rate risk associated with an FRM, most countries that offer them charge a prepayment penalty for the option to repay a loan early. The magnitude and term over which a prepayment penalty can be charged varies heavily from country to country. There are only two countries that have long-term FRMs with unlimited prepayment options for borrowers – Denmark and the U.S. In both these countries, almost all of the FRMs are financed in the capital markets, where sophisticated investors can price and manage the prepayment risk. Figure 3.1 provides some sense of the risk management challenges associated with an FRM by showing how the cost of the prepayment option (this cost is embedded in the mortgage rate) can vary substantially over time.

Figure 3.1: Cost of the FRM Prepayment Option Over Time



Source: Cello, Citigroup

A variation on the standard FRM can be found in Canada and a number of European countries where the dominant instrument is a rollover or short-term FRM. The rate of these instruments is initially fixed and adjusts periodically (for example, every one to five years). The borrower is subject to a prepayment penalty for repayment during the fixed-rate term but can typically make partial prepayments without penalty.

extension risk for lenders.

3.1.2 Adjustable-rate Mortgages

Adjustable-rate mortgages (ARMs) are the most popular mortgage instrument in the world because they manage to strike somewhat of a balance in meeting the differing needs of borrowers and lenders. They are initially affordable for borrowers because of their relatively low starting rate and also offer less interest rate risk for lenders because the contract rate adjusts to changes in inflation and real interest rates. Still, borrowers have to gamble that their income can keep up with payment changes, which may be quite challenging in a highly inflationary environment.

As the name suggests, an ARM has a note rate that periodically resets over the term of the mortgage depending upon where interest rates are. Once the note rate adjusts, the borrower's new principal and interest payments are determined by re-amortizing the mortgage based on the current principal balance, the original maturity date, and the new note rate. Initial rates on ARMs are usually lower than on fixed-rate mortgages (FRMs) because the ARM borrower is taking on interest rate risk. More precisely, these borrowers are taking on the risk that their mortgage payments might increase in the future. While this simple description captures the essence of an ARM contract, there are many variations in the types of ARMs available to consumers and understanding these variations requires familiarity with a fair amount of terminology. Fundamentally, this terminology describes the "how" and the "when" of the homeowner's note rate and payment change. We now provide a guided tour through this forest of jargon.

The starting point of determining a borrower's ARM rate is an **index**.³ The index is a financial market rate that reflects the prevailing cost of borrowing money. Changes in this cost will lead to movements in the index which will in turn lead to adjustments in the borrower's note rate. The most common indices in use in the U.S. are 1-year LIBOR and the 1-year Constant Maturity Treasury (CMT) index. The 1-year CMT is the average yield of Treasury securities with 1-year remaining to maturity and is published each Monday in Federal Reserve Release H.15. Other popular indices include the COFI Index (Cost of Funds Index)⁴ and the MTA Index.⁵

³In some countries (the U.K, for example), the rate on an ARM is changed according to the lender's discretion. Typically, the lender changes the rate according to changes in its cost of funds. This can be advantageous to borrowers if the lender primarily raises funds through retail deposits. These liabilities adjust less rapidly than money-market rates and shield borrowers somewhat from interest rate fluctuations; however, in less competitive or developed markets this instrument can create problems for borrowers if lenders abuse their privilege by manipulating rates. Indices are supposed to be beyond the control of lenders (the LIBOR-fixing scandal provides a compelling counterexample) – but they can create complexity for borrowers and may lead to greater payment volatility if the indices are more volatile than retail funds.

⁴The COFI Index is determined by the monthly weighted average cost of funds for the member institutions of the 11th District of the Federal Home Loan Bank (FHLB). The COFI for a given month is released on the last business day of the following month by the FHLB of San Francisco. For example, COFI for February is released on the last business day of March.

⁵The MTA (Monthly Treasury Average) Index is the 12-month average of the monthly average yields of U.S. Treasury securities adjusted to a constant maturity of one year. It is calculated by averaging the previous twelve monthly values of the 1-year CMT. The MTA Index is calculated on a monthly basis. The value for a particular month is generally available on the first Monday of the following month when monthly values of 1-year CMT are published in the H.15 release.

To capture the cost of originating a loan and to build in a profit margin, lenders add a **margin** to the index rate to determine the final loan rate: **Loan Rate = Margin + Index**. The margin is constant over the life of a mortgage loan and is typically 275 bps over 1-year CMT or 225 bps over 1-year LIBOR. For certain types of ARMs, lenders will frequently offer an initial discount on the **fully-indexed interest rate** (the sum of the index plus the margin) to attract borrowers – this introductory rate is called a **teaser rate** and can last from origination to when the borrower's rate first adjusts.

The speed with which the borrower's ARM note rate reacts to changes in the underlying index depends upon the length of the **reset period**. The reset period governs how often the loan rate on an ARM can change and is typically divided into two distinct intervals: the time to the first note rate change (the **first reset period**)⁶ and the time interval between all subsequent note rate changes (the periodic reset period). As the distinction between the two periods suggests, the length of the first reset period can be different from that of the periodic reset period and typically ranges from 1 month to 10 years. The length of the periodic reset period is more constrained and ranges from 1 month to 12 months. For example, in a **5/1 Hybrid ARM**, the mortgage rate is initially fixed for five years and then resets annually after that. The newly reset rate on the 5/1 Hybrid ARM is equal to the index (1-year CMT or 1-year LIBOR) plus the margin (275 bps over 1-year CMT or 225 bps over 1-year LIBOR).

When the note rate on an ARM resets, a **lookback** period is used to determine the appropriate value of the index to use for setting the rate – this period is equal to the number of calendar days that are counted back from the interest rate change date to determine the new index value. The lookback period is usually 30, 45, or 60 calendar days.⁷ For most ARMs, the monthly payment usually adjusts a month after the note rate change.

To reduce the interest rate risk that the borrower is exposed to, ARMs come with **interest rate caps** that govern the maximum level to which the loan rate can reset. Typically, there are two types of rate caps: a **periodic cap** and a **lifetime cap**. Lifetime caps are required by U.S. law and limit rate changes over the life of the loan. Periodic caps limit rate changes between adjustment intervals. The periodic cap that applies to the first adjustment interval is often referred to as the **first reset cap** or the **initial reset cap**.

The market convention is to quote the cap structure in a specific format: **first reset cap/periodic cap/lifetime cap**. Thus, a 5/2/5 cap structure implies a first reset cap of 5%, a periodic cap of 2%, and a lifetime cap of 5%. A first reset cap of 5% implies that the borrower's mortgage rate cannot increase by more than 5% above the initial mortgage rate on the day of the first reset. Similarly, a periodic cap of 2% means that in any given year (or other specified periodic reset period), the borrower's mortgage rate cannot go up by more than 2% above the previous year's mortgage rate. Finally, a lifetime cap of 5% limits a borrower's rate from increasing by more than 5% above the initial mortgage rate over the entire term of the loan.

Typically, the ARM's cap structure also acts as a **floor** and restricts the maximum amount

⁶The first reset period is also known as the **initial period**.

⁷The interest change date is not included in the calculation of the lookback period.

rates can fall from the pre-adjustment rate level. For example, for a 5/2/5 cap structure, the maximum that rates can fall at first reset is 5% from the initial mortgage rate. Similarly, for each subsequent reset, the borrower's rate cannot decrease by more than 2% from the previous period's rate. There is no lifetime floor on the mortgage rate.

Another contractual feature sometimes present in ARMs that helps decrease the borrower's exposure to interest rates is the presence of **payment caps** that specify the maximum amount that the monthly payment can increase by. These limits are set as a percentage of the previous monthly payment. The presence of payment caps can lead to a situation in which the borrower's note rate adjusts upward without a corresponding increase in the payment amount. The capitalization of any unpaid interest in the scheduled monthly payment into the borrower's principal balance is referred to as **negative amortization**. Negative amortization usually occurs due to upward interest rate adjustments without a corresponding increase in the payment amount. This might happen either because of a mismatch between payment and rate adjustment frequencies or the existence of payment caps. Only a few types of ARMs allow negative amortization with the most popular such mortgage being an **Option ARM**.

The total amount of negative amortization possible on an ARM is limited in order to protect the lender. In particular, the increase in the outstanding balance of the loan is capped at 110% to 125% of the original balance. When this limit is hit, the payment is immediately **recast** – the monthly payment is raised to amortize the loan completely over the remaining term of the loan. In other words, a payment recast defines the payment restructuring of a negatively amortizing ARM to force the eventual payoff of the principal balance to zero. Payment caps are not applicable in this situation.

A **convertible** ARM gives the borrower an option to convert their loan to a fixed-rate mortgage after an initial lock-out period. As one might expect, the borrower does have to pay for this option and all else being equal, rates tend to be higher on convertible ARMs. Sometimes, the lender will also charge the borrower a fee when the loan converts although many lenders offer free conversions. While different conversion windows are offered by lenders, the borrower usually has to exercise their conversion option between year one and year five of the mortgage loan. To illustrate, consider a convertible 3/1 Hybrid ARM with a conversion window between years one and five. At any time between year one and three, the borrower has the option to convert from their initial fixed rate (which will convert to a floating rate at the first reset date) to a new 30-year fixed-rate mortgage for the remaining life of the loan. Also, after the end of year three and before the end of the conversion period, the borrower can convert the adjustable-rate loan to a fixed-rate loan at their discretion.

Table 3.2 collects some of the terminology discussed above.

Homeowner Note Rate Changes on ARMs

A useful way to contextualize some of the above terminology is to walk through the mechanics of a note rate and payment change on an ARM. This exercise also gives us a foundation for

better understanding the cash flows on an ARM MBS.

An ARM has specific dates when the mortgage rate to the borrower changes (**interest change dates**). Interest must be calculated at the new note rate beginning on the interest change date. The actual monthly payment change will typically occur a month after the interest rate change.⁸ The basic calculation to determine the new mortgage rate is to:

1. Identify the new index level;
2. Add the index and margin;
3. Round off (typically to the nearest eighth of a percent); and,
4. Check for cap restrictions.

The payment change date is the first date that the monthly payment changes to reflect the note rate change. For example, suppose the interest change date specified in the borrower's note is February 1 with the old note rate being 6% and the new rate being 7%. The situation is illustrated in Figure 3.2. Then, the January interest collected in the February 1 monthly payment is calculated at 6%, while amounts due after February 1 are based on the 7% interest date. The payment change date in this case is March 1.

As itemized above, to calculate the new note rate on February 1 we need to go through a four step process. The new index level in Step 1 is determined by “looking back” a pre-specified number of calendar days (30, 45, or 60). Figure 3.2 shows that with a 45 day lookback, we would use the most recently published value of our index as of December 15th in order to set the note rate on February 1. For example, if the calculation uses an index published once a month at the end of the month, the index value published at the end of November would be used.

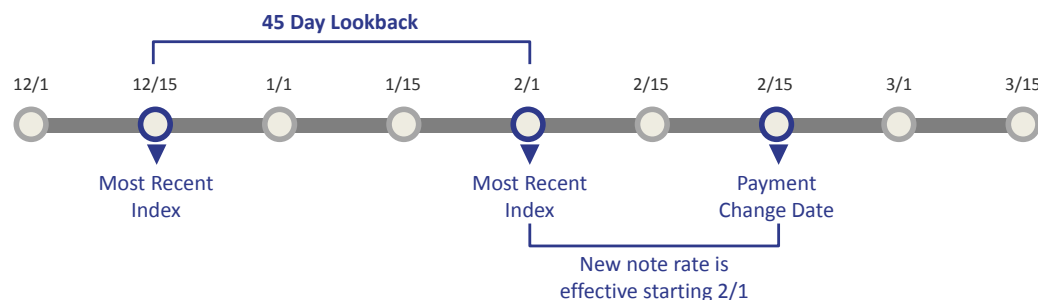
3.1.3 Indexed Mortgages

These mortgages are designed for high and volatile inflation environments. They attempt to reduce the impact of inflation on nominal interest rates in order to make loans initially more affordable. They also index payments to inflation or income in an attempt to make loans more affordable over time. The most popular types of such mortgages are the **price-level adjusted mortgage (PLAM)** and the **dual index mortgage (DIM)**.

The PLAM is an FRM in real terms – the rate is set at the beginning of the contract and fixed for the life of the loan, and principal balance and payment are adjusted periodically for changes in a price index. Typically, the balance is adjusted frequently (monthly) and the payment less frequently (annually), leading to negative amortization on the loans. PLAMs have been used successfully in Chile since the early 1980s but this is because the country

⁸As discussed previously, Option ARMs are a counterexample. The note rate on an Option ARM resets every month while the payment change frequency is annual.

Figure 3.2: Timeline for ARM Note Rate Changes



Source: Freddie Mac, Cello

has benefited from a stable and declining inflation environment since widespread use of the instrument began. The experience with PLAMs in other markets has not been as satisfactory. Commercial banks in Mexico were using PLAMs in the early 1990s. The sharp devaluation of 1994 (the “Tequila crisis”) led to a rapid increase in inflation (from 10% to 50% and higher within a year.) The resulting payment shock led to massive borrower defaults (over 50%).

The DIM attempts to address the affordability problem by indexing payments to wages by allowing the accrual rate on the loan to vary with inflation or a nominal interest rate. Once again, the basic mortgage design dilemma arises – in an attempt to maintain affordability for the borrower over time, a new problem is created: if the wage and rate indices diverge for a period of time the loan may not amortize. This happened to several vintages of DIMs in Mexico during the 1990s because of the Tequila crisis and aggressive initial terms.

Another indexed instrument gaining popularity in relatively high interest rate environments are loans indexed by foreign exchange (FX). These loans adjust balances and payments to changes in the exchange rate or are denominated in foreign currency. FX-indexed loans have been especially popular in transition economies (over 50% of loans in Poland, Romania). These loans carry great risks for borrowers and lenders. A sharp devaluation can lead to a payment shock for borrowers whose incomes are in domestic currency. Similarly, a devaluation can lead to significant losses for unhedged lenders.

3.1.4 Interest-Only Mortgages

Interest Only (IO) and Option ARMs are “affordability” products that reflect relatively recent product innovations in the mortgage market. The IO mortgage allows borrowers to make only interest payments on their mortgage until the IO term (typically 3, 5 or 10 years) ends, after

which the loan switches to being fully amortizing for the remainder of the mortgage term.⁹ In an **Option ARM**, the interest rate adjusts on a monthly basis while the monthly payment adjusts annually. The borrower has one of four different payment options each month: They can make a minimum monthly payment that is determined by an initial teaser rate and the payment change cap, an interest only payment, a 30-year fully amortizing payment, or a 15-year fully amortizing payment. If the borrower makes the minimum payment then the loan can experience negative amortization. Every five years or when the loan amount hits the negative amortization limit (110%/115%/125% of the initial loan value), the loan is **recast**. Option ARMs can come with a prepayment penalty with 1-year and 3-year penalties being the most common.

3.1.5 Reverse Mortgages

A **reverse mortgage** is a loan advance to senior citizens against the value of their homes. The reverse mortgage allows the homeowner to convert their home equity into cash without selling the house or making monthly payments. The reverse mortgage is primarily designed for “house rich, cash poor” elderly homeowners. A borrower has the option of choosing one of the following payment plans: take a lump sum payment, establish a line of credit to tap into, receive fixed payments from the lender over the course of their life, or receive fixed payments over a pre-selected term. A borrower does not need to repay the loan until they die (in which case the borrower’s estate pays off the loan), move out of their house for a period of twelve months, or sell the house.

Many institutions, including the U.S. federal government, some local and state governments, and banks and mortgage companies are currently involved in issuing reverse mortgages. One of the largest programs created for the purpose of facilitating the origination of reverse mortgages is the **Home Equity Conversion Program (HECM)** program backed by the FHA. The HECM program has a greater than 90% share of the total U.S. reverse mortgage market and is the only program in which the reverse mortgage is insured by the federal government. The program provides a guarantee to borrowers that lenders will meet their obligations¹⁰ while simultaneously insuring lenders against borrower default.

3.2 Borrower Mortgage Choice

As we can see from the listing of mortgage products in the previous section, the modern day menu of mortgage products offers a number of different options for potential borrowers. Borrowers can choose between ARMs with various reset terms (1/1s, 3/1s, 5/1s, 7/1s, 10/1s) and FRMs with different terms (10-, 15-, 20-, and 30-years). Before the advent of the Housing

⁹Formally, the **IO term** or **IO period** is defined to be the interval between the first payment due date and the last IO payment date.

¹⁰Besides servicing the loan, lenders are also responsible for sending cash advances (either monthly or lump sum payments) to reverse mortgage borrowers.

Crisis of 2007-2009, borrowers could also choose between fully-amortizing, interest-only (IO) or even negatively-amortizing variants of the above mortgages. Interestingly, the borrower's choice of a particular mortgage from the menu of available product types allows us to make inferences about the borrower on a number of different fronts including:

1. The borrower's current financial situation and their expectations for future income growth;
2. How long the borrower intends to stay in the house (**tenure horizon**);
3. How comfortable the borrower is with the mortgage payment changing and;
4. The cost of the mortgage relative to other available mortgage products.

The crucial concept behind the mortgage choice of borrowers is that of **self-selection**. Let's take a closer look at how this concept plays out in terms of a borrower choosing between ARMs and FRMs. There are two key differences between ARMs and FRMs: (1) Rates on ARMs are adjustable and (2) ARM borrowers are charged a lower rate because the yield curve is typically upward sloping (rates on ARMs are indexed to short-term rates) and because the value of the prepayment option on an ARM is less valuable than the one available to fixed-rate borrowers. The two mortgage products thus pose very different financial risks for the homeowner. For example, an ARM holder is exposed to the risk that their income may not keep pace with increases in real interest rates and/or inflation while these risks are not an issue for FRM holders. On the other hand, fixed-rate mortgagors are exposed to the risk that rates may decrease although they can refinance into a lower rate by paying a transaction fee of about 1-2 points (1-2% of the loan balance).

Fundamentally then, ARM borrowers are homeowners who are willing to make the gamble that over the period of their expected tenure, they will pay less on their monthly payments than if they took a fixed-rate mortgage and that increases in their mortgage rates will not outstrip increases in income.¹¹ For example, if a borrower knew with a high degree of certainty that they would be moving within the next five years then they would always pick a 5/1 Hybrid ARM mortgage over a 30-year mortgage since 5/1 Hybrid ARM rates are typically lower. This ability to extract information about borrower tenure expectations and risk tolerances from their choice of mortgage product provides us with a conceptual foundation for understanding the prepayment behavior of ARM borrowers versus FRM borrowers.

3.3 Mortgage Mathematics

It is often important in many contexts to be able to work out the monthly payment corresponding to a mortgage. In this section we use the formula for an annuity to derive some of the important payment formulas. An **annuity** is a contract that pays the holder (the

¹¹For further details on how ARM borrowers can differ from FRM borrowers see [Gardner87].

annuitant) a periodic sum, based on a predetermined schedule or formula, for a fixed period of time. Annuities are important because several fundamental cash flows that occur in fixed income space are structured as annuities. In the specific example that is of interest to us, a loan amount is repaid through a series of periodic payments (weekly, biweekly, monthly, ...).

We begin by deriving the formula for the present value (price) of an annuity. Suppose that we have an annuity that pays an amount M at the end of every period and that the per period interest rate is r/m , where r is the annual interest rate and m is the number of compounding periods in a year. Then, if the annuity makes N payments, the present value P of these cash flows is equal to:

$$P = \sum_{k=1}^N \frac{M}{(1 + \frac{r}{m})^k} \quad (3.1)$$

If we use the formula for the sum of a finite geometric series we obtain:

$$P = \frac{M}{\frac{r}{m}} - \frac{M}{(\frac{r}{m})(1 + \frac{r}{m})^N}. \quad (3.2)$$

In the situation that is of interest to us, we know the value of P and actually want to solve for M :

$$M = \frac{\frac{r}{m}(1 + \frac{r}{m})^N P}{(1 + \frac{r}{m})^N - 1}. \quad (3.3)$$

The calculation for M defines the periodic payment over a specified period of time that is equivalent (under the assumed interest rate) to an initial payment of P . This process of substituting periodic payments for a current obligation is known as **amortization**. For example, suppose that we borrow \$1,000 and that the terms of the loan are that the yearly interest rate is 12% compounded monthly and that we have to make equal monthly payments at the end of every month so as to repay (amortize) the loan over 5 years (60 monthly payments). We can use formula for M to calculate the monthly payments:

$$\frac{\frac{12\%}{12}(1 + \frac{12\%}{12})^{60} \$1000}{(1 + \frac{12\%}{12})^{60} - 1} = \$22.24 \text{ per month.}$$

We can think of the fixed-monthly payment M as consisting of both a principal and an interest component. In other words, part of M is used to reduce the principal balance on the loan and the remaining part to pay off the periodic interest that is due on the balance of the loan. For convenience, let us denote the initial balance of the loan by $B = B(0)$.¹² In the applications we are interested in, we want to find the outstanding loan balance, $B(k)$, at the end of every period k , where $k = 1, 2, \dots, N$. To do this, let $M = P(k) + I(k)$, where $P(k)$ is the portion of M that is used to pay down principal at the end of every period k and $I(k)$ is the interest due at the end of every time period. The decomposition of M is implicitly pointing out the fact that even though the monthly payment is fixed, the fraction of it that is used to pay off principal versus interest varies as a function of time.

¹²Note that B is playing the role of P in the above formulas.

First, notice that the interest due at the end of every time period $I(k)$ is given by:

$$I(k) = \frac{r}{m}B(k-1) \quad (3.4)$$

We also have:

$$\begin{aligned} B(k) &= B(k-1) - P(k) \quad (k = 1, 2, \dots, N) \\ &= B(k-1) - (M - I(k)) \\ &= (1 + r/m)B(k-1) - M \end{aligned}$$

It is now easy to derive a general formula for $B(k)$ by using this recursive expression (use the formula for a finite geometric series):

$$B(k) = (1 + r/m)^k B - \left[\frac{(1 + r/m)^k - 1}{r/m} \right] M \quad (3.5)$$

Since the loan is paid off after N periods, we know that $B(N) = 0$ and this gives us a formula for M (which is of course identical to the one we derived before):

$$M = \frac{\frac{r}{m}(1 + \frac{r}{m})^N B}{(1 + \frac{r}{m})^N - 1}. \quad (3.6)$$

We can simplify this expression a little by dividing the numerator and denominator by $(1 + r/m)^N$ and setting $d = 1/(1 + r/m)$:

$$M = \frac{\frac{r}{m}B}{1 - d^N} \quad (3.7)$$

We can use the same substitution to simplify the formulas for $P(k)$, $I(k)$, and $B(k)$.

$$\begin{aligned} P(k) &= M - I(k) \\ &= M - \frac{r}{m}B(k-1) \\ &= (1 + \frac{r}{m})^{k-1} (M - \frac{r}{m}B) \\ &= Md^{N-k+1} \end{aligned} \quad (3.8)$$

$$\begin{aligned} I(k) &= \frac{r}{m}B(k-1) \\ &= M(1 - d^{N-k+1}) \end{aligned} \quad (3.9)$$

$$\begin{aligned} B(k) &= (1 + r/m)^k B - \left[\frac{(1 + r/m)^k - 1}{r/m} \right] M \\ &= B \left(\frac{1 - d^{N-k}}{1 - d^N} \right) \end{aligned} \quad (3.10)$$

Generally speaking, these derivations use the following rules which apply to any type of mortgage. The only slight difference in what follows is that we allow for a non-constant contract interest rate, r_k , and a non-constant monthly payment $M(k)$, instead of the fixed M and r associated with a level-payment mortgage.

1. $I(k) = \frac{r_k}{m} B(k-1)$
2. $P(k) = M(k) - I(k)$
3. $B(k) = B(k-1) - P(k)$
4. $B(0) = P, B(N) = 0$

3.3.1 APR and Points

In practice, there is an origination fee associated with every mortgage that is typically quoted in **points**, with 1 point equal to 1% of the initial principal balance of the loan. The actual loan amount received by the homeowner has the points subtracted from it. In other words, if the initial loan principal is \$100,000 and there is an origination fee of 2% (= \$2,000), the actual loan amount received is \$98,000. Essentially, the homeowner is paying a higher interest rate which can be calculated by solving for r in the equation that relates M , P , and r . Let's work through an example. Assume that a lender is offering a rate of 5% on a 30-year mortgage with a balance of \$100,000. There is also 2 points of origination fees associated with the mortgage. The monthly payment on the mortgage is given by:

$$\frac{\frac{5\%}{12} (1 + \frac{5\%}{12})^{360} \$100,000}{(1 + \frac{5\%}{12})^{360} - 1} = \$536.82 \text{ per month.}$$

To determine the effective mortgage rate r' that the borrower is paying after taking fees into account, we solve for r' in the following equation:

$$\frac{\frac{r'}{12} (1 + \frac{r'}{12})^{360} \$98,000}{(1 + \frac{r'}{12})^{360} - 1} = \$536.82 \text{ per month.}$$

A numerical algorithm tells us that $r' = 5.175\%$.

We can think of r' as a comprehensive measure of the cost of credit to the borrower that includes loan fees and consequently allows borrowers to comparison shop. r' is often referred to as the **annual percentage rate (APR)**. The higher the loan fees, the greater the APR. If there are no loan fees, the APR equals the original mortgage rate.¹³ The one drawback of the APR is that it divides the fees over the future term of the mortgage. A more reasonable calculation would only extend to the borrower's expected holding period for the house.

¹³ Assuming that the mortgage rate stays constant over the term of the mortgage.

Table 3.2: Understanding Common ARM Terminology

Terminology	Description	Comments
Index	Financial market rate that is indexed to cost of borrowing; used by lender as basis for loan rate	Moves in tandem with the yield curve; volatility of different indices can be quite dissimilar
Margin	Set percentage that lender adds to index rate to calculate the final loan rate	Up to 3.5% on an individual loan; stays the same during life of the loan
Teaser rate	A discounted rate that is lower than the fully indexed rate	Introductory rate to attract borrowers; lasts from a month to a year
Reset period	Governs how often the loan's rate can change	Can range from one month to 10 years; the new rate equals index plus margin
Lookback	Number of calendar days that are counted back from the mortgage rate change date to determine the new index value	The lookback period is usually 30, 45, or 60 calendar days
Interest rate caps	The maximum level to which the loan rate can reset	Lifetime caps, required by law, limit rate changes over the life of the loan; periodic caps limit rate changes between adjustment levels; cap levels vary
Payment caps	Maximum amount that the monthly payment can increase	Limit set at a percentage of previous payment; can result in negative amortization
Negative amortization	The capitalization of any unpaid interest in the scheduled monthly payment into the borrower's principal balance	Most mortgages include a cap on negative amortization; the cap typically limits the total amount you can owe from 110% to 125% of the original loan amount.
Conversion option	The option to convert to a fixed-rate mortgage after an initial lock-out period.	The borrower usually has to exercise their conversion option between year one and year five of the mortgage loan.

Source: Cello

Chapter 4

The U.S. Residential Mortgage Market

Now that Chapter 3 has given us an idea of the many different types of mortgages that are available, we turn to the specifics of the U.S. housing finance system. Many of the important characteristics of the U.S. mortgage market have already been collected in Chapters 1 through 3, but we reprise them here for convenience using Figure 4.1 as a visual guide. As before, the focus is on the residential mortgage market. Turning to the figure, the idea behind it is to identify the key counterparties in the U.S. housing finance system and the interactions that take place amongst them. The chapter itself is primarily concerned with the activities that occur in the areas labeled ① and ② in the figure, namely in the primary mortgage market. The secondary mortgage market will be discussed in further detail in a future chapter that focuses on the U.S. MBS market. The descriptions in the itemized list below are keyed to the corresponding numbers in the figure.

- ① The U.S. has a very large mortgage market with outstanding residential mortgage debt of approximately \$10 trillion as of Q1 2014.¹ The home ownership rate averaged 66% over 2010-2014 and roughly two-thirds of households have a mortgage. Fixed-rate mortgages with a term of 30-years are the dominant product and homeowners have the right to prepay these mortgages without penalty. A borrower's ability to prepay play a key role in determining the value of a mortgage, as shall be discussed in detail in Chapter 5.
- ② The U.S. mortgage market is bifurcated into the primary and secondary mortgage markets. The **primary mortgage market** is where mortgages are originated by depository institutions such as commercial banks or thrifts who have expertise in underwriting new mortgage loans.
- ③ The **secondary mortgage market** is where newly originated mortgages are sold to an end-investor. The basic idea behind a secondary mortgage market is to tap the

¹See Table L.218 in the *Flow of Funds* report produced by the Federal Reserve.

capital markets as a long-term funding source for the mortgage market. Secondary mortgage market transactions are primarily accomplished by combining purchased mortgages to form a mortgage pool (this process is called **securitization**) and then selling pro-rated shares in the cash flows of this pool as mortgage securities (hence the term **mortgage-backed security** or **MBS**). Securitization is the main channel for funding mortgages with approximately 85-90% of all mortgages funded through the capital markets. The remaining non-securitized mortgages are typically held in portfolio by commercial banks and funded through deposits and other sources of funding. There is little to no use of covered bonds currently for funding mortgages.

Government agencies play a critical role in both the primary and secondary U.S. mortgage markets. Explicitly government-backed organizations that are active in the primary mortgage market include: the Federal Housing Administration (FHA) and the Department of Veteran's Affairs (VA). Quasi-governmental organizations such as Fannie Mae, Freddie Mac, and the Federal Home Loan Banks also play a very significant role. These institutions are considered to be implicitly backed by the U.S. government and are known as the Housing **Government Sponsored Entities (GSEs)**.² Ginnie Mae is a fully government-backed institution that is also deeply involved in the secondary market.

- ④ Broker-dealers act as market makers for all types of MBS. They can also create new types of MBS called Collateralized Mortgage Obligations (CMOs) by redistributing the cash flows generated by underlying pools of mortgages. CMOs are created to meet the needs of certain types of institutional investors who require a very specific cash flow profile from their assets.
- ⑤ Investors in the MBS market span the entire range of institutional investors – foreign central banks, insurance companies, commercial banks, mutual funds, hedge funds etc. In recent years, both the U.S. Treasury and the Federal Reserve have also been active participants in the MBS market.

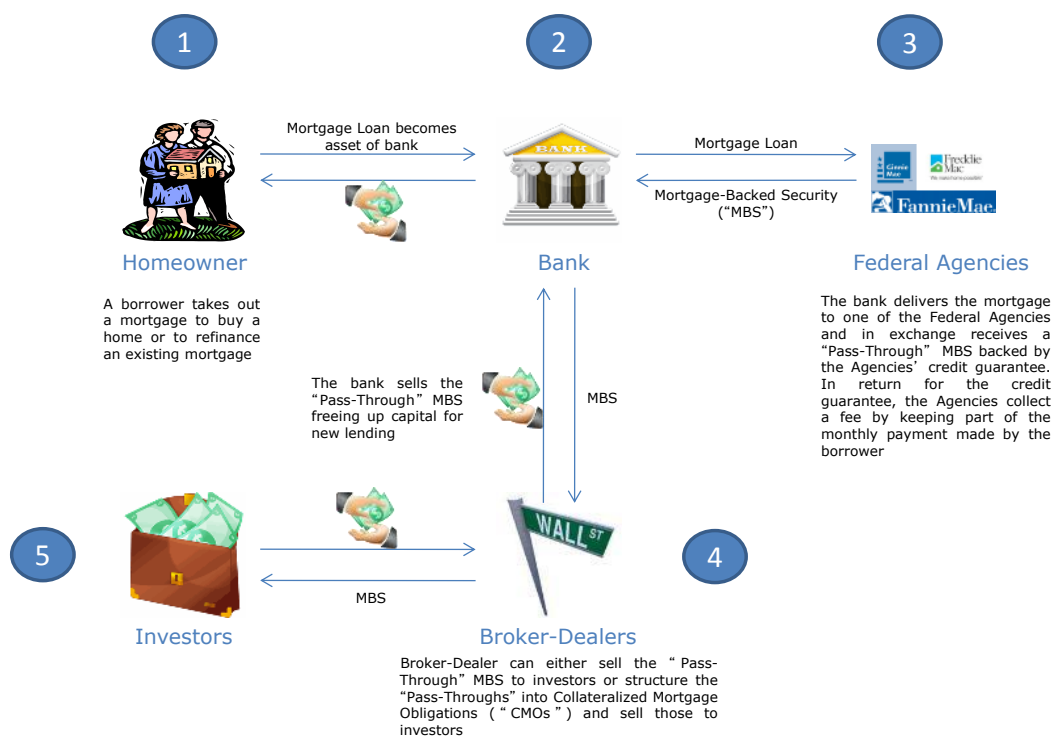
As one would expect, this bird's eye view of the U.S. mortgage finance system does not do full justice to its complexity. We now discuss certain aspects of the primary mortgage market in further detail. Of particular importance is the origination process from the vantage point of the borrower, mortgage funding, mortgage servicing, the segmentation of the mortgage market into different sub-sectors depending on the characteristics of both the borrower and the loan, and the U.S. laws governing mortgages.

4.1 The Origination Process

The mortgage origination process commences when a borrower files a loan application either to refinance a previous loan or to purchase a new home with a lender. The “lender” itself

²As noted previously, in the housing context, the GSE term is often used to refer only to Fannie Mae and Freddie Mac and this is the convention we will adhere to.

Figure 4.1: A Snapshot of the U.S. Mortgage Finance System



Source: Cello

can come from one of the following **origination channels**:

- **Retail Channel.** In the retail channel, the originator is directly involved with the applicant in the underwriting process via telephone, the Internet, and/or traditional brick and mortar outlets. All other things being equal, retail originations are considered to be less risky (from the vantage point of credit risk) than originations through other channels, as the originator was directly involved with the applicant during the underwriting process.
- **Wholesale Channel - Brokers.** In the wholesale channel, a **mortgage broker** solicits the loan and processes a consumer application. Mortgage brokers act as agents and simply bring together originators with borrowers. They are independent contractors and are not employees of the lender. Brokers often work with more than one lender, and large lenders can work with thousands of brokers. Brokers do not control the underwriting process, nor do they have any source of liquidity for funding the origination. The mortgage lenders brokers work with are called **wholesale lenders** and are often divisions of larger depository institutions. Generally, a wholesale lender requires a broker to enter into a wholesale lending agreement before the broker may originate loans on the lender's behalf. For their service, brokers command a fee that is passed on to the borrower.

In a variant of standard wholesale mortgage originations, some brokers **table fund** loans. In such a transaction, the mortgage broker closes the loan as the lender of record and then assigns the loan to a purchaser at or immediately after the closing. The loan purchaser provides the funding for the loan but the documents name the mortgage broker as the creditor.

- **Correspondents.** Correspondent lending is a blend of retail channel and wholesale channel lending that often features smaller institutions acting as **correspondent lenders**. Correspondent lenders are the primary interface with consumers, conducting all the steps in the mortgage origination process and funding their own loans. They generally originate and deliver loans³ pursuant to underwriting standards set by other lenders or investors, usually larger depository lenders who act as **aggregators**. In addition to soliciting consumers directly, correspondent lenders may receive applications and mortgage documents from mortgage brokers. Correspondents draw on warehouse lines⁴ from financial institutions for liquidity.

Regardless of the origination channel, it is useful to conceptualize the origination process as consisting of four stages: (1) Pre-Application; (2) Application; (3) Approval; and (4) Settlement.

³Including **servicing rights**. The next section provides some details.

⁴A **warehouse line** is a short-term revolving line of credit from a warehouse lender to fund closings of mortgages that will then be sold into the secondary market. Warehouse loans typically expire within 30 days.

4.1.1 Pre-Application

At this stage the borrower essentially makes the decision to take out a mortgage (whether for purchasing a home or refinancing an existing mortgage) by viewing mortgage rates advertised by various lenders and/or mortgage brokers. While the Internet has made this type of comparison shopping quite straightforward at some superficial level more rigorous comparisons still represent a challenge since there is not necessarily a standardized breakdown of the costs associated with closing a mortgage.⁵ We will skip some of the nuances involved with the choice of mortgage product by the borrower,⁶ and assume that they have decided on a 30-year mortgage. In this case, they would see a **Mortgage Rate/Point** menu similar to Table 4.1. The points show in the table are expressed as a percentage of the loan amount and the table assumes that there is a standard upfront closing cost of 2% of the loan balance (2 points).⁷

Table 4.1: 30-Year Mortgage Rate/Point Menu

Base Rate	Upfront Origination Points	Discount Points	APR (%)	Comments
4.00%	2%	0%	4.17%	Par mortgage
3.75%	2%	1%	4.00%	Discount mortgage
4.50%	0%	0%	4.50%	Premium no-cost mortgage

Source: Adapted from [Kalotay09]

How does a borrower decide which mortgage is the most cost effective one? As discussed in Chapter 3, the **Mortgage APR** was introduced to address this problem and gives a borrower a sense for the all-in cost of their mortgage, *assuming that they hold the mortgage to term*. In this case, as per the table, the best mortgage for the borrower is the discount mortgage. However, in reality, the vast majority of borrowers do not hold their mortgage to term. In this case, the appropriate calculation for them to look at is the APR over their tenure horizon. We call this calculation the **Effective APR** and the results for various borrower horizons are displayed in Table 4.2. Now, we can see that the **no-cost** mortgage is the best fit for borrowers with a short horizon whereas borrowers with longer horizons should pick either a par or discount mortgage. The intuition is that the longer the horizon, the more the present value of savings generated by the lower monthly payments offsets the upfront costs.

There are still some over-simplifications to consider. First, third-party fees such as title fees, attorney fees etc are not normally included in the APR. Second, this calculation does not

⁵There has been substantial progress on this front recently with the 2010 government-mandated introduction of a new set of forms that provide considerable transparency to consumers with respect to all the costs associated with a mortgage transaction.

⁶See the section on Borrower Mortgage Choice in Chapter 3 for further details.

⁷The table makes some significant simplifications. First, mortgage rates are typically quoted in increments of 0.125%. Second, the “Origination Points” number is deceptive in the sense that it only includes fees charged by the originator and does not include “third-party” fees such as title insurance, mortgage recording taxes etc. Third, closing costs can show significant variation by borrower and loan characteristics.

Table 4.2: Effective APRs for Different Borrower Tenure Horizons

Mortgage	Tenure Horizon (Years)			
	2	5	10	30
Par	5.47%	4.64%	4.48%	4.17%
Discount	5.95%	4.71%	4.47%	4.00%
Premium	4.50%	4.50%	4.50%	4.50%

Source: Adapted from [Kalotay09]

value the prepayment option written to the borrower. The value of the borrower prepayment option depends upon, among several other factors, the “moneyness” of the mortgage (par/premium/discount) and the expected volatility of interest rates.⁸ Tax considerations are also important because of the different treatment of discount points versus origination points. The choice of the mortgage should be made on an after-tax basis. Still, most of these considerations are academic because there are no widely available tools for mortgage borrowers to compute their Effective APR, let alone the after-tax prepayment option-adjusted APR.⁹

If the borrower reaches out to a specific originator, they are often shepherd through the initial process by a loan officer who can describe the different programs available, explain the general nature of the application process, and recommend the necessary documents. The loan officer may also provide the borrower with an initial indication of the maximum mortgage amount they may qualify for based on the borrower’s verbal assertions of how much the property is worth, their monthly income etc. This initial quote is called a **pre-qualification** and does not constitute a firm offer (or rejection) of credit.

4.1.2 Application

This step often involves assembling a comprehensive package of documents, which the lender uses to determine whether the borrower qualifies for mortgage credit. Virtually all borrowers fill out a **Uniform Residential Loan Application**¹⁰ In addition to this paperwork, borrowers are required to provide financial information including income history for the past two years, creditor information, employment history for the past two years, a list of liquid and non-liquid assets, and a list of all other investments that may be used for reserves or closing costs. Borrowers who are self-employed or paid on a commission basis, have a history of credit problems, or who already own property are required to provide additional supporting documentation.

⁸We will explore these points in further detail in Chapter 5.

⁹An interesting example of a mortgage refinance decision calculator is <http://analytics.kalotay.com/refival/init.do>. However, given the “black box” nature of some of the calculations, it is clearly not intended for the average consumer.

¹⁰See <http://www.freddiemac.com/uniform/pdf/form65.pdf>.

4.1.3 Approval

The standard approval process involves:

- A **credit check**, which involves reviewing the applicant's mortgage payment history and obtaining their credit report.
- **Qualification**, or determination of the **ability to pay**. This involves (i) verifying income and financial assets by means of W2s, pay stubs, tax returns, and bank statements; (ii) determining liabilities, typically through the credit report; and (iii) calculating **qualifying ratios**, namely computing the ratios of monthly mortgage debt payment to monthly income and total monthly debt payments to monthly income.
- A determination of property value, typically by means of an appraisal.

Based on the results of this approval process, borrowers ultimately may not qualify for the loan, or for the rate that was initially offered. In addition, borrowers have the option to turn down the loan offer, for example, because another lender has offered better loan terms. Assuming the approval does successfully go through, the result is a **mortgage commitment** that is valid for a specified number of days (typically 30 to 60 days). This period is known as the **rate lock** period. Borrowers have the option of opting for a better rate if interest rates fall over the rate lock period or if they receive a better quote. As such, many loan applications do not result in closed loans. This **fallout ratio** fluctuates over time and present a risk for originators that they attempt to hedge through the financial markets.

4.1.4 Settlement

The **settlement** process allows the lender to ensure that the borrower has legal title to the property, and complete other formalities related to funding the mortgage loan. The various steps in the settlement process include ordering a title search, obtaining title insurance, deed preparation and notarization, obtaining a survey, procuring loan documentation, and completing the closing with the borrower, which entails paying off the existing loan and disbursing funds. Finally, lenders require the establishment of an **escrow** account. Lenders use this account to ensure that borrowers pay off their real estate tax bills, hazard, and flood insurance premiums, and mortgage insurance premiums (if applicable) on a timely basis.

While the loan approval process is carried out or coordinated by the lender based on documents submitted by the borrower with the mortgage application, the activities under the settlement process are orchestrated by a Title or Escrow company, or by a real estate attorney, depending on the state.

4.2 Funding

As previously mentioned, originators have a variety of alternatives to fund their loans: they can securitize them in an Agency MBS or in the private-label MBS market, sell them as whole loans, or keep them on their balance sheets. In most of our discussions, we will implicitly assume that loans are securitized in an Agency pool, meaning that this option either dominates or is equally profitable compared to its alternatives for the originator.

A key feature of Agency pooling is that principal and interest payments in these securities are guaranteed by the GSEs in exchange for a fee (known as a **guarantee fee** or **g-fee**) that is collected partly out of the interest payments of the loan and partly as an upfront charge paid by the originator.¹¹ Through this process the credit risk of the mortgage is passed on to the GSEs. The effective g-fee is not fixed but has both seller- and loan specific components. Each seller has a seller-specific g-fee (generally expressed in basis points to be collected over the loan's life) and loan-specific components that are added on as upfront surcharges known as **loan-level price adjustments (LLPAs)** for short). The average size of the overall effective g-fee was approximately 20 basis points prior to 2007 and increased to about 63 basis points by 2014:Q4 because of the repricing of credit risk in mortgages during the housing crisis and in its aftermath.

4.3 Mortgage Servicing

When a lender originates a mortgage, it actually has two different assets that can be transferred and sold: the loan itself and the rights to service the loan (referred to as **mortgage servicing rights** or **MSRs**). In many cases, the entity that owns the loan is not necessarily the entity that services the loan.

Mortgage servicers are responsible for the day-to-day management of a loan and administer the loan until it is either paid off or transferred to another servicer. Major duties of servicers include collecting and crediting borrower monthly loan payments, operating a call center to answer borrower inquiries, remitting payments to investors, administering escrow accounts, and handling collection and loss mitigation activities in the event of borrower default.

Mortgage servicers may hold the MSRs but subcontract out the servicing function or portions of the servicing function to an outsource provider, a **subservicer** or, in the case of default, a **special servicer**. Loan servicers are governed by investor guidelines; state, federal and local laws; insurers and guarantor requirements, borrower expectations and their own standards. Note that loan servicers may service for many different investors such as Fannie Mae, Freddie Mac, Ginnie Mae, private investors and their own company.

¹¹This corresponds to ③ in Figure 4.1.

4.4 Subsectors of the Mortgage Market

One of the most confusing aspects of the U.S. single-family residential mortgage market is its segmentation into a number of different subsectors with no obvious guiding principle at work in terms of how these subsectors are defined. Loans are mapped into different categories depending upon their loan balances (**Conforming** versus **Jumbo**), borrower credit (**Prime**, **Alt-A**, **Subprime**), or lien position (**first-liens** versus **home equity** loans). In general, these naming conventions have been developed by participants in the primary and secondary markets although they are not always rigorously applied. In part, this is because there is no ironclad definition of what constitutes a prime borrower versus a subprime borrower. In particular, at the point of origination, a borrower does not necessarily know that they are applying for a prime loan versus a subprime loan.

4.4.1 Conventional Conforming

These loans “conform” to Fannie Mae and Freddie Mac underwriting standards and are therefore eligible to be pooled into Fannie Mae or Freddie Mac-backed MBS. The underwriting standards that define what loans conform have evolved over time but generally derive from a combination of Congressional charter-based and traditional risk-based criteria.

According to Congressional charter, single-family, conventional mortgage loans are subject to maximum original principal balance limits (**conforming loan limits**), which Fannie Mae and Freddie Mac’s regulator, the Federal Housing Finance Authority (FHFA), adjusts annually based on the average price of such residences. In 2014, the general loan limit was \$417,000, but it may be greater in higher-cost areas.¹² The GSE’s charter also establishes that any conventional single-family that it purchases or securitizes has an LTV ratio over 80% must have credit enhancements.¹³

In addition to these charter-based, mortgage eligibility criteria, the GSEs also rely on traditional risk-based criteria – the so-called “3 C’s of mortgage underwriting” – **collateral**, **capacity**, and **creditworthiness**. Collateral is property pledged to secure debt and is measured by the LTV ratio. Capacity represents the borrower’s ability to make payments on their monthly debt load; the ratio between the payments that the borrower must make on monthly basis to repay outstanding debts and the borrower’s monthly income is the most frequently used measurement of capacity. Creditworthiness is an assessment of the borrower’s probability of repaying a debt and is measured by the borrower’s credit score. Typically, the Enterprises stipulate a maximum LTV of 95%; a debt-to-income ratio between 36% to 45% (depending on compensating factors); and to have a minimum credit score of 660 for loans with an LTV above 75%.¹⁴

¹²Most high cost areas have maximum loan limits around \$625,500 but there are areas of the country that have limits up to \$721,050.

¹³Three types of credit enhancement are available to the GSEs in order to meet this requirement – mortgage insurance, repurchase agreements, or lender participation in the mortgage.

¹⁴If the LTV is less than or equal to 75%, the minimum credit score is 620.

4.4.2 Government (FHA/VA)

These loans are insured or guaranteed by the following the federal housing institutions: the Federal Housing Administration (FHA), the Department of Veteran's Affairs (VA), the U.S. Department of Agriculture (USDA), and the Office of Public and Indian Housing (PIH). Compared to loans pooled in Fannie Mae or Freddie Mac-guaranteed MBS, these loans are characterized by lower borrower credit scores and higher LTV ratios. FHA/VA loans are almost entirely securitized in Ginnie Mae MBS.

Borrowers who take out these loans would typically have problems getting their mortgage loans funded in some of the other sub-sectors because of their high LTV ratios, which are typically in excess of 95%, along with having credit scores that are often on the lower side. In many cases, these borrowers consist of first-time homebuyers or have a low- to medium-income profile and consequently do not have the financial wherewithal to put down a large downpayment on a home. In order to understand why these borrowers have to rely on FHA or VA insurance for lenders to extend them a mortgage, it is helpful to review how mortgage insurance works in the U.S.

Borrowers with higher LTV ratios typically have a higher propensity to default and the less equity a borrower has in their home, the higher the potential losses that a lender could incur in foreclosing and selling the property. Consequently, mortgage lenders protect themselves by requiring mortgage insurance (MI) when a borrower has an LTV ratio greater than 80%. In fact, as discussed above, by charter, Fannie Mae and Freddie Mac are not allowed to purchase a mortgage with an LTV over 80% without credit enhancement. Generally, the higher the LTV, the greater the amount of insurance charged. Almost all insured single-family loans are insured by the FHA, VA, or private mortgage insurers. The insurance programs offered by these different institutions differ principally in the following dimensions:

- Maximum mortgage amounts and LTV ratios allowed;
- Underwriting standards;
- Funds required at closing for the down payment and closing costs; and,
- The percentage of the loss on a foreclosed loan that is covered by each organization.

The most important distinction for the purposes of our discussion is that FHA-insured and VA-guaranteed loans can have higher LTV ratios and are underwritten to less stringent credit guidelines than are loans covered by private mortgage insurance (PMI). The private-mortgage insurance industry is typically not very active in LTVs between 97%-100% because of the high risk associated with these loans. On the other hand, FHA allows LTVs as high as 97%-98%, while VA allows LTVs as high as 100%. In addition, the borrower's ability to finance closing costs and insurance premiums can sometimes increase the effective LTV beyond 100%.

4.4.3 Jumbo (Conventional Non-conforming)

Jumbo mortgages are high-quality mortgages that meet stringent underwriting guidelines, similar to those set for Agency mortgages by Fannie Mae and Freddie Mac. These mortgages tend to fall into the Non-agency market because their loan balances are greater than those allowed by Fannie and Freddie for conforming loans. From a demographic perspective, borrowers who take out Jumbo mortgages are typically more affluent and financially sophisticated compared to the average conforming borrower.

4.4.4 Alt-A

Alt-A loans are taken out by borrowers who have prime A-quality credit but who, for one reason or another, are unwilling or unable to meet all the criteria to qualify for a conventional or jumbo prime loan. Instead, they borrow under the expanded underwriting guidelines of lenders with alt-A programs. Possible reasons why a homeowner may borrow under an alt-A program include:

- Inadequate documentation of assets or income to qualify for the conventional or jumbo prime programs;
- A blemished credit history that disqualifies the borrower from obtaining a conventional or jumbo prime loan in spite of an adequate FICO score;
- The loan-to-value (LTV) ratio for the property or occupancy type falls outside of standard agency underwriting guidelines; or,
- A loan-to-value ratio over 80% without primary mortgage insurance.

A rigorous definition of what constitutes an alt-A loan is difficult given the multiple factors that are taken into account in the underwriting process and the idiosyncratic variation in underwriting guidelines from program to program. Matters are further complicated by the fact that in certain cases it is more economic for an originator to securitize a conforming balance loan through the alt-A sector because of the high guarantee fee charged on these loans by the agencies.

The two originators behind the inception of the alt-A sector were Indy Mac and Residential Funding Corporation (RFC) through their RAST and RALI private-label MBS shelves respectively. The “alt-A” or “Expanded Criteria Underwriting” programs of both originators were largely focused on providing mortgage credit to borrowers who represented exceptions to the standard Agency and Jumbo underwriting matrices. The alt-A sector has been in a moribund state ever since the 2007-2009 Housing Crisis.

4.4.5 Subprime

Subprime mortgage lending is the origination of residential mortgage loans to customers with impaired credit histories. Typically, these borrowers have lower credit scores and/or other credit deficiencies that prevent them from qualifying for prime mortgages. To compensate subprime originators for greater default risks and higher servicing costs, subprime mortgagors pay higher interest rates, points and origination fees. Often termed “B&C” lending, subprime lending is very different from other non-conforming sectors such as Alt-A or Jumbo, with subprime borrowers exhibiting increased credit risk and reduced prepayment risk. After origination, the majority of subprime loans were securitized and sold in the capital markets as part of the “Home Equity” ABS sector. While “Home Equity” is used to classify a large gamut of securities backed by residential mortgage loans from prime high LTV mortgages to HELOCs, the vast majority of this sector consists of securitized subprime mortgages.

The underwriting process for subprime loans is not standardized and underwriting criteria vary widely among issuers. Subprime mortgages are “risk priced,” with less credit worthy borrowers being charged much higher interest rates. Finally, subprime lenders were also typically completely dependent upon the securitization and secondary whole loan markets for liquidity.

In the aftermath of the 2007-2009 Housing Crisis, subprime lending is a minor fraction of what it used to be.

4.4.6 Home equity

Home equity loans can be classified as **closed-end (piggybacks)** or **open-end (HELOCs)** second loans. Closed-end loans are often securitized in private-label MBS, while HELOCs are mostly held in portfolio by the originator.

A closed-end loan is just a regular fixed-rate mortgage so we’ll discuss HELOCs here. A HELOC is a credit line against the equity in the borrower’s home. Funds can be drawn against this credit line based on borrower needs up to a maximum amount (that depends upon the borrower’s equity). The lending institution will typically set a specified **draw** period during which the borrower is allowed to make withdrawals. Typical draw periods range from 5 to 10 years. During the time that borrowers are able to draw down funds, they usually pay a monthly interest charge on the outstanding balance on a HELOC.¹⁵ At the end of the **draw period**, the borrower enters a **repayment period** and must pay off the outstanding balance in regular periodic payments of principal and interest over a fixed term ranging from 10 to 20 years.¹⁶ HELOCs are often, but not always, in a junior lien position. HELOCs usually have an adjustable interest rate that changes over time, so the consumer’s payments may not be the same from month to month.

¹⁵A minimum credit line maintenance charge is imposed regardless of withdrawals.

¹⁶Some HELOCs may require a full repayment of principal at this point.

4.5 Laws affecting mortgages

There are three main groups of laws that affect mortgages: (i) foreclosure and bankruptcy laws; (ii) tax laws; and (iii) laws governing lending practices. To differing extents, each of these laws has an impact on the origination and repayment of mortgages. In particular, the impact of regulation on homeowner prepayment behavior is explored more fully in Chapter 5.

4.5.1 Foreclosure and bankruptcy laws

Foreclosure laws govern the process by which a lender can seize property which secures a loan. These laws, broadly speaking, govern the rights of the lender and the borrower after the borrower fails to make three consecutive monthly mortgage payments. At this point, the lender can begin foreclosure proceedings against the borrower; the mortgage is said to be 90+ days delinquent or in foreclosure (in many statistics, the two categories are combined into a single category labeled as **seriously delinquent**). In practice, in addition to a couple months of delinquency, commencing foreclosure proceedings requires an inability to come to terms on a **workout plan**¹⁷, or unsatisfactory performance by the borrower on a current workout plan. The foreclosure process concludes by auctioning the borrower's property (the **foreclosure sale**) in order to repay the loan. This sale itself does not necessarily result in the repayment of the outstanding loan if the winner of the auction is the lender/servicer, which is what frequently happens in practice. If the lender/servicer is the highest bidder at the auction, the property becomes part of their **real estate owned (REO)** portfolio and the liquidation of the REO property is what leads to repayment of the loan.

Foreclosure laws vary widely by state and the variation generally occurs along the following three dimensions:

- The foreclosure procedure followed: **Judicial**, **power-of-sale**, or some other type of process;
- The availability of deficiency judgments; and,
- The availability and duration of the statutory redemption period.

Judicial foreclosure is available in all states and is the only option in nearly half of these states. In such foreclosures, a judge hears the court case that the servicer has filed against the borrower. If the judge rules for the lender, the date for a court-supervised sale of the house is set. In a power-of-sale foreclosure, the servicer or trustee usually proceeds directly with the sale.¹⁸

¹⁷Once a borrower falls behind on their mortgage payments, the mortgage servicer will customarily attempt to search for cost-effective alternatives to the very expensive option of foreclosing the borrower's mortgage. This activity of pursuing alternatives to foreclosure by "working" with the borrower is often generally referred to as **loss mitigation** or **workout**.

¹⁸The trustee or servicer's right to sell the property without going through a court gives them "power-of-sale" and thus explains the nomenclature for this foreclosure method.

An important legal right that the borrower possesses prior to the foreclosure sale is called the **equity of redemption**. This right means that at any point prior to the sale, the borrower can halt the foreclosure process by paying the amount of outstanding mortgage debt (plus late payments and legal fees) in full. This right is available to the borrower in all states. However, the practical utility of it is somewhat circumscribed by the fact that the borrower has to come up with a significant amount of funds and/or a new financing arrangement, and may not have much time to accomplish all of this in the relatively speedy power-of-sale process. There is also another form of legal protection available to the borrower before the foreclosure sale – **arrearages** legislation. This basically says that the borrower can reinstate their mortgage (cure the default) by making up the missed payments (plus other penalty fees). Some states limit the number of times a borrower can reinstate a loan.

Foreclosure sale proceeds are used to pay off all the mortgage holders in order of their priority. In the unlikely event that there is a surplus (most borrowers would not default if they had equity in their property), it goes to the borrower. In many cases, the sale amount will fall short of the outstanding loan amount (plus foreclosure-related expenses). The difference between the outstanding mortgage debt and the foreclosure sale price plus default-related costs is called the **deficiency amount**. In theory, the borrower is personally liable for this amount in many states although lenders will rarely pursue borrowers. The borrower also potentially owes taxes on the amount of forgiven debt.

Bankruptcy law governs the rights of lenders when loans are not secured by real property, that is, for loans other than mortgages. Most often, households file for bankruptcy in order to cancel unsecured loans such as credit card debt. Mortgage loans cannot be canceled in the same way.

In a typical bankruptcy proceeding, a borrower's unsecured debts are discharged; however, his/her creditors seize all non-exempt assets. Which assets are exempt, and what their dollar value is, varies considerably from state to state. Delaware, for example, allows all debtors to exempt a little over \$5,000 of any property. Connecticut, by contrast, allows homeowners to exempt up to \$75,000 in home equity alone. The statutes governing treatment of home equity are known as the **homestead exemptions**. Several states in the Midwest, as well as Texas and Florida, have unlimited homestead exemptions. Bankruptcy reform legislation enacted in October 2005 imposes a national cap on the amount of home equity that can be exempted. The borrower can freeze the foreclosure process through the **automatic stay** by filing for bankruptcy and consequently such filings often increase the time spent in the foreclosure process.

4.5.2 Tax laws

The U.S. tax code privileges owner-occupied housing-related investment over other types of debt- and equity-related investment in several ways: the deductibility of mortgage interest and property taxes, the deductibility of certain costs associated with originating a mortgage, and the exclusion of home sales from the capital gains tax. Taken as a whole, owner-occupied housing-related taxation is viewed as being regressive through favoring higher-income

homeowners, and as subsidizing owner-occupied housing.

Imputed Rent. The general income tax framework in the U.S. aims to tax all forms of income while allowing for the deduction of costs incurred in earning that income. However, people who own and occupy their homes can deduct mortgage interest and property taxes from income while the rental value of the home (**imputed rent**) is excluded from taxable income.¹⁹ The key point is that owning a house provides a flow of housing services that otherwise would have to be paid for, and therefore should be included in gross income.²⁰ Net rental income would then be gross rent less maintenance and operating expenses, mortgage interest payments, property taxes, and a depreciation allowance. The deductibility of mortgage interest is considered a tax benefit only because the current tax code does not require imputed rent to be included in gross income. If imputed rent were to be included then mortgage interest would be a legitimate business expense, just as it is for landlords. It is also worth noting that prior to the Tax Reform Act of 1986, many forms of consumer interest payments were tax-deductible. The Act ensured that only interest paid on mortgages used to acquire a primary residence as well as home equity loans remained tax-deductible.

Mortgage Interest. The **mortgage interest deduction (MID)** provides a significant tax break to homeowners, one which is estimated to average \$80bb a year in foregone tax revenue.²¹ Under current law, homeowners who can itemize can deduct the interest on up to \$1 million of debt used to buy, build, or substantially improve their primary residence and one other home (defined as “acquisition debt”). In addition, homeowners can deduct the interest on a home equity line of credit (HELOC) that is used for any purpose up to \$100,000 of debt secured on the homes. To be deductible, the amount of the HELOC cannot exceed the difference between the market value of the house and the acquisition debt on the house.²²

When a mortgage is refinanced, the tax treatment of interest on the new loan depends on whether the replacement loan has the same balance as the outstanding mortgage or a higher balance (cash-out refinance). In a replacement mortgage, essentially the standard MID framework applies, provided that the total acquisition indebtedness is no more than \$1 million. With a cash-out refinancing, the tax treatment depends on how the excess cash is used. When deployed towards home improvements, it's considered to be acquisition indebtedness and the interest is deductible (subject to the \$1 million limit). If the excess cash is used for some other purpose such as buying a car or paying tuition, it's considered home equity debt subject to the \$100,000 limit.²³

¹⁹[Ozanne12] quantifies how the different tax treatments of owner-occupied and rented houses affect the relative costs of owning versus renting.

²⁰The value of these services is equal to the rental income the homeowner would earn if they were to rent the home to another person.

²¹Based on the 2014-2018 forecast produced by the Joint Committee on Taxation.

²²Some further restrictions have recently (for tax years beginning in 2013) been applied to the mortgage interest deduction for married taxpayers filing a joint return. In 2015, a couple with adjusted gross income (AGI) of more than \$309,900 generally must reduce their itemized deduction by 3% of the amount by which their AGI exceeds \$309,900 or 80% of their allowable itemized deductions, whichever is less.

²³Note that there are no tax implications associated with a cash-out refinance as there is no change to the homeowner's net worth. For example, suppose a homeowner had a mortgage of \$110,000 on a house that is worth \$200,000. In this case, assets total \$200,000, debts \$110,000, and net worth (equity) is \$200,000 -

Mortgage insurance premiums became temporarily tax deductible as of 2007 as a response to the housing crisis. The tax break has been extended through 2016 but only applies if the AGI is less than \$109,000 (married filing jointly).

Tax benefits that are realized from the MID are considered regressive on two accounts. First, homeowners in higher tax brackets benefit more. Every \$1,000 in deduction is equivalent to \$396 in tax savings in the top marginal tax bracket and only \$100 for the lowest bracket (10%). Second, the tax benefit only accrues to homeowners who itemize expenses but lower income homeowners are less likely to itemize since their deductible expenses tend to be less than the standard deduction.

Property taxes. Homeowners who itemize are allowed to deduct property taxes levied by state and local governments.²⁴ As is the case for mortgage interest, this deduction would be a legitimate business expense rather than a tax expenditure if owners were required to include their imputed gross rent as income.

Origination and Discount Points. For a house purchase, origination points and closing costs are deductible from income for tax purposes when paid in cash (these can also be rolled into the balance of the loan). However, on a refinanced loan, origination points and fees are not deductible when paid upfront but discount points (which qualify as prepaid interest) are. The discount point deduction must be amortized over the term of the loan. In the case where a homeowner finances their closing costs through taking on a higher interest rate, the additional interest is considered to be interest in general and is, as such, deductible.

To consider an example involving discount points, suppose that a \$250,000 mortgage is refinanced into a new 30-year loan with 2 discount points (\$5,000). Even though the points are paid upfront, they are deducted over 30 years at a rate of \$167 per year (first- and last-year deductions will be smaller).

If a homeowner is already amortizing points and refinances with a *new* lender – the unamortized balance can be deducted in the year of the refinance. However, when refinancing with the *same* lender, the unamortized points from the old loan must be added to any points paid on the new loan and the resulting total deducted over the life of the loan.

Capital Gains. When people sell an asset for more than the purchase price (the “basis”), they generally realize a capital gain that is subject to taxation as income.²⁵ Long-term capital gains – currently defined as those realized on assets held for more than a year – are taxed at various rates, most of which are below the rates applied to other types of income. However, the tax code allows taxpayers to exclude some of the capital gains attributable to owner-occupied housing. The current exclusion is limited to gains from the sale of a primary residence of up to \$500,000 for a couples filing joint tax returns and \$250,000 for most other

\$110,000 = \$90,000. Now suppose the homeowner took out a second mortgage (or did a cash-out refinance) that raised the mortgage amount to \$150,000. Then, total assets are \$240,000 (cash + house value), while debts have increased to \$150,000. Net worth remains the same at \$90,000.

²⁴Not including homeowners who are subject to the alternative minimum tax (AMT). The AMT does not allow the deduction of property taxes by homeowners.

²⁵The basis includes the costs of purchasing the property, such as the fee for a title registration.

tax payers. The exclusion is further restricted to houses that have been primary residences for two of the last 5 years. The capital gains exclusion can also be used repeatedly.

The current treatment of capital gains on house sales dates back to the Tax Reform Act of 1997. Until 1997, realized capital gains were generally not taxed because there was a “rollover” provision and a one-time \$125,000 exemption for those over the age of 55. However, the rollover proposition meant that younger taxpayers could only avoid gains by constantly trading up because sales proceeds that were not deployed towards purchasing more expensive homes were subject to taxation at the capital gains tax rate.

4.5.3 Laws regulating lending practices

The regulatory framework of the mortgage industry is established through acts of Congress and was, until recently, enforced by a number of different federal agencies including the Federal Reserve Board (FRB) of Governors, the Federal Deposit Insurance Corporation (FDIC), the Office of the Comptroller of the Currency (OCC), the Office of Thrift Supervision (OTS), and Housing and Urban Development (HUD).²⁶ The Housing Crisis led to a Congressional rule-making response with significant implications for the mortgage market, namely the passage of the Dodd-Frank Wall Street Reform and Consumer Protection Act (DFA). In particular, DFA consolidated compliance and oversight of 15 different regulations into one entity: the Bureau of Consumer Financial Protection (CFPB). The CFPB has taken the lead in terms of implementing aspects of DFA that are mortgage-specific.

We now overview the points of emphasis for mortgage regulation. One particular focus is the desire to prohibit discrimination in housing-related transactions. The Equal Credit Opportunity Act (ECOA, 1974) bans discrimination on ethnic or racial grounds. The Home Mortgage Disclosure Act (HMDA, 1975) requires most mortgage lending institutions to file a report on each application for a mortgage and the outcome of that application (for example, whether the application was approved or denied, and whether the borrower accepted the offer). Over time, information from the HMDA reports has become a useful tool for monitoring general mortgage market trends. The Community Reinvestment Act (1977), or CRA, broadly evaluates how well depository institutions serve the credit needs of households in the same geographical area as their deposit market. The CRA places particular emphasis on loans made to lower-income borrowers, or to borrowers in lower-income neighborhoods.

Another set of laws protects consumer privacy and regulates the way in which financial institutions can collect, use and share information about consumers. These regulations include the PATRIOT Act, Fair Credit Reporting Act (FCRA), and the Gramm-Leach-Bliley Financial Modernization Act (GLBA).

²⁶There is also extensive regulation of mortgage lending at the state, city and county level. An area of particular focus for these more localized laws is tackling **predatory lending** – a broker and/or a lender taking advantage of a borrower through deception, fraud, aggressive sales tactics, or taking unfair advantage of the borrower’s lack of understanding of loan terms. See [Bostic08] for more details on why states have chosen to become involved in this area and for a sense of the variation in approaches across different states.

A third set of laws protects consumers by attempting to increase the transparency of complex mortgage transactions. In particular, consumers lack an understanding of some of the trade-offs between mortgage products and, after having chosen a mortgage product, don't always understand the full spectrum of costs associated with taking out a mortgage or how to compare offerings across several lenders.²⁷ The federal government has attempted to guide consumers through these complexities by mandating that lenders provide timely information about the time and costs of mortgage credit during the loan application process. These disclosures are regulated through the Truth in Lending Act (TILA) and the Real Estate Settlement Procedures Act (RESPA) passed in 1968 and 1974 respectively. In 2010, the Dodd-Frank Act (DFA) transferred rulemaking authority under TILA and RESPA to the CFPB. The CFPB has collapsed four different disclosures into the two key ones:

- **Loan Estimate:** This disclosure is required to be provided within 3 days of application and represents the lender's best estimate in "good faith" of fees charged for the origination of the mortgage. The itemized set of costs includes legal fees, title search fees, appraisals etc.
- **Closing Estimate:** The disclosure is made available to the borrower 3 days prior to closing and covers details of the closing costs and a summary of the transaction.

One of the most significant laws affecting mortgage origination in recent years is the introduction of the **Qualified Mortgage (QM)** rule as mandated by the DFA. The Act requires lenders to ensure that borrowers have the ability to repay their loans. The idea was to protect financially unsophisticated consumers from being steered into loans that could turn out to be unaffordable because of payment shock. The creditor is required to consider the consumer's credit history, current income, employment status, and other financial resources. The creditor must also measure the ability to repay using a payment schedule that fully amortizes the loan over the scheduled terms. A lender may presume a loan has met the ability-to-repay test if it is a **qualified mortgage (QM)**.

In addition to the above restrictions, QMs must also satisfy the following requirements:

1. The loan has a term less than or equal to 30-years with no negative amortization;
2. Points and fees are capped at 3% (with some higher amounts for loans less than \$75,000);
3. Prepayment penalties are limited to fixed-rate loans and to 3% in Year 1, 2% in Year 2, 1% in Year 3; and 0% thereafter; and,
4. The loan cannot have a balloon payment (with an exception for small portfolio lenders).

Due to liability issues, there is unlikely to be lending outside of the QM box. Each time a lender is sued for a violation of the "ability-to-repay" statute, it will cost \$70,000 to \$100,000

²⁷Some of these nuances associated with a mortgage rate quote were discussed in the Origination Process section of Chapter 4.

to defend the case. Moreover, a lending institution losing a case will be subject to “enhanced damages.” Enhanced damages include actual damages, twice the finance charge (includes interest points and fees), all the fees the consumer paid, statutory damages up to \$4,000, court costs, and reasonable attorney’s fees associated with the enforcement action.

Chapter 5

Prepayments on Residential Mortgages

5.1 Introduction

Before making the transition from mortgages (Chapter 3) and primary mortgage markets (Chapter 4) to MBS (Chapter 6) and MBS markets (Chapter ??) we will take a detour towards developing a solid foundation in homeowner prepayment behavior. From an investor standpoint, understanding homeowner prepayment behavior is the prerequisite for grasping the investment characteristics of MBS.

What drives homeowners to prepay their mortgages? Broadly speaking, the difficulty in answering this question arises from the inherent complexity involved in understanding how a typical household values the complex bundle of rights and obligations that come along with a mortgage. In addition to mortgage rates, some of the factors that play a role in the prepayment decision include home values, the wealth of the borrower, their employment status, their income prospects, their age, the size of their family etc. Additionally, since a mortgage is just one item in a household's "balance sheet," which will generally consist of a number of assets (cash, mutual funds, stocks, housing) and liabilities (credit cards, mortgages, other loans), the decision to prepay the mortgage will also be influenced by the composition of this balance sheet. For example, a homeowner carrying credit card debt with a high APR may be more motivated to refinance their mortgage and use it as a vehicle for debt consolidation. Unfortunately, the hurdles in understanding prepayment behavior are raised by the fact that most prepayment data sets are bereft of household-level data that could potentially influence the prepayment decision.

Still, we can get some substantial traction on the causative factors for prepayments because of the availability of a substantial amount of time-series data on homeowner prepayment rates that stretches back to the late 1970s/early 1980s. A large number of mortgages have been securitized in mortgage-backed securities (MBS) over this time period and prepayment rates on these MBS constitute the source for all of our prepayment data. Correlating prepayment

rates on MBS with interest rates, home prices and the characteristics of the loans backing the MBS allows us to grasp many of the forces driving the prepayment behavior of homeowners.

The goal of this chapter is to provide an introduction to the different sources and motivations for prepayments along with a discussion of the typical “patterns” we see in prepayment behavior. Part of our discussion is largely qualitative and seeks to give readers a down-to-earth feeling for the multi-faceted nature of homeowner prepayment behavior. We also mention the explanatory factors used in our prepayment model and illustrate some of the response functions in the process of going through some of the practical details of how a prepayment model is built.

The plan of this chapter is as follows. To understand the themes underlying homeowner prepayment behavior, we begin by defining what a prepayment is and by describing the different ways that homeowners can prepay their mortgages. Next, we provide an explanation for why prepayments play such an integral role in determining the risk/return profile of an MBS. After this, we provide a guided tour of the conventions used for quoting and measuring prepayment rates. Before we get into some of the minutiae associated with understanding homeowner prepayment behavior, we walk through a detailed example to understand some of the typical patterns we see in homeowner prepayment behavior. The next step is to delve into the individual components of prepayment behavior (home sales, refinancings, curtailment and defaults) and discuss their causes. Our discussion then proceeds by going into the technical details of how prepayment models are built and validated. A concluding section examines the uses and limitations of prepayment models.

5.2 What is a Prepayment?

As we know from Chapter 3, a mortgage comes equipped with a repayment schedule which establishes a sequence of monthly¹ payments through which homeowners can payoff their debt. A **prepayment** is any activity by which the homeowner accelerates this schedule and extinguishes the mortgage before its final payment date. In the U.S., homeowners have the right to prepay their mortgage at any point in time, although there are transaction costs associated with this activity.²

To understand some of the nuances associated with the definition of prepayments, it is useful to think about the homeowner’s choices when the monthly payment on their mortgage comes due. In this situation, they have the following mutually exclusive options:

1. Make the scheduled monthly payment; or
2. Payoff the mortgage in full; or
3. Send more than the scheduled monthly payment; or

¹Mortgages with a bi-weekly payment schedule also exist in the U.S. but are not very common.

²These costs will be discussed in detail later on.

4. Not make the scheduled monthly payment.

Options 2 through 4 would all be classified as a prepayment. Option 2 corresponds to situations where the homeowner sells a mortgaged house, refinances an existing mortgage, or simply decides to extinguish their debt; Option 3 is called a **curtailment** or a partial prepayment; and Option 4 is called a **default** or **involuntary prepayment**.³ We now explore some of the details associated with these different types of prepayments.

5.2.1 Home Sales

Prepayments due to home sales are called **turnover** or **mobility-related** prepayments. The sale of a mortgaged house will almost always result in the prepayment of the underlying mortgage. This is because most mortgage contracts have a “due-on-sale” clause embedded in them. Essentially, this clause stipulates that the lender demand payment of the remaining balance of the loan when the house is sold.⁴ Mortgages which contain no due-on-sale provision are called **assumable** and allow the new borrower (the person who is buying the home) to “assume” the payments on the existing mortgage. Government loans (FHA- and VA-insured mortgages) are assumable but it is now quite rare to find a conventional mortgage that is assumable.⁵

The importance of the due-on-sale clause becomes clear when we think about the situation where the home for sale carries a mortgage with a coupon (interest rate) that is significantly below prevailing mortgage rates. In this case, a potential buyer would clearly prefer to assume the existing mortgage (assuming that they could afford the downpayment) rather than take out a new mortgage. The seller would also benefit in this situation since potential buyers

³Strictly speaking, one would have to not make the scheduled payment for a number of months before triggering a default.

⁴A due-on-sale clause is actually of more general applicability: whenever a mortgage has such a clause attached to it, the lender can demand full repayment when a property transfers through sale, inheritance, or other means. However, the Garn-St. Germain Act from 1982 provides an exemption for cases in which the transfer is to a spouse or child who will live in the residence, a surviving joint tenant, or to a spouse or a child in a court-ordered divorce settlement. In these cases, the new owners are allowed to assume the mortgage.

If the mortgage has a due-on-sale clause attached to it and the new owner of the property does not fall into one of the Garn-St. Germain exemption categories, they can still try to get permission from the lender to assume the loan or take out a new mortgage to payoff the old one. Finally, they could continue making payments on the old mortgage without assuming it but the lender retains the right to enforce the due-on-sale at any time.

One of the things that falls out of this discussion is the fact that the death of a homeowner will frequently not result in any kind of prepayment, as long as an heir or surviving joint tenant continues to make payments on the existing mortgage.

⁵FHA loans closed before December 14th, 1989 and VA loans closed before March 1st, 1988 are assumable by any buyer without restriction. For FHA/VA loans originated after these dates, the assumption process is a little more onerous. First, the buyer has to be approved by the relevant agency in the same manner that a new borrower is approved. Next, there are fees associated with the assumption: FHA allows lenders to charge a \$500 assumption fee and a fee for the credit report. VA allows a \$255 processing fee, a \$45 closing fee, and a funding fee of 0.5% of the loan balance.

would find the home more affordable. The only party that stands to lose from this situation is the originator who would not be able to collect origination fees on a new loan.

There is some interesting history behind assumable mortgages and the due-on-sale clause. The due-on-sale clause first came into the spotlight in the late 1970s/1980s, when a sustained increase in interest rates led originators to insert this clause in their mortgage contracts. Needless to say, consumers rebelled and after a protracted legal dispute, the issue was resolved in favor of the lenders by the Garn-St. Germain Act of 1982. Basically, the Act provides for the federal pre-emption of any due-on-sale restrictions imposed by state law. As such, it essentially codifies the enforceability of the due-on-sale clause.

5.2.2 Refinancing

In refinancing, a homeowner replaces the mortgage financing their current property with a new one that frequently (but not always) carries a lower interest rate. Homeowners have various reasons for refinancing including obtaining a lower interest rate (**rate refinance**), shortening or lengthening the repayment period (**term refinance**), converting from an adjustable-rate mortgage to a fixed-rate mortgage or vice-versa, and extracting equity from a property (**cashout refinance**). In addition, borrowers with a poor credit history who typically pay higher mortgage rates can often qualify for a lower rate after making timely payments on their mortgage obligation over a prolonged period of time (**credit refinance** or **credit curing**).

5.2.3 Curtailment

Sending in a monthly payment that is larger than the one required will (eventually) result in paying off a mortgage before it reaches its termination date. Curtailments therefore correspond to partial prepayments of the outstanding mortgage principal.⁶

5.2.4 Full Payoff

This corresponds to the situation where the homeowner simply decides to retire their mortgage by paying off the existing mortgage balance.

5.2.5 Default

A mortgage default is defined as the failure to fulfill any of the terms of the mortgage contract including being late on monthly payments, failing to pay property taxes, failing to pay

⁶Here, it is worth pointing out that making a curtailment does not release a borrower from the obligation of making the full amount of their scheduled payments in the future. In other words, the amount of the curtailment cannot be used to reduce future monthly payments.

insurance premiums etc. In theory, even being one month overdue on the monthly payment would constitute a default but, in practice, mortgage default usually corresponds to persistent non-payment by the borrower.

For Fannie Mae, Freddie Mac and Ginnie Mae securities, loans that have been delinquent for three to four consecutive months are typically removed from the pool.⁷ Since the various agencies guarantee full and timely payment of principal and interest on their MBS, the investor sees this as a prepayment. For private-label securities, the usual convention is to **advance** monthly payments of principal and interest until the ultimate disposition of the mortgaged property. Any principal losses that are incurred in the sale of the property are passed onto bond holders depending upon their position in the cash flow waterfall.

The above discussion shows that mortgage default typically results in the investor receiving early payment of their principal⁸ and demonstrates why defaults are often called **involuntary prepayments**. In general, when discussing prepayments on Agency MBS, we will include defaults in the definition of prepayments even though we are being somewhat imprecise when we do so.

5.3 The Importance of Prepayments

If homeowners were not allowed to prepay their mortgages, the pricing and analysis of MBS securities would be a relatively straightforward exercise. The cash flows on an Agency-guaranteed MBS would be the same in all interest rate environments and would consist of the principal and interest payments made by the underlying mortgagors less a servicing and guarantee fee. In this situation, MBS holders would simply worry about the credit risk posed by the underlying corporations who were collecting and forwarding the mortgage payments (Fannie Mae, Freddie Mac, and Ginnie Mae). In other words, an MBS would be the same as a **non-callable** bond.

The complexity of MBS arises from the homeowner's right to prepay their mortgage at any time. Since this prepayment rate will depend upon interest rates, among other factors, the cash flows on the MBS will vary based on interest rates. Thus, it is imperative to be able to forecast prepayment rates in different interest rate environments in order to be able to accurately estimate the cash flows generated by MBS investments in these same environments.

In order to develop a more precise idea for how prepayment rates will depend upon mortgage rates, we need a more nuanced understanding of the homeowner's right to prepayment. The homeowner's right to prepay their mortgage through **voluntary** means (home sales, refinancing, curtailment) can be thought of as equivalent to the homeowner purchasing an option when they enter into a mortgage contract. More precisely, the voluntary prepayment option is like a **call option** with strike price equal to par (plus the transaction costs associated

⁷In the case of the GSEs (Fannie Mae and Freddie Mac), the current convention is to automatically transfer the loan to the retained portfolio once the delinquency trigger is hit. For Ginnie Mae, the loan is removed from the pool by the servicer and held in their portfolio.

⁸It may not be the full amount of the principal for private-label securities.

with refinancing). The voluntary prepayment option resembles a call option in that the borrower has the right to “call” back the balance of the loan at any time. One situation in which they would consider “exercising” this option is when the market value of their mortgage is greater than par. Since the market value of the mortgage is the present value of the remaining payments discounted at prevailing mortgage rates, the lower the current mortgage rate is, the higher the market value of the mortgage. In down to earth terms, the lower the current mortgage rate is, the more the homeowner can save on their monthly payments by refinancing their mortgage. As a result of this equivalence between the right to prepayment and option contracts, market participants often use jargon borrowed from the options market to describe the incentives faced by the homeowner. Specifically, the homeowner’s prepayment option is said to be **in-the-money** whenever market rates are lower than the rate on the existing mortgage, **at-the-money** when these rates are (approximately) equal, and **out-of-the-money** otherwise.⁹

The dependence of home sales on interest rates is also fairly straightforward: a homeowner with a below-market mortgage rate has to face the additional frictional cost of increasing their monthly payments when they get a mortgage on the new house. Thus, higher mortgage rates tend to deter home sales.

It is worth pointing out that the voluntary prepayment option has some aspects to it that one does not find in call options. When the house is sold, the homeowner must prepay their mortgage, whether it appears to be “economic” or not (i.e., prevailing mortgage rates may be higher than the rate on their existing mortgage). But, note that this exercise of the option in “out-of-the-money” situations may not be sub-optimal because homeowner may be trying to maximize the return on the property (among other factors) which may or may not have the same objective function as trying to exercise the prepayment option optimally.

Since the homeowner also has the right to default (**involuntary prepayment**) on their mortgage, they also purchase a put option (in addition to the call option) when they enter into a mortgage contract. The right to default is like a put option because the borrower can turn over the house (“put the house”) to the lender in exchange for foregoing future payments on the mortgage. The value of this put option is driven by the market value of the mortgage and the value of the house, which are in turn respectively driven by interest rates and trends in home prices. The value of the home being less than the market value of the mortgage¹⁰ is not a sufficient reason for default because the borrower also has to weigh the high costs associated with defaulting: moving expenses, a damaged credit record, the expected cost of deficiency judgments etc. Consequently, in addition to negative equity, borrowers typically have to be subject to some **trigger event** in order to default – a shock to their income resulting from sustained unemployment, a divorce, an unexpected increase in health-related costs, or some other form of liability.

To summarize the above discussion, in essence, the investor in an MBS has short position in a call and a put option and a long position in a non-callable bond with the same payment

⁹A more precise definition of “moneyness” would also consider the costs of refinancing a mortgage. These costs typically range from \$1,500 to \$3,000 for a \$200,000 loan; Table 5.2 provides further details.

¹⁰In this situation, a homeowner is said to have “negative equity.”

schedule as the MBS.¹¹ In equation form:

$$\begin{aligned} \text{Value of MBS} &= \text{Value of Non-Callable Bond} \\ &\quad - \text{Voluntary Prepayment Option} \\ &\quad - \text{Involuntary Prepayment (Default) Option} \end{aligned} \tag{5.1}$$

The value of the MBS investment thus depends on the value of these options. The goal of prepayment modeling is to allow for precise quantification of the value of these options by building a predictive model for homeowner prepayment behavior.

5.4 Measuring Prepayment Rates

Since all of our data on homeowner prepayments comes from the MBS market, the prevailing conventions for calculating prepayment rates are strongly linked to the mechanics of the MBS market. Concretely, in a typical MBS structure, a group (“pool”) of residential mortgage loans is sold to a trust which in turn issues securities and makes monthly payments to investors out of the payments made by the underlying homeowners. The monthly payments passed through to investors consist of scheduled principal, interest, and **unscheduled principal**. This unscheduled principal amount is what corresponds to homeowner prepayments.

To convert the dollar amount of the unscheduled principal to a prepayment rate, we first need a way to calculate the scheduled balance of the pool based on the regular amortization of principal. This is straightforward since, given data on a mortgage pool consisting of its coupon, remaining term, and months elapsed since origination, we can compute the scheduled balance of the pool for each remaining month using a formula. This balance can then be compared to the actual balance of the pool and the difference between the two numbers corresponds to unscheduled payments of principal (or **prepayments** for short.) In equations:

$$\text{Prepayment} = \text{Scheduled Principal Balance} - \text{Actual Principal Balance} \tag{5.2}$$

This equation gives us the dollar amount of prepayments. To convert this dollar amount into the prepayment rate per dollar of principal over a month, we calculate the **Single Monthly Mortality (SMM)** rate. The SMM in any month for a pool of mortgages is given by the expression below. Given a collection of loans that is outstanding at the beginning of a month, we can think of the SMM as measuring the fraction of these loans that terminate because of prepayments.¹²

$$\text{SMM} = \frac{\text{Prepayment}}{\text{Scheduled Principal Balance}} \tag{5.3}$$

For example, consider a pool which has an outstanding balance of \$1.1 million at the beginning of July and suppose that we are expecting scheduled principal payments of \$100,000 on the

¹¹We need to be a little more precise in the case where the MBS is backed by the Agencies. In this case, the relevant Agency guarantees full and timely payment of principal and interest so the investor in the MBS is not on the hook for any credit-related losses.

¹²The calculation implicitly assumes that we can think of each dollar of principal as corresponding to a single loan. Thus, all prepayments are whole prepayments on \$1.00 unit loans within the pool.

pool for the month. Now, suppose that at the end of the month we measure the total amount of principal payments received and we find the amount to actually be \$110,000. Then:

$$\begin{aligned}\text{Prepayment} &= \text{Scheduled Principal Balance} - \text{Actual Principal Balance} \\ &= \$1,000,000 - \$990,000 \\ &= \$10,000\end{aligned}$$

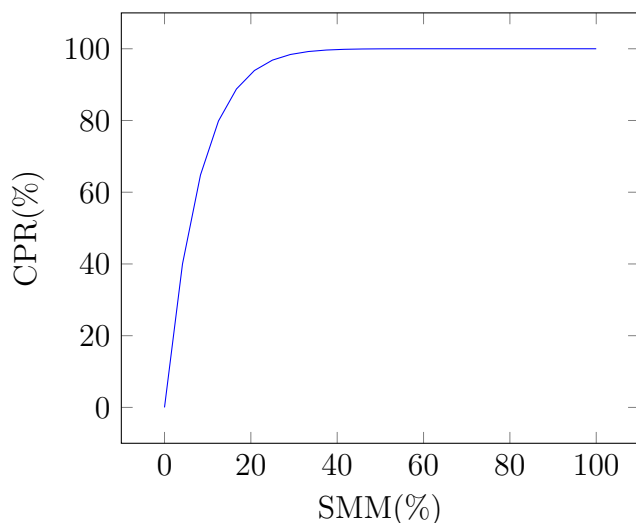
$$\begin{aligned}\text{July SMM} &= \frac{\$10,000}{\$1,000,000} \\ &= 1\%\end{aligned}$$

The most common convention used to quote prepayment rates in the market is the **Constant Prepayment Rate (CPR)**. The CPR calculation is simply an annualized version of the SMM calculation, assuming that the SMM remains constant for twelve months. One can think of the CPR as calculating what fraction of the loans outstanding at the beginning of the month will terminate over the next year because of prepayments. SMM and CPR are related by the following expression:

$$\text{CPR} = 1 - (1 - \text{SMM})^{12} \quad (5.4)$$

Figure 5.1 uses the above formula to plot the relationship between SMM and CPR for different prepayment speeds. The third widely used convention for quoting prepayment speeds that

Figure 5.1: The Relationship between CPR and SMM

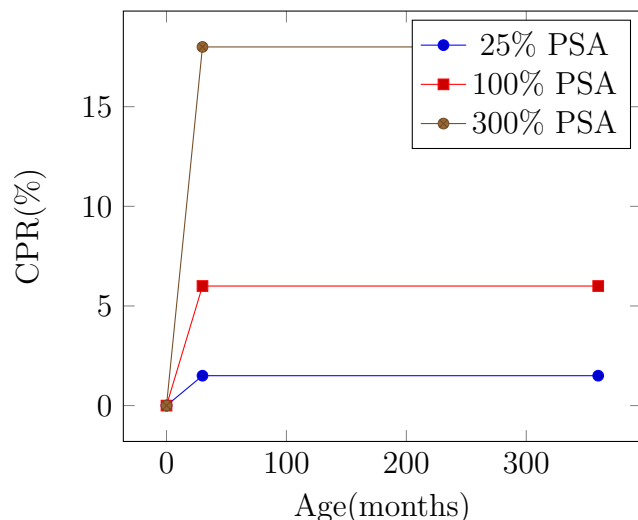


Source: Cello

is prevalent in the market is the **Public Securities Association (PSA) Curve**. Unlike CPR and SMM, the PSA convention calculates prepayment speeds by comparing them to a reference prepayment curve. This benchmark prepayment curve, also referred to as the

100% PSA curve, defines a vector of prepayments that is indexed to the age of the MBS. The vector has speeds increasing linearly from 0% CPR at an age of 0 months to 6% CPR at an age of 30 months, and then remaining constant at 6% CPR as a function of age after that.¹³ A speed of 300% PSA means that prepayment rates as a function of age would be 3 times the CPRs given by the 100% PSA curve. Similarly, speeds corresponding to the 25% PSA curve would be one fourth of the speeds given by the base curve. The 100%, 300%, and 25% PSA curve are plotted in Figure 5.2.

Figure 5.2: The Relationship between CPR and PSA



Source: Cello

Even though the convention for quoting speeds through SMM and CPR seems very different from the PSA convention, it is fairly easy to move from one measure to another.¹⁴ The only additional information required is the age of the pool. For example, a pool with an age of 20 months that is prepaying at 300% PSA has a prepay speed of 12% CPR (see Figure 5.2), which in turn corresponds to an SMM of 1.06%.

Given that one can move from PSA to CPR and back, the question arises as to what purpose these different prepayment quoting conventions serve. One obvious difference is that the CPR quoting convention has no dependence on age, making it difficult to compare prepayment speeds on two MBS pools with different ages since speeds on more seasoned pools will generally be higher than on new ones.¹⁵ On the other hand, the PSA convention gives us a direct way to compare speeds on pools with different ages. For example, consider two pools with ages of 20 and 30 months respectively, both of which are prepaying at 12% CPR. Although both pools are prepaying at the same CPR, the PSA on the pool with an age of 20

¹³This ramp was originally developed using historical prepayment data dating from the early 80's to capture the so-called "seasoning" effect commonly observed in mortgage securities.

¹⁴The conversion between monthly CPR and PSA is straightforward. Things become a little more complicated if we want to compare these metrics over longer periods of time (3 months, 6 months, etc.).

¹⁵We discuss the "seasoning effect" in MBS prepayments in detail later on in this chapter.

months is 300% PSA, whereas it is 200% PSA on the 30 month seasoned pool. Thus, one could conclude that the first pool is prepaying faster than the second one after controlling for seasoning. It is important to keep this in mind that although PSA percentages provide a way of comparing prepayment speeds on pools, it is not advisable to assume that speeds will actually follow the PSA prepayment ramp in future months.

Although CPR is the preferred metric for quoting prepayment speeds, the PSA convention is still sometimes used for quoting the speed assumptions used to price Agency and Non-agency CMOs. This is primarily because the PSA curve incorporates a seasoning ramp similar to the ones observed on mortgage pools. Another commonly used prepayment curve in the Non-agency sector is the **PPC** or **Prospectus Prepay Curve**. Unlike the PSA curve, the PPC curve is not a standard curve and is deal-specific. It is arrived at by using the collateral characteristics underlying the deal and is typically defined in the prospectus of the deal.

5.5 A High-Level Overview of Homeowner Prepayment Behavior

Now that we have some idea of what prepayments are, why they are important, and how they are measured, it is useful to get a broad sense of how they can vary as a function of interest rates. The prepayment speeds released on individual MBS pools can be very noisy because of the relatively small number of homeowners backing a pool so the market convention is to look at prepayment rates on a collection of pools grouped together by pool coupon and the year that the loans were originated in. This grouping ensures that all borrowers in the pools we are tracking pay more or less the same mortgage rate and have held their mortgages for approximately the same amount of time. Figure 5.7.2 shows prepayments on 7% pools (7s for short) originated in 1996 (average borrower mortgage rate = 7.61%). We will take a relatively detailed tour through the trends in the figure.

Let us start by getting a perspective for how high and low prepayment speeds can get. Prepayment speeds started out quite low on the 7s but ramped up to 7% by December 1997. The mortgage holders in the pool did not really have an opportunity to refinance over this period since mortgage rates ranged between 7.50% to 8.50% over the 1996-1997 time frame. This gradual ramp up of prepayment speeds over time is typical behavior for mortgage loans and is called the **seasoning ramp**.

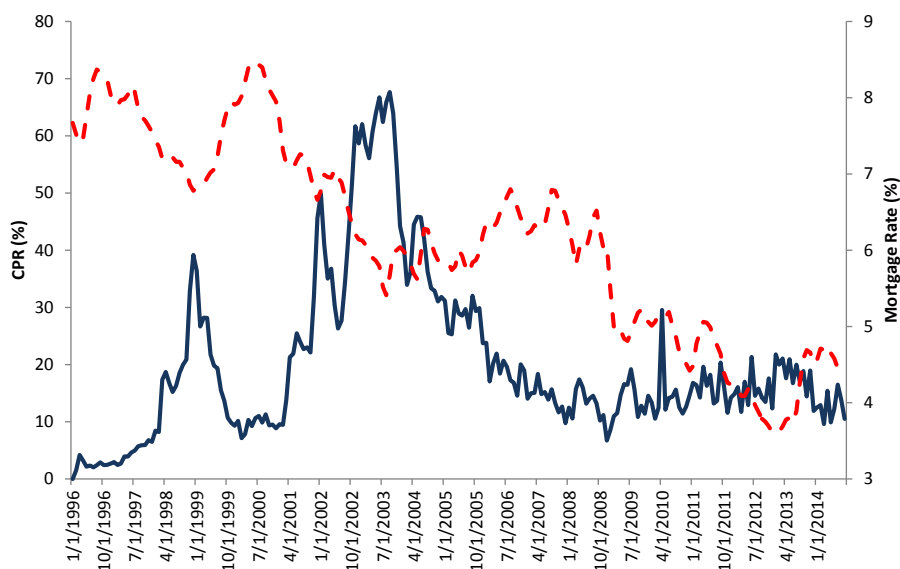
As rates fell to around 6.70% in October 1998, we see a dramatic spike in prepayment speeds to 40% CPR in December 1998. At this juncture, mortgage rates had fallen to levels not seen since 1993 resulting in a massive increase in refinancing activity for most coupons. This type of explosion in refinancing activity is typically referred to as a **refinance wave** by market participants. Speeds gradually declined as mortgage rates began to rise but never really fell below 9%-10% CPR over 2000 even though mortgage rates climbed all the way up to 8.50%.

The market saw a succession of refinance waves as rates fell almost continuously from a peak of 8.54% in May 2000 to a multi-decade low of 5.16% in June of 2003. It is interesting to

note that at this point, speeds on the 7% pools reached “only” 66% CPR. We say “only” because even though prevailing mortgage rates were some 2% lower than what the borrowers in the pools were paying, some homeowners in these pools did not prepay. The takeaway is that even at very high levels of rate incentive, prepayment rates on mortgage pools do not necessarily reach 100% CPR.

The prepayment behavior of 7s over the remaining portion of the graph highlights a very important phenomenon: the **burnout** of mortgage pools. As the terminology suggests, after an MBS pool has been through a refinancing episode, exposure to the same or greater refinancing incentives does not lead to the same refinancing response. For example, even with rates reaching an all-time low of 4.22% in November 2010, prepayment speeds on 7s barely responded and have remained in the teens.

Figure 5.3: Prepayment Rates on 7% Pools Originated in 1996



Source: Cello

While the discussion above helps in building an overall perspective for the total variation we can expect to see in prepayment speeds and for some general patterns that are typical of MBS prepayment rates, it does not really give us a feeling for what causes these patterns and variations, or how the individual components of prepayments (home sales, refinancing, curtailments, default) contribute to this picture. The goal of the next few sections is to delve into the details of each of the different components of prepayments and fill out some of the missing elements of this picture.

5.6 The Components of Prepayments: Home Sales

The first component of prepayments that we will analyze in detail is home sales (or **turnover**). Since home sales result from people moving out of their homes (the academic term for this is **residential mobility**), understanding what makes households move is very useful prelude to developing a model for turnover.

5.6.1 Why do people move?

Across homeowners and renters, there are generally three main reasons for moving that hold across all countries:¹⁶

- **Housing-related reasons:** Desire to change tenure status (from renting to owning or vice-versa), wanting a better house/apartment, seeking a better neighborhood (less crime, better schools etc.).
- **Family-related reasons:** Changes in marital status, desire to establish your own household.
- **Job-related reasons:** Starting a new job, transfer of an existing job to a new location, looking for work, easier commuting, retirement.

In general, this data is consistent with academic theory which views mobility as a mechanism that households use to adjust their consumption of housing services. Some of the reasons for this adjustment in consumption can be a change in the household's financial circumstances or an increase in the size of the family.

However, it should be kept in mind that borrowers cannot always adjust their housing consumption “instantaneously” because of the transaction costs associated with moving. These costs include both search costs (the hassle involved in trying to find a suitable new dwelling) and broker costs (fees paid to the real estate agent for selling the home). Furthermore, interest rates and house prices will also play an important role in the borrower's ability to move particularly if (a) The market rate for a new mortgage is higher than the rate on the borrower's current mortgage; or (b) The borrower does not have positive equity in the home which will make it difficult to repay the underlying mortgage by selling the house. Thus, both a below-market mortgage rate and negative equity restrict the mobility of the homeowner: this is the so-called **lock-in effect**.

So far, we have discussed the determinants of mobility at the local household level, but there are also “macro” factors that can play a significant role. One such factor is the housing policy posture of the government. In general, governments tend to promote homeownership through giving it favorable tax-treatment versus renting. Changes in tax law (such as the taxation of realized capital gains on home sales) can act as a deterrent or spur to housing mobility.

¹⁶[Sanchez11]

Another very significant factor is the availability of mortgage credit since most houses need to be financed. Tighter lending standards will tend to depress mobility while looser lending standards reduce the frictional cost of moving.

5.6.2 Homeowner Mobility Rates in the U.S.

Now that we have a sense for why people move, it is also helpful to get some perspective for the average annual mobility rate in the U.S, and how much it varies from year to year. One important source of data on homeowner mobility rates is compiled by the U.S. Census Bureau and is presented in Figure 5.4. The data suggests that the average annual mobility rate for homeowners has ranged between 5% to 9% over the past 15 years and has been declining gradually over the years. More comprehensive data sets that go further back in history also lead to the general conclusion that Americans were considerably more mobile in the past. The reasons adduced for this phenomenon include:¹⁷

1. The early history of the U.S. was characterized by a large number of “great migrations” (from settled states to the frontier, farms to the cities) which have now ended.
2. Over time, *forced* moves have become less necessary. An expanding social net, greater earnings, and home insurance have made it easier to withstand idiosyncratic shocks to household wealth.
3. Daily travel is easier than before and has made it much more practical to live several miles from a job. Thus, a job change does not necessarily have to result in a relocation.

Multi-generational demographic and social trends that effect housing mobility over long periods of time are very difficult to forecast and are usually left unmodeled. This decision is somewhat mitigated by the fact that most of the year-on-year changes in the overall mobility rate are relatively small and only make their presence felt when accumulated over a long period of time.¹⁸

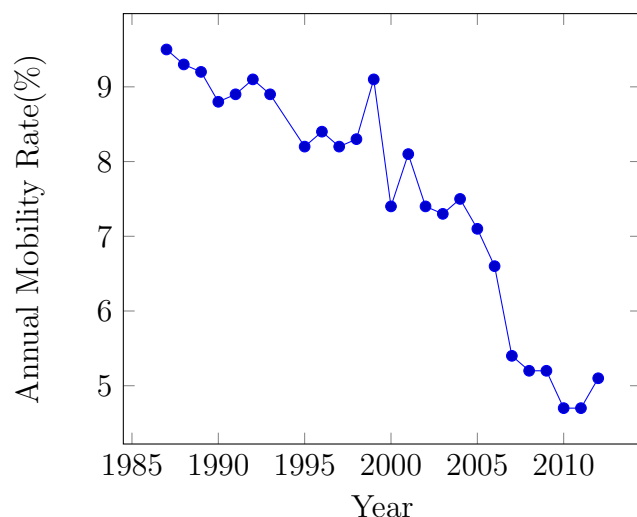
5.6.3 Explanatory Factors for Modeling Mobility

The above discussion gives us some sense for why people move and this in turn gives us a potential list of viable candidates for modeling turnover-related prepayments. Unfortunately, as remarked earlier, the data currently available on MBS do not include demographic variables such as the age of the homeowner or the size of their family. Thus, in many cases, we have to use proxy measures to attempt to capture some of the variables that effect mobility.

¹⁷Further details can be found in [Fischer02]. More recent declines (post 2007) can likely be attributed to a tightening of underwriting standards and a sharp fall in house prices.

¹⁸Note also that the weighted-average life of a mortgage pool is typically less than 7 years.

Figure 5.4: Annual Residential Mobility for Homeowners



Source: U.S. Census Bureau

Loan Characteristics

Some of the characteristics of the loan can provide important information about the borrower's propensity to move:

- **Mortgage Rate.** The borrower's mortgage rate will be important since, as we have already pointed out, borrowers with below-market mortgages face a disincentive to move.
- **Number of Points Paid at Origination.** Borrowers who pay relatively greater origination fees in order to reduce their mortgage rates probably intend to stay longer in their homes.¹⁹
- **Spread between FRM and ARM Rates at Origination (or ARM share at origination.)** In a flat yield curve environment, borrowers who would prefer to hold an ARM may end up choosing a fixed-rate mortgage. Since ARM borrowers are more mobile,²⁰ all else being equal, turnover-related prepayments on these fixed-rate pools will be higher than for the same pools originated in a steep yield curve environment.²¹

¹⁹Recall that the origination fee associated with every mortgage is typically quoted in **points**, with 1 point equal to 1% of the initial principal balance of the loan.

²⁰As per our discussion of Borrower Mortgage Choice in Chapter 3, mortgage holders will attempt to match their **tenure horizon** (how long they plan to be in the house) with the loan that they select. See [Campbell03] for more on this topic.

²¹To make this statement a little more rigorous, we also have to account for the fact that in the past borrowers would often finance home purchases with an ARM because they needed the lower mortgage rate to satisfy the qualification ratios. However, purchase mortgage borrowers can have different expected holding periods than refinancers! Thus, the ARM share or the mortgage yield curve slope needs to be measured

Housing Market and Economic Conditions

Labor market conditions are important in determining mobility but are rarely used in prepayment models since these conditions are hard to forecast over the entire term of the mortgage. Home prices are used in prepayment models although one could argue that they are equally challenging to forecast. Basically, home prices affect mobility by determining the value of the home and the amount of equity the borrower has. Increasing home prices result in increased home equity which can be used as a source for the downpayment on the larger home that a household desires. Also, rapid increases in home prices can also encourage turnover through speculative activity (**property flipping**). On the other hand, a decrease in home prices erodes equity and constraints the ability of the household to trade up. It may also result in an inability to sell the house.

A cumulative rate of home price changes over some suitable period (say two years) can also serve as a valuable index of a key determinant of mobility rates: the availability of mortgage credit. Most houses will need to be financed and the availability of financing will thus clearly have an impact on mobility rates. A sustained increase in house prices is usually associated with looser lending standards, while sharp decreases in home values lead to originators being much more conservative in terms of their underwriting.

Time and Seasonal Indicators

The age of the household is an important index for one of the primary drivers of mobility: housing dissatisfaction. Age is correlated with wealth and the size of the family, hence the link between age and the desire to own better housing. Given the absence of demographic variables in our MBS data set, one often uses the age of the mortgage loan as a proxy for the length of time-in-residence. Of course, the age of the mortgage is also correlated with the age of the borrower. Finally, the age of the mortgage is also linked to the household's willingness to move. As the mortgage "seasons," the typical household will be more receptive to a move as they recoup the costs they incurred by moving into their current house. Seasonal adjustments are also important since moves are more likely in certain times of the year compared to others.

5.6.4 Patterns in Mobility-related Prepayments

The previous paragraphs provided us with a set of explanatory variables that are linked to the likelihood of mobility. Our goal now is to explore the variation between mobility rates and these variables. All of our analysis is based on prepayment rates observed on 30-year Fannie Mae and Freddie Mac pools.

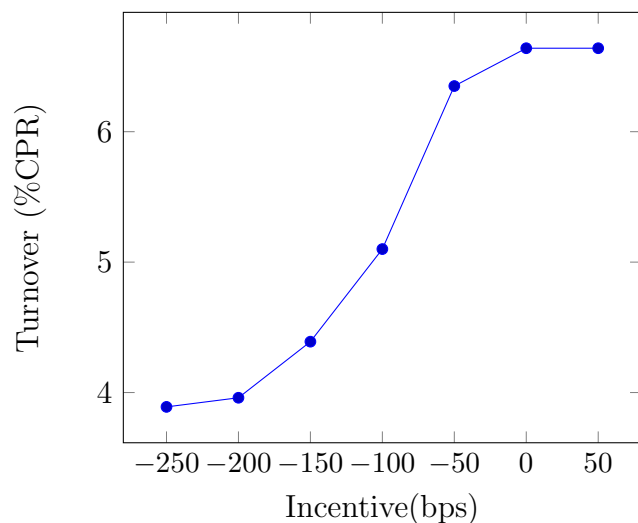
relative to the share of purchase mortgages in origination volumes. Further complicating matters is the fact that underwriting standards for ARMs have evolved sharply over the past few years and borrowers are now typically qualified on the fully-indexed rate rather than the initial "teaser" rate.

The Lock-in Effect

The data presented in Figure 5.5 illustrates turnover-related prepayments on 30-year MBS pools as a function of the MBS pool's "incentive." The incentive is calculated here as the difference between the pool's weighted-average coupon (WAC) and prevailing mortgage rates – more negative values of this metric correspond to situations where prevailing rates are higher than the borrower's rate. The relationship between turnover rates and incentive depicted in the figure is intuitively reasonable. First, since the fundamental drivers of mobility are housing dissatisfaction and job changes, interest rates don't really affect a borrower's ability to move when the pool is "in-the-money." The figure shows that the long-run average of the mobility rate of the 30-year mortgage holder stabilizes between 6%-7% as the incentive gets more positive.

On the other hand, as prevailing mortgage rates rise above the pool's WAC, the homeowner finds it increasingly difficult to move. Mobility rates steadily decrease as the amount of "disincentive" increases up to a certain juncture beyond which further increases in mortgage rates do not have an effect on borrower moves. The reason for this is probably that a certain amount of "forced" job-related moves occur regardless of interest rates. Note that even if mortgage rates are very high, the borrower does have the option of moving to rental housing in this situation.

Figure 5.5: The Lock-in Effect



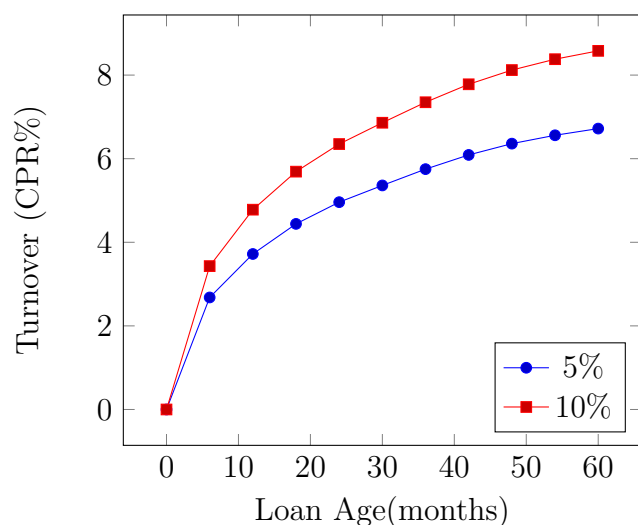
Source: Cello

A similar relationship holds between the turnover rate and the LTV ratio of the homeowner – as LTVs increase above 100% the homeowner gets progressively more locked-in since they cannot sell the house without incurring an out-of-pocket cost that increases with the LTV.

The Seasoning Ramp

The seasoning ramp for MBS captures the fact that most households get progressively more “dissatisfied” with their housing as their income trends up and the size of their family increases. In addition, the sunk costs associated with the initial move become less of a factor with time. Figure 5.6 illustrates a number of seasoning ramps for 30-year mortgage pools indexed by the average 2-year change in home prices experienced by the borrower over the observation period. Notice the distinct dependence of the steepness of the seasoning ramp on the health of the housing market. Homeowners experiencing an average 2-year rate of appreciation of 10% relocate at a rate that is almost 2% CPR greater than those experiencing a 2-year appreciation rate of 5%.

Figure 5.6: The Seasoning Ramp for Turnover



Source: Cello

Seasonal Factors

Home sales exhibit a pronounced seasonal pattern with the seasonal highs in sales activity occurring in the summer and the lows in winter. The driving factors behind these seasonal patterns include the school year calendar and the weather. Figure 5.1 summarizes the seasonal factors calibrated from a study of turnover-related prepayments. It indicates that the prime months to move occur in the middle of summer (May through August) with activity gradually tapering down into the winter months. January and February are the worst months to move.

Table 5.1: Seasonal Factors for Housing Turnover

Month	Seasonal Adjustment	Pct. Change from Previous Month
January	0.70	-19%
February	0.72	3%
March	0.97	35%
April	0.97	0%
May	1.16	19%
June	1.28	11%
July	1.22	-5%
August	1.21	0%
September	1.01	-17%
October	1.00	-1%
November	0.90	-10%
December	0.86	-4%

Source: Cello

5.7 The Components of Prepayments: Refinancing

Refinancings constitute the most important component of prepayments and are the most challenging to predict because refinancing rates can go from 0% to nearly 70%-80% on a pool of mortgages depending upon where mortgage rates are.

5.7.1 Why do People Refinance?

Homeowners have various reasons for refinancing including obtaining a lower interest rate, changing other terms of their mortgage (converting from an adjustable-rate to a fixed-rate mortgage or shortening/lengthening the repayment period), and liquefying equity. In addition, borrowers with poor credit who pay higher rates (since such borrowers have a higher propensity to default) can often qualify for a lower rate after making their monthly payments in a timely manner over a prolonged period of time. This phenomenon is known as **credit curing**. Credit curing is still on the rare side for borrowers in Fannie Mae and Freddie Mac pools since most of these homeowners have pristine credit histories. Most residential mortgages do not have a penalty for refinancing although a typical refinance still remains fairly expensive – as much as 2-3% of the balance of the loan.²²

A study by the Federal Reserve²³ surveyed homeowner refinancing activity over 2001 and early 2002 and provides a breakdown of the relative importance of the different reasons for refinancing. While the results are somewhat specific to this time period, they still provide

²²See Table ?? for a detailed itemization of the transaction costs associated with a typical refinancing.

²³[Canner02]

a useful illustration of the principal underlying rationales for refinancing. As one would expect, the overwhelming motivating factor behind refinancing is obtaining a lower rate and thus reducing the homeowner's monthly debt service burden. 96% of all homeowners who refinanced over the survey period obtained a lower rate. A small number (approximately 10%) of refinancing homeowners shifted from fixed-rate to adjustable-rate mortgages (ARMs) when they refinanced. Since ARMs have lower rates in upward sloping yield curve environment, they may be attractive to refinancers who plan to move in the near future or who expect future interest rates to remain stable or decline further.

Refinancers can also lower their monthly payment by lengthening the term to maturity on their debt. In the survey, after refinancing, 74% of homeowners had mortgages with a longer maturity because they chose 30-year mortgages and extended the term of their mortgage by about 6 years on average. 17% of homeowners chose 15-year mortgages and shortened their maturity by an average of 7.5 years, while the remainder kept the maturity roughly the same.

A significant number (45%) of homeowners who refinanced over the survey period used the opportunity to extract some of their home equity. This cash is used for various purposes including debt repayment, home improvements and consumer expenditures. Only 27% of homeowners who liquefied equity had a lower payment as a result of taking on more debt.

While the core drivers behind refinancing decisions remain the same in today's housing markets, the prevalence of cash-out refinancings has sharply diminished after the housing crisis of 2007-2009 given the substantial decline in household levels of home equity and underwriting standards that are significantly more stringent.

5.7.2 Explanatory Factors for Modeling Refinancing

There are a large number of borrower and loan characteristics that provide information about a loan's propensity to refinance. The list of factors below, while long, is by no means exhaustive, thus gesturing towards the complexity of modeling this activity.

Refinancing Incentive, Lag, and Costs

The refinancing incentive for a given loan (or pool of loans) can be defined as the difference between the mortgage rate of the loan (or WAC of the pool) and prevailing mortgage rates. Prevailing mortgage rates are calculated using the weekly Freddie Mac Survey rate (often referred to as PMMS Primary Mortgage Market Survey Rate), a widely used reference for the industry which applies to conventional conforming loans with LTVs of 80% or less. The refinancing incentive needs to be adjusted for **loan-level pricing adjustments** and charges related to mortgage insurance premiums, if any. The incentive also needs to be **lagged**: the current month's refinancing driven prepayment rates typically reflect rates from 4 weeks to 6 weeks ago since it takes about that much time to process a refinancing application.

Table 5.2 presents estimates of the typical fees charged by different states for a standard

refinancing.²⁴ Note that in addition to the standard fees, every borrower is expected by the lender to prepay up to two months of their insurance and tax premiums to ensure that the lender has enough money to make payments when these are due. There are some compensating factors: (1) Lenders sometimes waive some fees, particularly loan application fees, and (2) Borrowers often do not pay many of the costs (excluding prepaid escrow expenses) out of pocket. Instead, they either roll them into the new loan balance or, in the case of a no-point/no-cost mortgage, pay them over time through a higher note rate.

How do borrowers decide when to refinance? A standard heuristic calls for a refinance as soon as the incentive is at least 50bps (0.50%).²⁵ A somewhat more sophisticated decision process is the so-called **breakeven** method. Essentially, divide the cost of refinancing by the monthly payment savings and compare this breakeven period with the expected tenure in the house. Refinance if the expected tenure is greater than the breakeven period. Table 5.3 provides an example of this calculation in action for a \$200,000, 30-year fixed-rate mortgage with a current rate of 6% and assuming prevailing mortgage rates of 5%. We assume that the fees for the new loan are \$2,500, paid in cash at closing.

The calculation in Table 5.3 makes several simplifications which are probably ignored by the average borrower but can make a significant difference to the calculation. First, it does not take into account the time value of money; a more accurate calculation would present value the monthly savings (using the new mortgage rate to discount cash flows) over the borrower's tenure horizon and compare this value to the transaction costs of refinancing. Second, tax considerations play an important role in the mortgage refinance decision and are not fully incorporated in the analysis. In particular, the tax treatment of the loan fees and closing costs listed in Line 6 of the table is not straightforward. To the extent that no discount points are involved in these payments and they are paid upfront, the analysis stands. However, if the upfront fees involved include discount points then a small adjustment is needed since discount points in this case can be deducted over the term of the loan. Finally, if all the fees are financed in the balance of the loan then they are deductible as interest. Table 5.3 does incorporate the crucial consideration that the comparison of monthly payments should be done on an after-tax basis since mortgage interest is tax-deductible.²⁶

There is a deeper problem with both of these approaches in that they don't consider the value of the prepayment option that the borrower receives when they own a mortgage. In particular, these approaches ignore the trade-off between paying an above-market rate right now and the "time value" of the call option – it may be better to refinance in the future as opposed to now.²⁷ The **optimal refinancing** rule would say that one should refinance

²⁴The table also demonstrates the dependence of refinancing rates on **geography** because of the significant state-level variation in costs.

²⁵This is an evolution from an earlier rule which suggested a trigger of 2%; the evolution has been driven by the lower costs associated with refinancing a mortgage over time.

²⁶See the Tax Laws section in Chapter 4 for details.

²⁷"Serial" refinancing every time rates reach a trigger point doesn't work because, as we can see from Table 5.2, there are non-trivial costs associated with refinancing. Thus, we may not be able to recoup the costs associated with these multiple refinancings without a long enough holding period.

Table 5.2: Refinance Closing Costs by State

Loan Characteristics	CA	NY	TX	FL
Loan Type	30-Year Fixed			
Purchase Price	250,000			
Loan Amount	200,000			
Points & Lender Fees				
Lender Origination Fee	1,365	1,190	1,235	1,230
Third-party fees				
Flood Monitoring	17	17	17	17
Credit Report Fee	15	15	15	15
Title Closing Fee	675	1,190	530	238
Title Insurance ^a	425	356	882	323
Mortgage State Tax ^b	0	1,650	0	1,120
Mortgage Recording Fee ^c	175	650	180	260
Appraisal Fee	460	455	465	440
Tax Service Fee	84	89	89	79
Total third-party fees	1,851	4,422	2,178	2,492
Total Closing Costs	3,216	5,612	3,413	3,722
Additional Cash for Closing				
Prepaid Items				
Prepaid Int. (est 15 days)	318	318	318	318
Homeowner’s Ins. (1yr)	516	576	1,344	1,872
Escrow Account Funds				
Homeowner’s Ins. (2mos)	86	96	224	312
Property Taxes (2mos)	590	460	986	690
Total Additional Cash	1,510	1,450	2,872	3,192

^aTitle insurance estimates provided by Fidelity National Title for a non-cashout refinance executed 2 years after the original loan was taken out. It includes the Lender's portion of the Title Insurance and the Title Endorsement.

^bIn NY State, this amount can be lower for borrowers who take out a CEMA loan. For borrowers in counties with higher taxes (such as NYC) the mortgage taxes will be much higher, and applying for a CEMA loan would be most cost-effective.

^cSame as prior footnote.

Source: Morgan Stanley Research, Bank of America

Table 5.3: The Breakeven Period for a Refinancing

1. The current monthly mortgage payment	\$1,199
2. Subtract the new monthly payments	-\$1,073
3. This equals monthly savings	\$126
4. Subtract tax rate from 1.00	0.72
5. After-tax monthly savings	$\$126 \times 0.72$
6. New loan fees and closing costs	\$2,500
7. Total costs divided by monthly after-tax savings	$\$2,500 / 91$
8. Breakeven period	27.5 months

Source: Adapted from <http://www.federalreserve.gov/pubs/refinancings/>

when:²⁸

$$PV(\text{after-tax monthly savings}) = \text{Refinancing costs} + \text{Loss in option value} \quad (5.5)$$

There is a loss in the value of the prepayment option we receive when we exchange the old mortgage for a new lower one because the old call option is in-the-money and thus we could potentially derive further value from waiting. The new option would require more substantial decreases in rates to be exercised since it being issued either at- or out-of-the-money.

In general, borrowers rely on some variant of the breakeven rule in making the refinance decision but, since their tenure horizons are not directly observable, most prepayment models calculate a simple incentive measure to assess how attractive it is for borrowers to refinance at prevailing levels of mortgage rates.

Exposure to Refinancing Opportunities

The phenomenon of burnout (as mentioned in the discussion accompanying Figure) results from “borrower heterogeneity.” This phrase is shorthand for saying that mortgagors can exhibit significant diversity in their prepayment behavior. Most academic and Wall Street research indicates that these differences stem from one or more of the following factors:²⁹

1. Differences in the explicit transaction costs paid by mortgagors. For example, regional differences in the mortgage market may mean that two borrowers refinancing the same mortgage may face different levels of transaction costs.
2. Differences in “non-observable” transaction costs such as the search costs and credit barriers faced by different borrowers.

²⁸For further analysis, see [Kalotay08].

²⁹A good review of the academic literature on this topic can be found in [Hall00].

3. Differences in borrower level collateral and liquidity characteristics such as the amount of home equity available or the availability of cash to cover refinancing costs.
4. Differences in the personal attributes of the mortgagors such as knowledge, temperament, decisiveness or risk preference.
5. Differences in the expected holding period for the house.

Many of these differences are unobservable at the pool level, but nevertheless manifest themselves through the phenomenon of burnout in mortgage pools. To capture this heterogeneity, consider the following simple model which partitions the pool of borrowers into three groups – “slow”, “medium” and “fast” based on how efficiently they refinance their mortgages. We will explain later why this simple idea captures the dynamics of burnout when we discuss patterns of refinancing behavior.

Past History of Mortgage Rates

Empirical data shows that homeowner prepayment rates are sensitive to the past history of mortgage rates (see Figure 5.7.2). Specifically, homeowner refinancing rates appear to ignite when mortgage rates reach multi-year lows. This surge of refinancing activity results in a refinance wave (“refi wave” for short). This effect has been variously described as the **Media Effect**, the **Publicity Effect**, and the **New Rate Low Effect**. The terminology used to describe this phenomenon provides another likely cause for this effect: multi-year lows in mortgage rates typically spark extensive media coverage and proactive solicitation of refinancings by servicers who wish to protect their servicing portfolios thus resulting in lower transaction costs for refinancing. To model the publicity effect, we need to develop an index that provides a proxy for the strength of this effect. There are a couple of potential choices. For any given month (or week), one could calculate the number of months (or weeks) since rates have been this low. Another option would be to calculate the “in-the-moneyness” of the current mortgage universe. In general, the greater the number (or dollar amount) of refinaneable mortgages, the higher the value of the publicity effect.

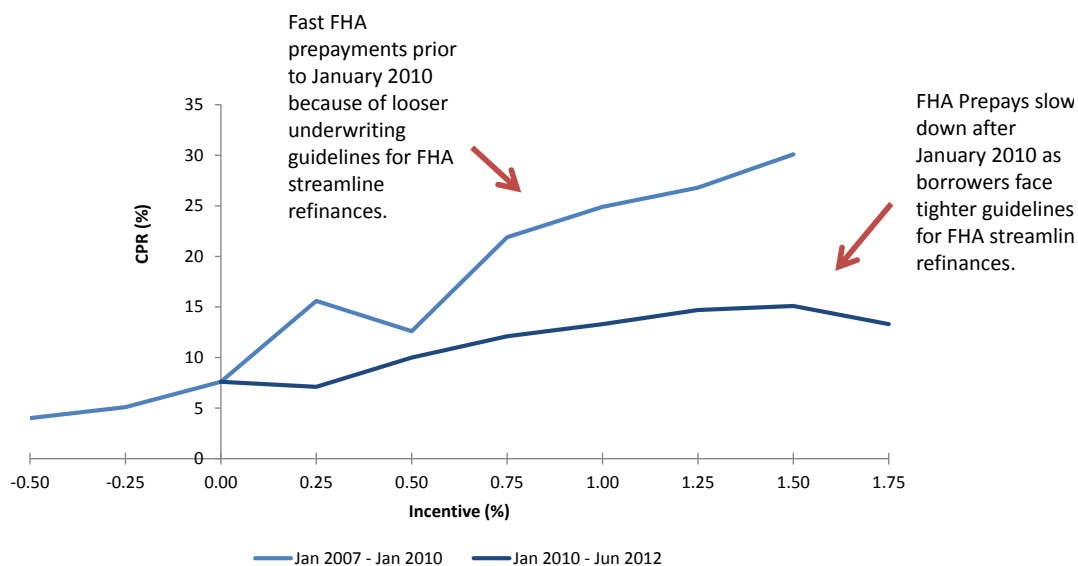
Structure of the Mortgage Origination Industry

Changes in the structure of the industry can have a profound impact on refinancing behavior (changes in refinancing because of regulatory changes are discussed below). For example, the mortgage broker and correspondent origination channels contracted sharply in the 2007-2009 Housing Crisis as mortgage volumes plummeted and the underwriting practices of these entities received significant scrutiny. Since these channels are usually associated with higher prepayment rates (further discussed below), the net impact was to significantly decrease prepayment rates for a given incentive level relative to the past.

Regulatory Policy

Changes in government housing policy can have a significant impact on prepayment rates. For example, Figure 5.7 graphs prepayment rates on FHA loans in two regimes. Regime 1 spans January 2007 to January 2010 and reflects a period of relatively stable underwriting criteria. In January 2010, the FHA substantially tightened underwriting guidelines: (1) Borrowers need to demonstrate “net tangible benefit” before refinancing; (2) Delinquent borrowers were disqualified from refinancing; and (3) Closing costs could no longer be added to the loan balance. The net impact of these changes can be seen in the Regime 2 line which tracks prepayment rates on FHA loans from January 2010 to June 2012. Clearly, there’s a significant change.

Figure 5.7: Underwriting Guidelines Changes Can Meaningfully Impact Refinancing Rates



Source: Cello

Loan Sizes

Many of the refinancing costs listed in Table 5.2 are independent of the size of the loan. Homeowners with larger loan sizes will therefore experience proportionately lower costs (relative to their loan balances) when refinancing. As a result, all else being equal, larger loans are expected to prepay faster than small loans.

LTV Ratios

Borrowers with 20% or more equity in their homes will typically have no trouble refinancing. Borrowers with higher LTV ratios generally receive much more scrutiny and may face challenges in refinancing unless there are other compensating factors.

Credit Scores

While credit scores ostensibly provide information about a borrower's willingness and ability to repay a mortgage, they also signal the financial sophistication of the borrower. All else being equal, borrowers with higher credit scores tend to be more efficient in exercising their refinance options.

Origination Channel

Loan correspondents and brokers receive origination fees and points from borrowers and earn their living through transaction volume. As such, they have economic incentive to "churn" the customer.³⁰ Unsurprisingly, loans that originate from one of these channels tend to refinance at higher rates when compared to the retail channel.

Servicer

Certain servicers are noticeably better than others in targeting borrowers in their portfolio to solicit a refinance.

Occupancy Status

Borrowers who take out a mortgage on a single-family property but who do not intend to reside in it face higher transaction costs because they are judged to pose more of a credit risk. Consequently, they will usually refinance more slowly than borrowers who are owner-occupants.

³⁰To some extent, this is mitigated by the fact that contracts between major lenders and their brokers and loan correspondents typically include non-solicitation clauses.

Seasonals

Since homeowners can prepay on any day of the month, the number of business days in the month will affect the total dollar volume of prepayments that is collected by servicers. All else being equal, months that consistently have a fewer number of business days (February) will correspond to lower prepayment rates. The holiday calendar also has an effect on refinancing rates since most homeowners will not be motivated to refinance over the last few weeks of December and the first week of January.³¹

5.7.3 Patterns in Refinancing-related Prepayments

Refinancing S-Curve

It is useful to begin our analysis of refinancing by looking at the simplest and most important aspect of the empirical data: the refinancing rate as a function of incentive. The results are presented in Figure 5.8. The distinctive shape of the refinancing response function is similar to that of an elongated “S”, hence the name **S-curve**. Notice that even when the mortgage is at- or out-of-the-money, the refinancing response is still not zero. This can be attributed to the prevalence of cash-out refinancing through which homeowners frequently raise their mortgage rate while using their mortgage as a debt consolidation vehicle. As the incentive increases, we notice an increase in refinancing activity with the slope of the S-curve steepest near the at-the-money zone and then gradually flattening out as incentive increases. The intuition is that at some threshold level, a small increase in incentive takes a number of potential refinancers off the fence. As the overall incentive increases, the marginal increase in refinancing activity because of an increase in incentive starts to diminish.

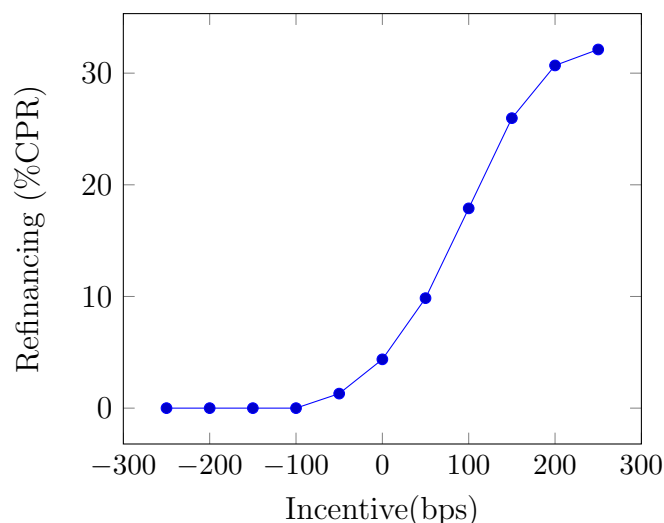
Burnout

The presence of borrower heterogeneity means that the refinancing propensities of a pool of mortgage borrowers range from “inefficient” to “ruthless.” Now, in the presence of refinancing incentives, the more “ruthless” refinancers will leave the pool earliest and leave behind a collection of relatively less efficient refinancers. Over time, this will cause aggregate prepayment speeds on the pool to gradually decay. Also, if rates rise and then fall again to the initial level that stimulated refinancings, the same pool will refinance at a lower rate than before since it has a smaller proportion of “ruthless” refinancers.

A simple model that captures these dynamics is presented in Figure 5.9. We assume mortgagors fall into three distinct groups when classified by their refinancing propensity: “Slow”, “Medium”, and “Fast.” The observed refinancing rate for the pool will be the average of the refinancing rates for the different borrowers, weighted by the proportion of the pool in that particular category for that particular month. Note that the differing refinancing

³¹This also applies to the mortgage originators who would be processing these applications.

Figure 5.8: The Refinancing S-Curve



Source: Cello

rates of the three classes dramatically change the composition of the pool over a period of two years in our example. In year one, “Slow” borrowers only make up 20% of the pool, but by year three they have more than doubled their representation of 46%. Also, for the same incentive, the pool prepayment rate has fallen from 58% CPR to 45% CPR because of the relatively quick exodus of “fast” refinancers.

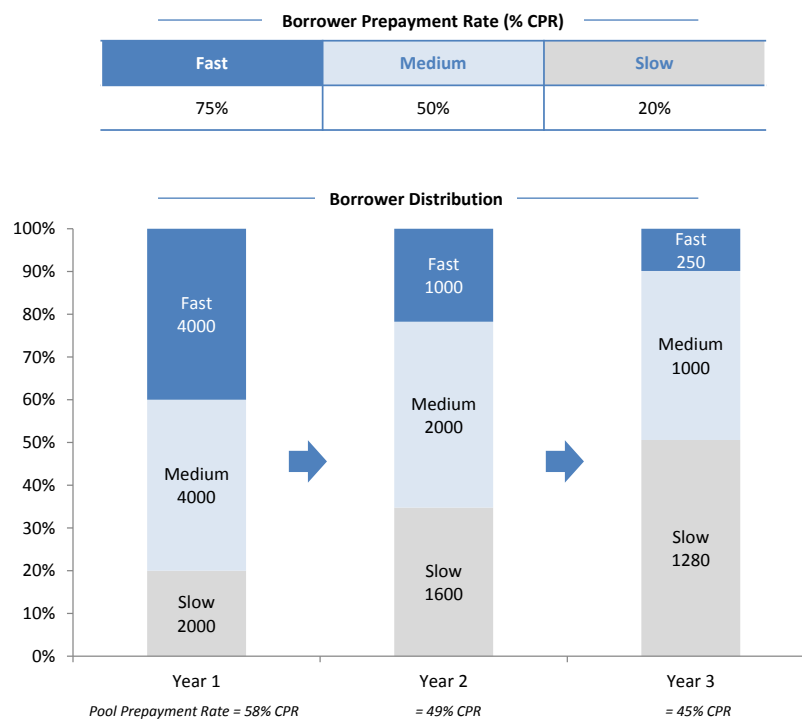
Media Effect

We start by creating an index that measures the overall refinanceability of the mortgage universe. In practice, we can implement a Publicity Effect in the model by defining an efficient refinancing S-curve (corresponding to a maximum level of the Media Effect) and a base refinancing S-curve (corresponding to a negligible value of the Media Effect). Figure 5.10 illustrates what these S-curves would look like in practice. Depending on the prevailing level of the refinanceability index, the S-curve is a blended average of the base curve and the efficient curve.

5.8 The Components of Prepayment: Curtailment and Full Payoff

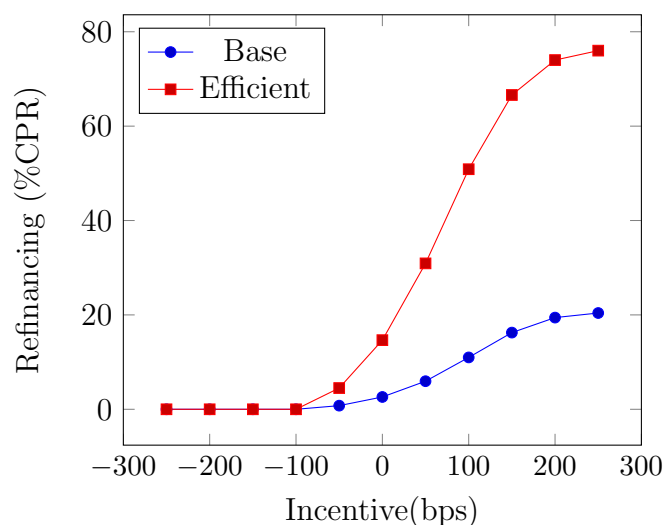
On the face of it, curtailment appears to be a sub-optimal strategy for homeowners. After all, if a mortgage is in-the-money, the homeowner should prepay the outstanding balance of the mortgage in full as opposed to in part. The weakness in this argument lies in the

Figure 5.9: Modeling the Burnout Effect



Source: Cello

Figure 5.10: The Refinance S-Curve in Different Media Effect Environments



Source: Cello

fact that the transaction and hassle costs associated with refinancing a mortgage can be significant. This is especially true in the situation where the mortgage carries a penalty for refinancing. If a borrower cannot refinance because of the costs involved, curtailment is the optimal strategy. Alternatively, some households may simply want to reduce the amount of their indebtedness by paying off as much of their mortgage debt as they can. Also, once the mortgage balance is below some threshold level, a homeowner may just decide to simply payoff the remaining amount (**full payoff**). This will typically only happen in the last few years before the maturity of the mortgage.

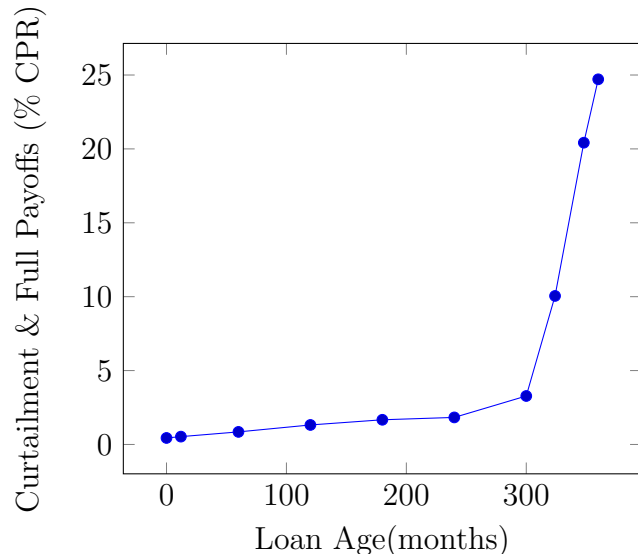
5.8.1 Patterns in Curtailment- and Full Payoff-related Prepayments

The empirical data on curtailment³² and full payoff rates are consistent with the discussion above. In general, these rates start out pretty low but can ramp up sharply towards the end of the life of the mortgage presumably because homeowners are now older and wish to be debt-free. Other studies suggest that curtailment rates on mortgages that carry a prepayment penalty or are second mortgages should be fairly high.³³ Figure 5.11 presents a curtailment and full payoff curve that corresponds to observed prepayments on 30-year Conventional MBS pools. The figure suggests that curtailment rates gradually increase from 0% to 1.8% over the first 20 years of the mortgage. We see a significant ramp up in prepayment rates over the last few years of the mortgage's maturity through a combination of curtailment and full payoffs.

³²[Fu97].

³³[Hayre91].

Figure 5.11: Curtailment and Full Payoffs on Conventional 30-year Mortgages



Source: Cello

5.9 The Components of Prepayment: Default

The homeowner's right to default can be viewed as them owning a put option on the house. In down to earth terms, a necessary condition for the borrower to default is if the value of the house is less than the market value of the mortgage.³⁴ The condition is not sufficient for default because the borrower has to weigh the costs associated with moving, the cost of a damaged credit record, and the expected cost of deficiency judgments.³⁵ In addition to negative equity, borrowers typically have to be subject to some type of **trigger event** to default – an income shock resulting from sustained unemployment, a divorce, an unexpected increase in health-related costs, or some other form of liability.

³⁴At least in principle, the market value of the mortgage is an important component of the homeowner's decision to default. A discount mortgage is more valuable to the homeowner than a premium one because, in the former case, they would potentially be forced to increase their mortgage rate when they moved. However, in practice the market value is unlikely to play a strong role in the decision making process since the homeowner is very unlikely to qualify for a new mortgage right after they default because of the damage to their credit rating.

³⁵In a **deficiency judgment**, creditors are allowed to collect an amount equal to the lender's foreclosure losses against the borrower's other assets. These judgments are rarely pursued but may be used as leverage against borrowers. Most states allow deficiency judgments but they are prohibited in 9 states (including California).

5.9.1 Explanatory Variables for Modeling Defaults

Credit Scores

A credit score (or FICO score) measures the willingness and ability of the borrower to pay their mortgage debt. Consumer credit scoring is now widely practiced in the U.S. and as a result the distribution of (initial) credit scores is available for most conventional MBS pools. Currently, these scores are not periodically refreshed which decreases their predictive power as the pool seasons.

Current LTV

The borrower's equity position is a very important determinant of default – a borrower will typically have to experience a negative equity position before considering defaulting. It should be kept in mind that in pool-level models we can only compute the equity position of the *average* borrower.³⁶ Hence, it is possible to see defaults on Agency pools with average positive equity.

Loan Age

Default rates will show some dependence on loan age since lenders will try not to underwrite borrowers who could potentially default immediately.

Delinquency Pipeline

Clearly a borrower who is currently delinquent has a greater chance of defaulting than a borrower who is making timely payments on their mortgage. Thus, any pool-level information on the delinquency distribution of the underlying mortgagors will be an important leading indicator of defaults.

5.9.2 Patterns in Default-related Prepayments

Default rates on mortgages show a distinct dependence on the age of the mortgage. The general pattern can be described as follows. Default rates start out low since lenders will not underwrite borrowers who they suspect will default immediately. Also, because mortgages with initial LTVs of 100% or greater are rarely originated in the U.S., most borrowers will have positive equity at origination. We start to see default rates ramp up gradually within the first two to three years after origination since it is over this period that borrowers are most vulnerable to losing all their equity if home values decline. After five or more years after

³⁶Unless we resort to using quartile-level data that is available on pools.

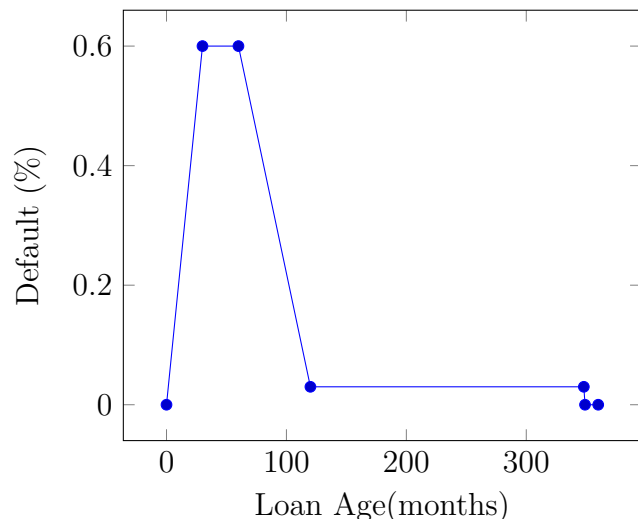
origination, default rates start to taper off since increases in home prices and an increasing amortization effect decrease the likelihood of negative equity.

The SDA Model for Defaults

The **SDA (Standard Default Assumption)** model is a benchmark curve for default analysis developed by the Bond Market Association. The curve is given by the following prescription:

- Beginning with a CPR of 0.02%, default rates increase by 0.02% CPR per month until month 30, when the peak value of 0.60% is reached.
- Defaults remain at that value for the next 30 months.
- Starting in month 61, defaults decrease by 0.0095% per month until month 120, when the tail value of 0.03% is reached. Defaults remain constant at this level for the remaining life of the security.
- By convention, the last 12 default rates are set to zero, assuming it takes twelve months to liquidate a property.

Figure 5.12: The SDA Curve



Source: Cello

Strengths and Weaknesses of the SDA Model

The shape of the SDA curve more or less captures the age-dependent nature of default rates described above. The curve (or some multiple thereof) tracks empirical data reasonably well

because age is strongly correlated with equity growth, which is one of the primary determinants of the decision to default. The weaknesses of the model are its lack of dependence on other macroeconomic (for ex., strength of the labor market) and microeconomic (the borrower's current LTV, credit score, and debt-to-income ratios) variables that are likely to have a strong impact on default rates. A more plausible model needs to take these factors into account.

The GSE Repurchase Option

One important nuance that must be considered in modeling default-related prepayments on GSE-backed pools is that the GSEs have the right to purchase mortgage (only at par value) from an MBS when they are significantly past due (typically when they are 120 days delinquent). For the MBS holder, these repurchases correspond to full prepayments. Thus, while the default-related timeline for a mortgage loan from the onset of delinquency through liquidation of the property can easily extend beyond a year, from the GSE MBS investor perspective, these payoffs are seen far earlier in the default process when the GSE repurchases the delinquent loan out of the pool.

5.10 Prepayment Models and Model Building

At a very high level of abstraction, a prepayment model simply tries to successfully predict all the different patterns of prepayment behavior we have seen in our discussion above. It is essentially an economic model that attempts to predict the behavior of homeowners by “fitting” the prepayment data on thousands of mortgage pools against factors that influence prepayment speeds. In previous sections we have already surveyed many of the collateral characteristics that are used as inputs to the prepayment model.

Based on our discussion so far, it should be evident that it will be quite challenging to find a fit that is both optimal (has the smallest prediction error) and parsimonious (utilizes the minimum number of variables) because of the following issues:

- The large number of predictive variables and the possible interactions between them raise the issue of which of these predictors and their interactions we should include.
- The non-linear nature of the relationship between prepayment rates and many of these variables mean that many of the standard techniques in the statistical tool kit for variable selection do not apply.

These difficulties can generally be lumped into the broad problem of **model selection**: finding the “best” model from a collection of competing candidates. Although there is no statistical consensus about the optimal way to tackle this problem, a general strategy that consists of two planks and that we follow is as follows:

1. Develop a **comparison strategy** that allows us to compare two models and determine which one is better.
2. Establish a **search strategy** that traverses the space of all possible models.

For the purposes of model comparison, various criteria have been proposed and a very partial list is as follows:

1. **Adjusted R-Squared:** Penalize bigger models since the R-Squared (R^2) of a model increases as the number of included variables grows.³⁷
2. **Akaike Information Criterion (AIC).** This measure of the relative quality of a model trades-off goodness of fit with the complexity of the model (as measured by the number of variables that the model has). Given a set of competing models, the preferred model is the one with the lowest AIC value.
3. **Cross-validation:** Partition the data into two complementary data sets: the **training** data set (typically 70-80%) of the data sample and the **holdout** data set ($\approx 20 - 30\%$). The family of competing models are estimated on the training data set and their performance is evaluated on the out-of-sample data set (by calculating the **root mean squared error (RMSE)** of the model predictions, for example). To reduce variability, we can perform multiple rounds of cross-validation (by randomly selecting training data sets from the overall population) and average results over different iterations.

In addition to these comparison strategies, it is often very useful to look at graphical data that examine the fit of the model across various explanatory variables, produce independent estimates of key behaviors being modeled (for ex., the baseline mobility rate projected by the model over time can be compared to a suitably normalized existing home sales series and/or mobility estimates from the U.S. Census Bureau), and compare durations being produced by an MBS pricing model that utilizes the prepayment model to either empirical durations or the results produced by the models in widely distributed MBS Analytics systems (for ex., Citigroup's *YieldBook*).

For the purposes of the search strategy, in a perfect world we would search the entire space of models (corresponding to all possible combinations of variables) and choose the one with the lowest (highest) score (as determined by the comparison metric). In practice, this is not computationally feasible and one follows a **stepwise** strategy based on **forward selection**: start the fitting process with no variables (just the constant term) and try out variables one after the other to see if they improve the fit. The order in which the variables can be tried out can correspond to the strength of their respective correlations to the dependent variable.

³⁷The R^2 of a model is often interpreted as the proportion of the variability in a data set that is explained by the model. An R^2 of 1 indicates that the fitted model explains all the variability in the data. An R^2 of 0.4 can be interpreted as saying that 40% of the variation in the data set can be explained by the explanatory variables. The remaining variation can be attributed to unaccounted variables or inherent variability. Adjusted R^2 adjusts for the number of explanatory terms in a model relative to the number of data points.

The non-linearity of the prepayment model with respect to explanatory variables itself poses separate issues in terms of specifying the model through expressing a functional relationship between prepayment rates and variables. Initially, one can let the data speak for itself by fitting splines. However, the optimization algorithms will sometimes generate parameter values that are economically indefensible. It is useful to have priors for functional forms based on economic theory as this ensures that the optimizer will produce sensible values. The behavior of the model for limiting cases of the explanatory variable (extrapolation behavior) also needs to be reviewed carefully. Once again, looking at pricing results produced by the MBS pricing model for OASs, Durations, and Convexities provides an important sanity check.

5.11 The Uses and Limitations of Prepayment Models

The oft-quoted aphorism “All models are wrong, but some are useful”³⁸ is eminently applicable to the case of prepayment models. It is clear that a good prepayment model is a foundational block for MBS analysis. In addition to allowing us to estimate the risk/return characteristics of specific MBSs, the model allows us to compare these characteristics for MBSs with differing collateral characteristics and structures in a consistent and coherent way. In this sense, a prepayment model can be viewed as a multi-dimensional interpolator that facilitates “relative value” analysis. The most useful prepayment models are dynamic: the volatility of MBS prepayments is not well-approximated by constant CPR and PSA assumptions and analyses that rely on these assumptions understate the risk of the MBS.

Of course, since life is rarely perfect, there are some complications associated with using prepayment models. The drawbacks associated with relying on a model are generally categorized under **model risk** – The risk that the model you are using is “wrong.” There are many ways to be wrong. For starters, the model may not be correctly specified. For example, we could model the refinancing response of the homeowner versus their refinancing incentive using a linear function when, as Figure 5.8 demonstrates, the appropriate function is S-shaped. Next, the model may not be correctly estimated. For example, it may have been estimated using bad data. Also, the model may not have been implemented correctly – there may be a software error.

Even assuming that we can traverse all of the above hurdles, the very strategy used to “fit” the model – regressing prepayment rates versus a collection of explanatory variables using historical data gives rise to problems because the past is rarely a good predictor of the future. Stated differently, by construction, a prepayment model is “backward looking” while homeowner prepayment behavior in the future can be quite different than the past. Concretely, homeowners may become more or less efficient in terms of exercising their prepayment options in the future as the cost structure and lending standards of the mortgage industry continue to evolve. It is very difficult to predict the pace and timing of this evolution.

Without going into the details, we will conclude with an outline of some of the popular approaches that correct for the backward looking nature of prepayment models. First, if we

³⁸Attributed to the industrial statistician George Box.

assume that the market prices of securities accurately reflect future prepayment expectations, we can calibrate the parameters of our model to these prices. This gives us a **market-implied** prepayment model. Alternatively, we can use an adaptive model that updates its parameters every month based on the difference between actual and projected prepayment rates. Finally, we can compute the **prepayment duration** of a MBS by looking at how the prices of the security in question are impacted by a change in the model. Computing the prepayment duration of the MBS thus gives us a feeling for how vulnerable our risk/return analysis is to an inaccurate prepayment forecast.

Chapter 6

Secondary Mortgage Markets and Securitization

Before we delve into the U.S. MBS market, it is useful to take a step back and understand some general concepts about securitizations. As mentioned in Chapter 2, the modern form of securitization (in the U.S.) was invented in the 1970s in order to facilitate the growth of the secondary mortgage market where newly originated mortgages are sold to end-investors from the capital markets. In this period, demographic trends resulting from the baby-boomers reaching home-buying ages and the disintermediation of the S&Ls created a shortage of funds for households looking to purchase a home and led to Wall Street responding by creating the first mortgage-backed security.¹ Working through a tangle of legal, accounting, regulatory, tax and other issues in order to broaden the appeal of these types of securities took a number of years and eventually led to a significant involvement on part of the GSEs. In order to make purchasing MBS more palatable to end-investors, the GSEs established standardized underwriting standards, monitored adherence to these standards on the part of originators, and offered quasi-government guarantees on securitized products at attractive yields. As a result, investors shifted their focus from the credit risk of the underlying loans to analyzing the interest rate and prepayment risk associated with the timing at which cash flows are received.

Let's take a closer look at the anatomy of a securitization to understand how illiquid assets that would only be attractive to a few buyers are transformed into standardized financial instruments that can trade in the global markets.

6.1 Securitization: An Overview

Fundamentally, securitization is a technology used in Finance in order to obtain funding for illiquid assets. The idea is to pool together “similar” types of loans (assets) and convert them

¹The first MBS was issued in 1977 by Bank of America through Salomon Brothers.

into a **security** – namely, a financial asset that can be traded in the markets. This security is often generally called an **asset-backed security** or **ABS** for short. Of course, when the underlying assets backing a security are mortgages, we get a specific type of ABS – an MBS.

Clearly, the cash flows generated by the securitization structure will depend on the timing and consistency of the cash flows being generated by the underlying loans/assets that have been pooled together.² The assets are transferred into a **trust** or **special purpose vehicle (SPV)**³ that is structured in such a way so as to be isolated from the bankruptcy of the entity selling the assets. The securitization vehicle in turn can issue multiple **classes** (or **tranches**) of securities representing the right to receive different “slices” of the payment streams arising from the pooled assets. The return profile for different tranches can vary significantly and can be enhanced through different forms of credit enhancement. An example of a basic securitization structure is shown in Figure 6.1. We will review the key entities that are part of a securitization.⁴

1. **Asset originator.** As the name suggests, this entity originates the loans and then transfers them to the securitization entity. It may continue to service the assets.⁵
2. **Issuer.** Issuers of ABS are typically passive SPVs or trusts, created for the limited purpose of acquiring the underlying assets and issuing securities. Most SPVs have no employees but have third parties acting as agents that perform the issuer’s necessary day-to-day functions.
3. **Credit Rating Agencies (CRAs).** CRAs are corporations that play a key role in the securitization process by providing advisory services on the creditworthiness of bonds. In order to determine this, they collect information about bond issuers and the bonds that they have issued and then publish **ratings** of the individual bonds. Essentially, the rating provides an indication of the likelihood of default of a given bond. All of the major CRAs use an alphabet-oriented rating scale. For example, the CRA Standard and Poor’s (S&P) uses the rating scale AAA, AA, A, BBB, BB, etc., down to C and then D, with pluses and minuses possible for all of the grades from AA to CCC. S&P describes AAA as indicating that the issuer has an “extremely strong capacity to meet financial commitments,” whereas D indicates “payment default on financial commitments.”

Many securitizations will contain multiple tranches, most of which will be rated by one or more rating agencies. By creating different tranches with different ratings, the securitization can appeal to a wider variety of investors. Pension funds and life insurance companies, for example, may be limited by regulations from purchasing non-investment grade securities.⁶

²The underlying loans/assets of a securitization are often referred to as **collateral**.

³Also referred to as a **special purpose entity (SPE)**.

⁴See [Telpner03] for further details.

⁵Recall from Chapter 4 that there are two assets associated with each originated loan – the loan itself and the right to service it.

⁶A security is considered investment grade if it is rated BBB- or higher by S&P.

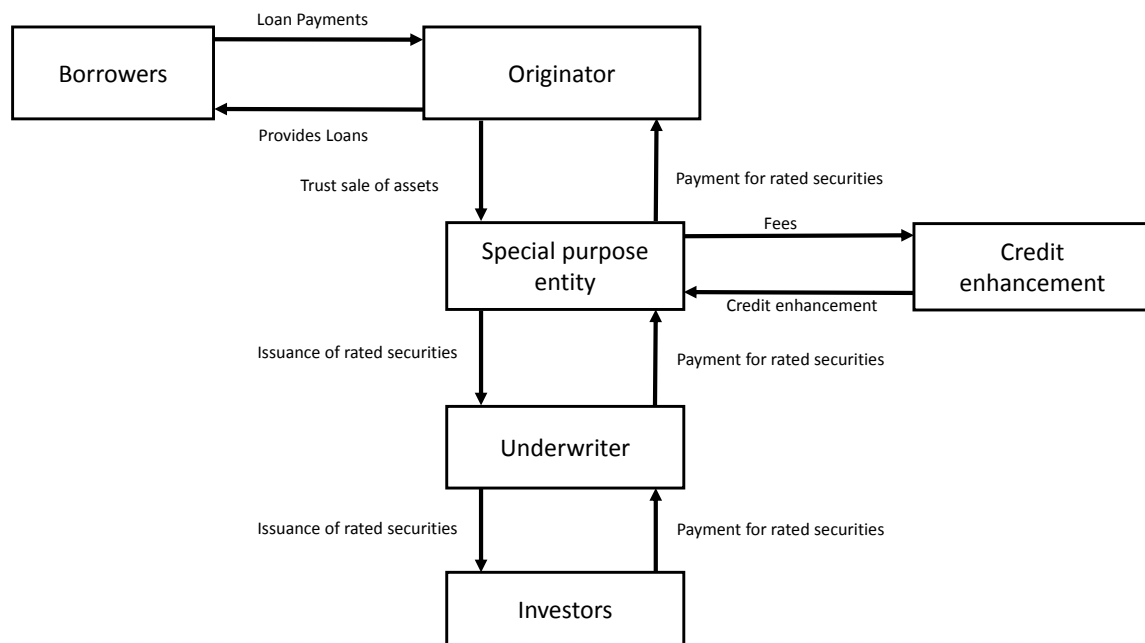
4. **Credit enhancement provider.** Depending on how a securitization is structured, highly rated credit enhancement providers are used to provide high credit ratings to the securitizations' senior tranches. A **credit enhancement** is protection, in the form of financial support, to cover losses in adverse conditions. Examples of credit enhancements include standby letters of credit, guaranties, and credit default swaps.
5. **Liquidity facility provider.** Certain securitization structures benefit from access to a liquidity facility provided by a financial institution in the form of a commitment to lend, a commitment to purchase assets or a letter of credit. Liquidity facilities are used in structures to cover potential time lags between inflows of revenue from the securitization's asset pool and its payment obligations under the ABS.⁷
6. **Underwriter.** The underwriter assumes responsibility for both pricing and marketing the tranches in a securitization. Accordingly, the underwriter takes a leading role in structuring securitizations based on what the underwriter believes will attract the largest number of potential investors.
7. **Servicer.** The servicer (who may also be the asset originator) is responsible for routine asset portfolio administration duties, such as making and processing collections, temporarily reinvesting asset proceeds and administering the day-to-day operations of the issuer.
8. **Administrator.** Administrators play an important role in certain types of securitizations such as Collateralized Debt Obligations (CDOs) and ABCPs. In a CDO securitization, the administrator actively manages and trades the underlying loan pool.
9. **Trustee.** The trustee's primary role involves holding the securitization cashflows in segregated accounts and notifying rating agencies and investors of events of default and covenant breaches.

For investors to be comfortable about investing in securitized products, several standards need to be met.

- **Standardized contracts.** The collateral backing loans in the pool needs to be available to creditors if the borrower/obligor were to default on their payments. This is only possible in a legal system which has well-defined mechanisms for repossessing collateral from the underlying borrower.
- **Underwriting.** The borrowers in the pool have been qualified based on a standard set of underwriting criteria – all borrowers have been vetted by their ability to make a downpayment, their cash flow liquidity, and their history of making timely payments on their existing debt burdens.

⁷An example of a securitization structure that uses a liquidity facility provider is an **asset-backed commercial paper (ABCP)** program. These are used to finance the assets by issuing short-term commercial paper but may hold assets with much longer maturities.

Figure 6.1: Overview of a Basic Securitization



Source: [Telpner03]

- **Historical data.** A database of historical statistics allows investors to subject the pool to a range of stress tests to determine how it would perform under differing conditions. Without getting a sense of this performance, the financial guarantees or excess collateral required to enhance the ratings to optimal levels for saleability in the market can be prohibitively expensive.
- **Standardization of applicable laws.** Since a securitization is a legal contract and institutional investors tend to be highly regulated entities, investors must be highly confident as to how federal, state, and local laws will affect their rights.
- **Standardization of servicer quality.** Standards specifying the quality of the servicers and the contracts under which they operate. The bankruptcy of the servicer of the sale of servicing rights cannot expose the investors to loss.
- **Supply of quality credit enhancers.** The rating on a security issue cannot be higher than the rating of the weakest link in the transaction (the so-called “weakest link” theory) and that is the quality of the credit enhancer. Each enhancer is rated individually, and if its rating changes, so may the ratings of all the issues it has guaranteed.
- **Hardware and software tools to analyze securitizations.** It should be relatively straightforward for investors to be able to track and project cash flows and monitor the many detailed elements in transactions on a daily and monthly basis.
- **Governance, Oversight and Fiduciary Responsibilities.** An independent party that represents the interests of all investors in the trust, subject to robust and continuous fiduciary duties of care and loyalty.⁸

6.2 The Pros and Cons of Securitization

Securitization achieves several objectives for the asset originator:

1. It removes the applicable financial assets and related indebtedness (financing costs) from the seller’s balance sheet;
2. It provides funding at lower costs than available through alternate means of financing (the debt markets) as a result of isolating the assets from the potential bankruptcy risk of the originator; and

⁸This requirement is relatively recent and arose from the collapse of the private-label MBS market after the Housing Crisis. Essentially, the existing framework for securitization suffered from a structural flaw in that there was no single party tasked with (1) providing fair protection to all investors in the trust; (2) adapting to events not specifically contemplated in the securitization contracts; and (3) offering transparency into any actions taken. This independent party is referred to as a **Deal Agent**. A framework of including Deal Agents as part of MBS transactions is still being worked on. See [Goodman16] for further details.

3. It improves the originator's liquidity including by means of expanding the range of investors available to the originator.

On the negative side, securitization can lead to **principal-agent** problems⁹ when the firm that originates the loans that are to be sold retains little or no interest in the pool of securitized assets. In this case, the originator does not have the same incentive to pay attention to the credit worthiness of its customers as would be the case if it had to portfolio the loans. The idea that securitization reduces the incentive to closely screen borrowers has often been cited as one of the major contributors to the collapse of the Subprime and Alt-A mortgage markets.¹⁰

6.3 Mortgage-backed Securities

Let's narrow our focus by looking at Figure 6.1 in the context of MBS. An MBS is a fixed-income security that represents a claim on a pool of mortgage loans, typically on residential properties. MBSs are created by aggregating a number of mortgages with similar amounts, terms, rates, and structure into a single pool. An MBS is issued with a face amount equal to the cumulative outstanding principal balance of the mortgages (original balance). Since the mortgages underlying a pool may not have identical characteristics, the properties of an MBS are often described using weighted averages: **WAM**: weighted average maturity; **WAC**: weighted average coupon, and **WALA**: weighted average loan age. The MBS **Coupon** is the interest rate paid on the MBS – the Coupon is lower than the WAC – the difference is equal to fees collected for servicing the underlying mortgages and any guarantee fee collected.

For an Agency MBS, the credit enhancement comes from the guarantee of the U.S. Agencies (Ginnie Mae, Fannie Mae, Freddie Mac) of “full and timely payment of principal and interest.” This essentially means that there is little to no credit risk in Agency MBS because these Agencies are considered to be backed by a full or implied government guarantee. Historically, credit enhancement in non-agency mortgage securitizations came from a number of different sources including corporate guarantees, letters of credit, surety bonds, pool insurance, spread accounts, reserve funds and subordinate classes. Over time, the provision of internal credit enhancement through the creation of **subordinate classes** has become the dominant way to shelter investors from the credit risk in a typical non-agency MBS transaction. In a nutshell, the subordinate class provides internal credit enhancement to senior classes in the structure by getting first priority on any losses experienced by the underlying collateral.

⁹The **principal-agent problem** or **agency dilemma** occurs when one person or entity (the “agent”) is able to make decisions that impact, or are on behalf of, another person or entity, the “principal.” There is a conflict of interest in that the agent's best interests may not be aligned with that of the principal's. For example, in the case of securitization, the originator securitizing their loans plays the role of the agent for the end-investor, the principal. In this case, the originator may be compensated purely on origination fees and the difference between where they bought the loans and were able to sell them, while the principal ends up with the credit and interest rate risk of the loans. The originator has an incentive to ramp up the flow of loans by cutting corners since they do not bear any of the credit risk which is of course opposed to the interests of the investor.

¹⁰See [Martin09].

The underwriter in a Non-agency MBS structures the MBS and places with resulting bonds with institutional investors. For Agency MBS, no need for additional structuring may be necessary since the significant majority of all Agency MBS consist of pools with pass-through cash flows. However, there is an active market in slicing and dicing Agency MBS pass-through cash flows to create Collateralized Mortgage Obligations (CMOs). We'll discuss this market in more detail in the next chapter.

6.3.1 MBS Cash Flows

It is instructive to understand the cash flows generated by an Agency mortgage-backed security to get a “live” example of what it would mean to invest in such an instrument. We assume that the Agency MBS is constructed from a \$100mm pool of mortgages with the following characteristics (see Table 6.1): The results are summarized in Table 6.2. The

Table 6.1: MBS Characteristics

Current Balance	\$100 million
WAC	8%
Coupon	7.5%
WAM	355 months
Prepay Assumption	150% PSA

Source: Cello

following assumptions are worth noting:

1. To generate the cash flows, we have to think of the MBS as a “giant” mortgage with a mortgage rate of 8%, a term of 355 months, and a starting principal balance of \$100 million. Only 7.50% of that 8% is paid out to MBS investors as interest, the remainder is collected by the servicer and by the relevant Agency. Both the WAC and the Coupon are quoted as annual rates and need to be converted to monthly rates since an MBS generates cash flows on a monthly basis.¹¹
2. The first month’s payment M is given by (see (3.6)):

$$M = \frac{\frac{8\%}{12} \cdot \$100 \text{ million}}{1 - 99.338\%^{355}} = \$736,267.83.$$

3. Of the first month’s payment of \$736,267.83, $\frac{0.50\%}{12} \cdot \$100 \text{ million} = \$41,666.67$ is the servicing fee, $\frac{7.50\%}{12} \cdot \$100 \text{ million} = \$625,000$ is MBS interest, and the **scheduled principal payment** is $M - \text{Interest} = \$69,601.16$.

¹¹Corresponding to the fact that homeowners make monthly payments on their mortgages.

4. Using the 150% PSA prepayment assumption and a starting age of 5 months, the prepayment rate for the first month (corresponding to an age of 6 months) is given by:

$$\begin{aligned}\text{CPR} &= 1.50 \left(\frac{6}{30} \right) 6\% = 1.8\% \\ \text{SMM} &= 1 - (1 - 1.8\%)^{1/12} = 0.15125\%\end{aligned}$$

5. Given the prepayment rate, the **prepaid principal** in the first month is \$151,146.65. (see (5.3)). This is computed as:

$$\begin{aligned}\text{Prepaid principal} &= \text{SMM} \times \text{Scheduled Principal Balance} \\ &= \text{SMM} \times [\text{Beginning Principal Balance} - \text{Scheduled Principal}] \\ &= 0.15125\% \times [\$100,000,000 - \$69,601.16] \\ &= \$151,146.65.\end{aligned}$$

6. Thus, for the first month, the MBS generates an estimated cash flow of \$845,747.81 (MBS Interest + Scheduled Principal + Prepaid Principal). The balance for the beginning of the next month is \$99,779,252.19 (Initial Beginning Balance - Scheduled Principal - Prepaid Principal).
7. The same sequence of calculations need to be gone through to calculate the next month's cash flows beginning with coming up with the monthly payment for a mortgage with a rate of 8%¹², a balance of \$99,779,252.19, and a term of 354 months.

¹²Under the assumption that the pool's averaged collateral characteristics have not "drifted" because of varying prepayment rates of borrowers in the underlying pool of mortgages.

Table 6.2: MBS Pass-through Cash Flows

Period	Actual Balance	Scheduled			Servicing Fee	MBS Interest	Scheduled		Total Principal	Cash Flow	SMM
		Principal Balance					Principal	Prepaid Principal			
0	\$100,000,000										
1	\$99,779,252.19	\$99,930,398.84	\$41,666.67	\$625,000.00	\$69,601.16	\$151,146.65	\$220,747.81	\$845,747.81	0.15125%		
2	\$99,533,099.37	\$99,709,292.99	\$41,574.69	\$623,620.33	\$69,959.20	\$176,193.62	\$246,152.82	\$869,773.15	0.17671%		
3	\$99,261,650.29	\$99,462,798.22	\$41,472.12	\$622,081.87	\$70,301.14	\$201,147.93	\$271,449.07	\$893,530.94	0.20223%		
4	\$98,965,033.34	\$99,191,023.60	\$41,359.02	\$620,385.31	\$70,626.70	\$225,990.25	\$296,616.95	\$917,002.27	0.22783%		
5	\$98,643,396.44	\$98,894,097.79	\$41,235.43	\$618,531.46	\$70,935.56	\$250,701.35	\$321,636.90	\$940,168.36	0.25350%		
:											
22	\$89,582,468.30	\$90,215,272.50	\$37,620.28	\$564,304.20	\$73,399.35	\$632,804.20	\$706,203.55	\$1,270,507.75	0.70144%		
23	\$88,857,032.03	\$89,509,097.90	\$37,326.03	\$559,890.43	\$73,370.39	\$652,065.87	\$725,436.26	\$1,285,326.69	0.72849%		
24	\$88,112,838.05	\$88,783,710.56	\$37,023.76	\$555,356.45	\$73,321.47	\$670,872.51	\$744,193.98	\$1,299,550.43	0.75563%		
25	\$87,350,374.62	\$88,039,585.50	\$36,713.68	\$550,705.24	\$73,252.55	\$689,210.88	\$762,463.43	\$1,313,168.67	0.78284%		
26	\$86,593,968.30	\$87,277,211.00	\$36,395.99	\$545,939.84	\$73,163.63	\$683,242.69	\$756,406.32	\$1,302,346.16	0.78284%		
:											
100	\$44,515,680.22	\$44,866,917.31	\$18,722.41	\$280,836.20	\$66,874.12	\$351,237.09	\$418,111.21	\$698,947.40	0.78284%		
101	\$44,100,922.71	\$44,448,887.28	\$18,548.20	\$278,223.00	\$66,792.94	\$347,964.57	\$414,757.51	\$692,980.51	0.78284%		
102	\$43,689,492.54	\$44,034,210.86	\$18,375.38	\$275,630.77	\$66,711.85	\$344,718.31	\$411,430.17	\$687,060.93	0.78284%		
103	\$43,281,363.58	\$43,622,861.67	\$18,203.96	\$273,059.33	\$66,630.87	\$341,498.10	\$408,128.97	\$681,188.30	0.78284%		
:											
200	\$15,978,415.73	\$16,104,488.44	\$6,734.88	\$101,023.21	\$59,225.01	\$126,072.70	\$185,297.72	\$286,320.93	0.78284%		
201	\$15,794,639.94	\$15,919,262.62	\$6,657.67	\$99,865.10	\$59,153.12	\$124,622.68	\$183,775.80	\$283,640.89	0.78284%		
:											
353	\$98,569.13	\$99,346.86	\$61.89	\$928.30	\$49,180.53	\$777.73	\$49,958.26	\$50,886.56	0.78284%		
354	\$49,061.20	\$49,448.30	\$41.07	\$616.06	\$49,120.83	\$387.10	\$49,507.93	\$50,123.99	0.78284%		
355	\$0.00	\$0.00	\$20.44	\$306.63	\$49,061.20	\$0.00	\$49,061.20	\$49,367.83	0.78284%		

Source: Cello

6.3.2 Engineering MBS Cash Flows

The cash flows listed in 6.2 are called **pass-through** cash flows since they represent a direct transfer of mortgage payments from homeowners to investors (less some fees). Investors who purchase an interest in an MBS pass-throughs share the principal and interest payments on a pro-rata basis. **Collateralized mortgage obligations (CMOs)** are formed by dividing the cash flows of an underlying pool of mortgages or an MBS issue into several classes. The different classes in a CMO are also referred to as **tranches**, from the French word for “slice.” The CMO market will be analyzed in further detail in the next chapter but for now we are most interested in seeing how financial engineering redistributes cash flows (at both the principal and interest level) to different classes (bonds) within a single securitization. We analyze two relatively simple examples of such cash flow engineering below.

Sequential Classes

A CMO with sequential-pay tranches is called a **sequential** CMO. Sequential CMOs are divided into bond classes (tranches) with different priority claims on the collateral’s principal. The tranche with the first priority claim has its principal paid entirely before the next priority class, which in turn has its principal paid before the third class, and so on. However, interest payments on the sequential tranches are not sequential and are made until the tranche’s principal is retired. Cash flows on the various tranches of a simple sequential CMO are shown in Table 6.3. This CMO consists of three tranches, A, B, and C, formed from the collateral in Table 6.1. We will assume the following principal balances for each tranche: Tranche A = \$50 million, Tranche B = \$30 million, and Tranche C = \$20 million. We also assume that the prepayment speed on the underlying collateral is 150% PSA.

The following priority disbursement rules apply:

1. Tranche A receives all principal payments from the collateral until its principal of \$50 million is retired. No other tranche’s principal payments are disbursed until the principal on A is paid.
2. After Tranche A’s principal is retired, all principal payments on the collateral are then made to Tranche B until its principal of \$30 million is retired.
3. Finally, Tranche C receives the remaining principal until it is paid off.
4. Although the principal is paid sequentially, each tranche does receive interest each period equal to its stated coupon (7.5%) times its outstanding balance at the beginning of the month.

Table 6.3: Sequential CMO Cash Flows

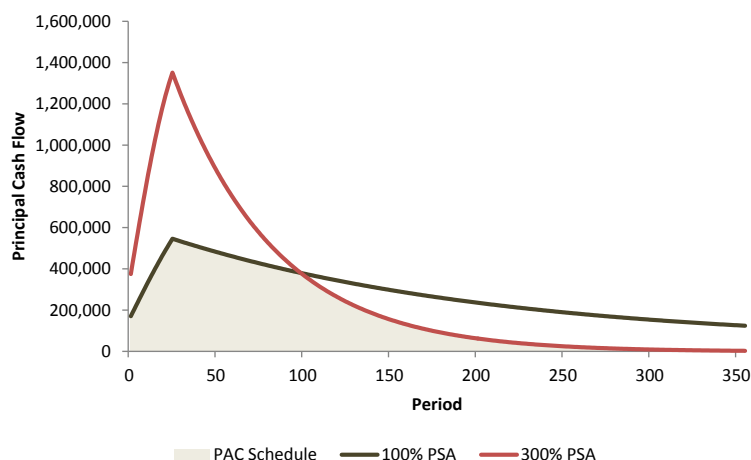
Period	Collateral Balance	MBS Interest	MBS Principal	Tranche A Balance	A Interest	A Principal	Tranche B Balance	B Interest	B Principal	Tranche C Balance	C Interest	C Principal
0	\$100,000,000.00			\$50,000,000.00			\$30,000,000.00			\$20,000,000.00		
1	\$99,779,252.19	\$625,000.00	\$220,747.81	\$49,779,252.19	\$312,500.00	\$220,747.81	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
2	\$99,533,099.37	\$623,620.33	\$246,152.82	\$49,533,099.37	\$311,120.33	\$246,152.82	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
3	\$99,261,650.29	\$622,081.87	\$271,449.07	\$49,261,650.29	\$309,581.87	\$271,449.07	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
4	\$98,965,033.34	\$620,385.31	\$296,616.95	\$48,965,033.34	\$307,885.31	\$296,616.95	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
5	\$98,643,396.44	\$618,531.46	\$321,636.90	\$48,643,396.44	\$306,031.46	\$321,636.90	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
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85	\$51,154,748.94	\$322,665.46	\$471,724.34	\$1,154,748.94	\$10,165.46	\$471,724.34	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
86	\$50,686,799.49	\$319,717.18	\$467,949.45	\$686,799.49	\$7,217.18	\$467,949.45	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
87	\$50,222,595.27	\$316,792.50	\$464,204.22	\$222,595.27	\$4,292.50	\$464,204.22	\$30,000,000.00	\$187,500.00	\$0.00	\$20,000,000.00	\$125,000.00	\$0.00
88	\$49,762,106.86	\$313,891.22	\$460,488.41	\$0.00	\$1,391.22	\$222,595.27	\$29,762,106.86	\$187,500.00	\$237,893.14	\$20,000,000.00	\$125,000.00	\$0.00
89	\$49,305,305.08	\$311,013.17	\$456,801.78	\$0.00	\$0.00	\$0.00	\$29,305,305.08	\$186,013.17	\$456,801.78	\$20,000,000.00	\$125,000.00	\$0.00
⋮												
177	\$20,650,839.00	\$130,466.76	\$223,841.95	\$0.00	\$0.00	\$0.00	\$650,839.00	\$5,466.76	\$223,841.95	\$20,000,000.00	\$125,000.00	\$0.00
178	\$20,428,822.73	\$129,067.74	\$222,016.27	\$0.00	\$0.00	\$0.00	\$428,822.73	\$4,067.74	\$222,016.27	\$20,000,000.00	\$125,000.00	\$0.00
179	\$20,208,617.76	\$127,680.14	\$220,204.97	\$0.00	\$0.00	\$0.00	\$208,617.76	\$2,680.14	\$220,204.97	\$20,000,000.00	\$125,000.00	\$0.00
180	\$19,990,209.82	\$126,303.86	\$218,407.94	\$0.00	\$0.00	\$0.00	\$0.00	\$1,303.86	\$208,617.76	\$19,990,209.82	\$125,000.00	\$9,790.18
181	\$19,773,584.76	\$124,938.81	\$216,625.06	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$19,773,584.76	\$124,938.81	\$216,625.06
⋮												
353	\$98,569.13	\$928.30	\$49,958.26	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$98,569.13	\$928.30	\$49,958.26
354	\$49,061.20	\$616.06	\$49,507.93	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$49,061.20	\$616.06	\$49,507.93
355	\$0.00	\$306.63	\$49,061.20	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$0.00	\$306.63	\$49,061.20

Source: Cello

Planned Amortization Classes

A **planned amortization class (PAC)** is structured so that there is a significant reduction in the variability of the principal and interest cash flows associated the bond. In other words, changes in prepayment speed assumptions will (subject to certain assumptions) not affect the principal and interest cash flows received by the PAC investor. In a PAC-structured CMO, cash flows generated by the underlying mortgages (i.e., the collateral) are divided into two general tranches: (1) The PAC, and (2) The **support class** (also called the **support bond** or **companion bond**). These two tranches are formed by generating two monthly principal payment schedules from the collateral: one schedule is based on assuming a relatively low PSA speed – **lower collar**, and the other on assuming a relatively high PSA speed – **upper collar**. The PAC bond is then set up so that it will receive a monthly principal payment schedule based on the *minimum principal* from these two principal payments. The shaded area in Figure 6.2 provides a visual depiction of how the PAC principal payment schedule is set up.

Figure 6.2: Creation of PAC Principal Schedule



Source: Cello

The PAC bond is designed to have no variation in its cash flows provided the actual prepayment falls within the minimum and maximum assumed PSA speeds. The support bond, on the other hand, receives the remaining principal balance and is therefore subject to significant

variability in its cash flows. Cash flows on the various tranches of a simple PAC CMO for a prepayment assumption of 150% PSA are shown in Table 6.4. This CMO consists of two tranches, PAC and Support, formed from the collateral in Table 6.1. Some noteworthy points about this CMO structure:

1. The first step is to calculate the total principal payments (scheduled and prepaid) for both collars. The principal payment schedule of the PAC is given by the minimum of the two payments. For example, in the first month, the principal payment is \$170,085 for the 100% PSA and \$374,456 for the 300% PSA; thus the principal payment for the PAC would be \$170,085. For the first 98 months, the minimum principal payment comes from the 100% PSA collar, and from month 99 onwards the minimum principal payment comes from the 300% PSA collar.
2. The sum of the PAC's principal payments is \$63.777 million. The support bond has a par value of \$36.223 million – the original face of the MBS less the principal balance of the PAC.
3. The PAC bond shows no cash flow variability as long as the actual prepayment speed is between 100% and 300% PSA.
4. As long as actual prepayments are between 100% to 300% PSA, the principal schedule for the PAC is fully deterministic. Principal cash flows for the companion are given by the MBS principal cash flows less PAC principal cash flows.
5. If prepayment speeds are greater than 300% PSA (not shown in the table), then at some point the companion is completely paid down and the PAC begins to receive the excess principal. In this case we say that the PAC band is **busted**.
6. If prepayment speeds are slower than 100% PSA (not shown in the table), then the PAC receives less than its scheduled principal payment and the companion receives no principal cash flows. This situation continues until the MBS principal payment is greater than or equal to the PAC's scheduled principal payment.

Table 6.4: PAC CMO Cash Flows

Period	Collateral Balance	MBS Interest	MBS Principal	PAC Principal Sch.	PAC Balance	PAC Interest	PAC Cash Flow	Support Principal	Support Balance	Support Interest	Support Cash Flow
0	\$100,000,000.00				\$63,777,029.85			\$36,222,970.15			
1	\$99,779,252.19	\$625,000.00	\$220,747.81	\$170,085.43	\$63,606,944.42	\$398,606.44	\$568,691.87	\$50,662.38	\$36,172,307.77	\$226,393.56	\$277,055.94
2	\$99,533,099.37	\$623,620.33	\$246,152.82	\$187,134.85	\$63,419,809.56	\$397,543.40	\$584,678.26	\$59,017.97	\$36,113,289.80	\$226,076.92	\$285,094.89
3	\$99,261,650.29	\$622,081.87	\$271,449.07	\$204,125.48	\$63,215,684.08	\$396,373.81	\$600,499.29	\$67,323.59	\$36,045,966.21	\$225,708.06	\$293,031.65
⋮											
98	\$45,355,282.89	\$286,126.13	\$424,898.49	\$381,870.80	\$21,256,120.18	\$135,237.44	\$517,108.25	\$43,027.69	\$24,099,162.71	\$150,888.69	\$193,916.37
99	\$44,933,791.42	\$283,470.52	\$421,491.46	\$379,498.54	\$20,876,621.63	\$132,850.75	\$512,349.29	\$41,992.92	\$24,057,169.79	\$150,619.77	\$192,612.69
100	\$44,515,680.22	\$280,836.20	\$418,111.21	\$372,970.33	\$20,503,651.31	\$130,478.89	\$503,449.21	\$45,140.88	\$24,012,028.91	\$150,357.31	\$195,498.19
101	\$44,100,922.71	\$278,223.00	\$414,757.51	\$366,552.18	\$20,137,099.13	\$128,147.82	\$494,700.00	\$48,205.33	\$23,963,823.58	\$150,075.18	\$198,280.51
⋮											
199	\$16,163,713.45	\$102,190.91	\$186,831.73	\$64,245.42	\$3,153,059.25	\$20,108.15	\$84,353.57	\$122,586.32	\$13,010,654.20	\$82,082.75	\$204,669.07
200	\$15,978,415.73	\$101,023.21	\$185,297.72	\$63,078.71	\$3,089,980.54	\$19,706.62	\$82,785.33	\$122,219.01	\$12,888,435.20	\$81,316.59	\$203,535.59
201	\$15,794,639.94	\$99,865.10	\$183,775.80	\$61,932.26	\$3,028,048.28	\$19,312.38	\$81,244.63	\$121,843.54	\$12,766,591.66	\$80,552.72	\$202,396.26
202	\$15,612,374.06	\$98,716.50	\$182,265.88	\$60,805.70	\$2,967,242.58	\$18,925.30	\$79,731.01	\$121,460.17	\$12,645,131.48	\$79,791.20	\$201,251.37
⋮											
353	\$98,569.13	\$928.30	\$49,958.26	\$2,627.17	\$5,052.56	\$48.00	\$2,675.17	\$47,331.09	\$93,516.57	\$880.30	\$48,211.38
354	\$49,061.20	\$616.06	\$49,507.93	\$2,559.46	\$2,493.10	\$31.58	\$2,591.04	\$46,948.47	\$46,568.10	\$584.48	\$47,532.95
355	\$0.00	\$306.63	\$49,061.20	\$2,493.10	\$0.00	\$15.58	\$2,508.68	\$46,568.10	\$0.00	\$291.05	\$46,859.15

Source: Cello

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